Table of Contents

SUPPLY MANAGEMENT SYSTEM

Functional Specification Document — Consolidated Master Edition

Version 4.0 | Full-Purpose Supply Chain Solution

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Prepared by	Business Analysis Team
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Scope	All Modules — End-to-End Supply Chain, including all approved addenda through Addendum 20
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Supersedes	Supply_Management_System_FSD.docx (v1),

Supply_Management_System_FSD_v2.docx (v2), Supply_Management_System_FSD_v3.docx (v3), and Addenda 5A, 8A, 17, 18, 19, 20 as standalone documents

Consolidation Note

This document merges the base Functional Specification Document (FSD v3 — itself a superset of v1 and v2, which are fully absorbed and retired) with all six approved FSD addenda into a single, non-duplicated master reference. No requirement, field definition, business rule, workflow step, lookup value, or acceptance criterion from any source document has been removed; content has only been re-sequenced into the module it functionally belongs to. Every addendum's own "purpose / document-control" framing has been condensed and its traceability information moved to Appendix B (Consolidation Source Map) at the end of this document, so that nothing is lost while the main body reads as one coherent specification rather than a stack of patches. Where an addendum corrected or extended a specific existing subsection, its content is appended immediately after that subsection using the same numbering convention (e.g. Section 6.7 onward for Addendum 17 items). Where an addendum introduced an entirely new, self-contained capability that sits between two existing modules, it is inserted as a lettered section (5A, 8A, 15A) immediately after the module it extends, so existing section numbers 1–17 are never disturbed and all pre-existing cross-references in this document remain valid.

1 Introduction

1.1 Purpose

This Functional Specification Document (FSD) describes the complete functional requirements, data models, business workflows, lookup collections, and role-based access controls for the Supply Management System (SMS). It serves as the authoritative reference for design, development, testing, and deployment teams.

1.2 System Overview

The SMS is a full-purpose, end-to-end supply chain platform covering eight integrated modules: Supplier Management, Demand Planning, Procurement, Inventory Management, Order Management, Warehouse Management, Logistics & Shipping, Finance & Payments, and Reports & Analytics.

1.3 Scope

- Multi-supplier, multi-warehouse, multi-currency support
- Full procure-to-pay lifecycle management
- Real-time inventory tracking with reorder automation
- Role-based access control with separation of duties
- Comprehensive audit trail and reporting

1.4 Definitions

PO	Purchase Order — a commercial document issued to a supplier

RFQ	Request for Quotation — sent to suppliers to obtain price quotes
GRN	Goods Receipt Note — confirms goods received from supplier
3WM	Three-Way Match — PO + GRN + Invoice reconciliation
SKU	Stock Keeping Unit — unique product identifier
CRUD	Create, Read, Update, Delete — standard data operations
AP	Accounts Payable
AR	Accounts Receivable
BOM	Bill of Materials
SLA	Service Level Agreement

2 User Roles & Permissions

The SMS enforces role-based access control (RBAC). Each user is assigned exactly one role. Permissions are defined at screen and action level.

2.1 Role Definitions

Role	Level	Responsibilities & Access
System Admin	Full access	Configure system, manage users, all modules, audit logs, global settings
Procurement Manager	Managerial	Approve POs, manage suppliers, RFQs, contracts, budgets
Purchase Officer	Operational	Create requisitions, issue POs, track deliveries, view inventory
Inventory Manager	Operational	Manage stock, adjustments, receipts, reorder rules, warehouse transfers
Warehouse Operator	Restricted	Receive goods, put-away, picking, dispatch, update stock locations
Finance Officer	Operational	Process invoices, payments, reconciliation, budget monitoring
Requester	Limited	Raise purchase requisitions, view own order status
Read-Only / Auditor	View only	View all records, export reports, no data modification

Separation of Duties rules:

- The user who creates a PO cannot be the same user who approves it.
- The user who receives goods (GRN) cannot process the payment for the same order.
- Supplier creation requires Procurement Manager approval.

Prior to Addendum 17, Roles referenced throughout Section 2.1 were a fixed, hard-coded list with no administrative screen. Section 2.2 below adds full CRUD management of Roles under the User & Access menu.

2.2 Item 3 — Role CRUD (New Feature) (Addendum 17)

#3	**NEW FEATURE**	**Role CRUD under User & Access menu**

2.2.1 Background

FSD v3 Section 2 defines 8 fixed roles (System Admin, Procurement Manager, Purchase Officer, Inventory Manager, Warehouse Operator, Finance Officer, Requester, Read-Only/Auditor) as a static reference list. There is currently no screen for an Admin to view, create, edit, or manage roles and their permission assignments. This addendum adds that screen.

2.2.2 Navigation

Menu path	User & Access › Roles
Route	/admin/roles
Visibility / RBAC	System Admin only — same restriction as User CRUD (FSD v3 Section 2)
Position in menu	Sibling entry alongside the existing Users link under the User & Access menu group

2.2.3 Role List Screen

Column	Source field	Behaviour
Role Code	role_code	Sortable, e.g. PROC_MGR
Role Name	role_name	Sortable, searchable, primary display
Description	description	Truncated with tooltip on hover
Active Users	Computed — COUNT(Users WHERE role_id = this row AND is_deleted = 0)	Read-only, links to filtered User list
Status	is_active	Toggle filter: All / Active / Inactive
Actions	—	Edit, Manage Permissions, Deactivate icons per row

2.2.4 Create / Edit Role Form

Field	Type	Required	Notes
role_code	Text (50)	Yes	Unique. Uppercase, underscore-separated. Immutable after creation.
role_name	Text (100)	Yes	Display name. Editable.
description	Text (500)	No	
permissions[]	Multi-select checklist	Yes (at least one)	Grouped by module (Procurement, Inventory, Finance etc.) for readability. Checking a permission creates a RolePermissions row.
is_active	Toggle	Yes	Default true on create.

2.2.5 Business Rules

- role_code cannot be edited once created — it is referenced by users, permissions, and historical audit records. Only role_name, description, permissions, and is_active are editable.
- A role cannot be deactivated while it has one or more active (non-deleted) users assigned. The system blocks deactivation and shows the count of affected users, directing the Admin to reassign those users first.
- Deleting a role outright is not permitted — only deactivation (is_active = false). This preserves referential integrity for historical audit and approval records that reference the role.
- The 8 roles defined in FSD v3 Section 2 (System Admin, Procurement Manager, Purchase Officer, Inventory Manager, Warehouse Operator, Finance Officer, Requester, Read-Only/Auditor) are seeded as system roles and cannot have their role_code changed, but their permission sets and descriptions remain editable by System Admin.
- Creating a new custom role beyond the original 8 is permitted — the role engine is not hardcoded to exactly 8 roles.

2.2.6 API Endpoints

Method	Endpoint	Description
GET	/api/roles	Paginated role list with active user counts
POST	/api/roles	Create new role with initial permission set
GET	/api/roles/{id}	Role detail including full permission list
PUT	/api/roles/{id}	Update role_name, description, is_active
PUT	/api/roles/{id}/permissions	Replace the full permission set for this role
PATCH	/api/roles/{id}/deactivate	Deactivate; blocked if active users exist
GET	/api/roles/{id}/users	List of users currently assigned this role

3 Module — Supplier Management

Manages the complete lifecycle of suppliers: onboarding, qualification, contract management, performance rating, and deactivation. All procurement activities originate from an approved supplier record.

3.1 Key Design Decisions

The following design decisions apply to the Supplier module and, as exemplary cases, establish the pattern for similar fields across all other modules in this system.

3.1.1 Preferred Payment Terms — Optional Field

The field preferred_payment_terms on the Supplier Master is intentionally non-mandatory. In real-world procurement, the actual payment terms for a transaction are negotiated and agreed at the time of Purchase Order issuance and may differ from any standing arrangement on the supplier's profile. Capturing a preferred value here is a convenience default only — it pre-fills the PO form but can always be overridden. Marking this field as required would force users to record inaccurate or speculative data during supplier onboarding, which degrades data quality. The same principle applies to all 'preferred' or 'default' fields across modules: they represent soft defaults, not contractual commitments.

3.1.2 Preferred Currency — Optional Field

Similarly, preferred_currency is non-mandatory. A supplier may transact in PKR for domestic orders and USD for imported goods within the same relationship. The actual transaction currency is confirmed at PO level, where it is mandatory. Storing a preferred currency on the supplier profile is informational only and must not be enforced as a constraint during PO creation. This pattern extends to any currency or financial preference field on master records across the system — always optional on the master, always confirmed on the transaction.

3.1.3 Supplier Type — Multi-Select with Junction Table

A supplier entity in the real world is not constrained to a single business type. A company may simultaneously operate as a Manufacturer (producing its own goods), a Trader (distributing third-party products), and an Importer. Forcing a single-select dropdown would require users to misrepresent the supplier's actual nature or create duplicate supplier records.

Design approach: The supplier_type field on the Supplier Master is replaced by a many-to-many relationship implemented via a junction table: SupplierTypeMapping. The UI renders this as a multi-select checkbox list or tag selector. The same design approach applies to any attribute where a real-world entity can legitimately hold multiple values simultaneously — for example, a warehouse that handles both standard and cold-chain goods, or a product that belongs to multiple categories.

3.1.4 Industry Category — Multi-Select with Junction Table

The same reasoning applies to industry_category. A supplier serving multiple sectors (e.g. Healthcare and Consumer Goods) must be discoverable under either category when procurement teams search or filter suppliers. A single-value field would make the supplier invisible to searches outside their primary category.

Design approach: Implemented via a SupplierIndustryMapping junction table. The UI renders as a multi-select with search. This ensures a supplier categorised as both 'Pharma' and 'FMCG' appears in filtered views for either category. All multi-select fields across the system follow this same junction table pattern.

3.2 Screens & Features

- Supplier List (searchable, filterable grid)
- Supplier Registration / Onboarding Form
- Supplier Profile (contacts, documents, bank details)
- Supplier Evaluation & Rating
- Contract Management
- Supplier Approval Workflow
- Supplier Portal (optional external login)

3.3 Supplier Master — Field Definitions

Field name	Data type	Required	CRUD	Lookup / values	Notes
supplier_id	Auto (UUID)	Yes	N/A	System generated	Primary key
supplier_code	Text (10)	Yes	Manual	Unique prefix e.g. SUP-0001	Business identifier
supplier_name	Text (200)	Yes	Free text	—	Legal trading name
supplier_type	Multi-select (junction)	Yes	Yes	Manufacturer / Trader / Service / Importer…	Multiple allowed — see SupplierType junction table
industry_category	Multi-select (junction)	Yes	Yes	IT / Pharma / FMCG / Construction…	Multiple allowed — see SupplierIndustry junction table
registration_no	Text (50)	Yes	Free text	NTN / SECP / CNIC	Business reg. number
tax_id	Text (30)	Yes	Free text	—	Tax identification number
country	Lookup	Yes	Yes	Pakistan / UAE / UK…	ISO country list
province_state	Lookup	Yes	Yes	Punjab / Sindh / KPK…	Dynamic by country
city	Text (100)	Yes	Free text	—	City name
address_line1	Text (200)	Yes	Free text	—	Street address
address_line2	Text (200)	No	Free text	—	Suite / Floor
postal_code	Text (20)	No	Free text	—	Zip / postal
phone	Text (20)	Yes	Free text	—	Primary phone
fax	Text (20)	No	Free text	—	Fax number
email	Email (150)	Yes	Free text	—	Primary contact email
website	URL (200)	No	Free text	—	Company website
primary_contact_name	Text (100)	Yes	Free text	—	Account manager name

Field name	Data type	Required	CRUD	Lookup / values	Notes
primary_contact_title	Text (100)	No	Free text	—	Job title
primary_contact_phone	Text (20)	Yes	Free text	—	Mobile number
primary_contact_email	Email (150)	Yes	Free text	—	Direct email
preferred_payment_terms	Lookup	No	Yes	Net 30 / Net 60 / COD / Advance	Optional default — actual terms agreed per transaction
preferred_currency	Lookup	No	Yes	PKR / USD / AED / EUR	Optional default — actual currency agreed per PO
bank_name	Text (100)	No	Free text	—	Beneficiary bank
bank_account_no	Text (30)	No	Free text	—	Account number
bank_iban	Text (34)	No	Free text	—	IBAN
bank_swift	Text (11)	No	Free text	—	SWIFT / BIC code
credit_limit	Decimal	No	Number	—	Max outstanding balance
lead_time_days	Integer	No	Number	—	Standard delivery days
status	Lookup	Yes	Yes	Active / Inactive / Blacklisted / Pending	See lookup below
rating	Decimal (1dp)	No	Auto	1.0 – 5.0 (system calculated)	Performance score
onboarding_date	Date	Yes	Date picker	—	When supplier was approved
last_review_date	Date	No	Date picker	—	Last evaluation date
approved_by	FK → User	Yes	Auto	System assigned	Who approved onboarding
notes	Text (500)	No	Free text	—	Internal remarks
created_at	DateTime	Yes	Auto	System generated	Record creation timestamp
updated_at	DateTime	Yes	Auto	System	Last update

Field name	Data type	Required	CRUD	Lookup / values	Notes
				generated	timestamp
created_by	FK → User	Yes	Auto	System assigned	Record creator

3.4 Junction Tables — Multi-Select Relationships

The following junction tables implement the multi-select behaviour for supplier_type and industry_category. These tables are maintained via the Supplier Profile screen. CRUD operations (add/remove associations) are performed by Procurement Manager and System Admin only.

Table: SupplierTypeMapping

Field	Data type	Required	Notes
mapping_id	Auto (UUID)	Yes	Primary key
supplier_id	FK → Supplier	Yes	The supplier being classified
supplier_type_id	FK → SupplierType	Yes	The type being assigned
is_primary	Boolean	No	Mark one type as the primary business type
assigned_by	FK → User	Yes	Who added this classification
assigned_at	DateTime	Yes	When the classification was added
notes	Text (200)	No	Optional context for this classification

Sample data — SupplierTypeMapping:

Supplier name	Type assigned	Is primary	Notes
Al-Fatah Enterprises	Manufacturer	Yes	Core manufacturing arm
Al-Fatah Enterprises	Trader	No	Also distributes third-party goods
MedPlus Pvt Ltd	Importer	Yes	Primarily imports from China
MedPlus Pvt Ltd	Trader	No	Resells imported pharma products
CitySupplies Co.	Service Provider	Yes	Maintenance and facility services
CitySupplies Co.	Trader	No	Also supplies consumables

Table: SupplierIndustryMapping

Field	Data type	Required	Notes
mapping_id	Auto (UUID)	Yes	Primary key

Field	Data type	Required	Notes
supplier_id	FK → Supplier	Yes	The supplier being classified
industry_id	FK → IndustryCategory	Yes	The industry sector being assigned
is_primary	Boolean	No	Mark one industry as the primary sector
assigned_by	FK → User	Yes	Who added this classification
assigned_at	DateTime	Yes	Timestamp
notes	Text (200)	No	Optional context

Sample data — SupplierIndustryMapping:

Supplier name	Industry assigned	Is primary	Notes
PharmaCo Ltd	Pharmaceuticals	Yes	Core business is pharma
PharmaCo Ltd	FMCG	No	Also supplies consumer health products
TechTrade Pvt	IT & Technology	Yes	Primary sector
TechTrade Pvt	Electronics	No	Also handles consumer electronics
BuildRight Corp	Construction	Yes	Primary sector
BuildRight Corp	Industrial / Manufacturing	No	Also supplies factory materials

3.5 Lookup Collections

Lookup: Supplier Type — Master (CRUD: Yes, Admin + Proc. Mgr)

Code / ID	Display value	Active	Notes
MFG	Manufacturer	Yes	Produces goods directly
TRADE	Trader / Distributor	Yes	Buys and resells
SVC	Service Provider	Yes	Provides services, not goods
IMPORT	Importer	Yes	Imports goods from abroad
GOVT	Government Entity	Yes	Public sector supplier

Lookup: Industry Category — Master (CRUD: Yes, Admin + Proc. Mgr)

Code / ID	Display value	Active	Notes
IT	IT & Technology	Yes	Software, hardware, telecoms
PHARMA	Pharmaceuticals	Yes	Medicines, medical

Code / ID	Display value	Active	Notes
			supplies
FMCG	FMCG / Consumer Goods	Yes	Fast-moving consumer products
CONST	Construction	Yes	Building materials and services
INDUST	Industrial / Manufacturing	Yes	Factory inputs and machinery
HEALTH	Healthcare	Yes	Hospitals, clinics, equipment
FOOD	Food & Beverage	Yes	Edible and packaged goods
ELEC	Electronics	Yes	Consumer and industrial electronics
LOGIST	Logistics / Transport	Yes	Freight and courier services
FIN	Financial Services	Yes	Banking, insurance, fintech

Lookup: Supplier Status (CRUD: Yes)

Code / ID	Display value	Active	Notes
PENDING	Pending Approval	Yes	Submitted, awaiting review
ACTIVE	Active	Yes	Approved, can receive POs
INACTIVE	Inactive	Yes	Temporarily disabled
BLACKLIST	Blacklisted	Yes	Permanently blocked
SUSPENDED	Suspended	Yes	Under investigation

Lookup: Payment Terms (CRUD: Yes)

Code / ID	Display value	Active	Notes
NET30	Net 30 Days	Yes	Payment due 30 days after invoice
NET60	Net 60 Days	Yes	Payment due 60 days after invoice
NET90	Net 90 Days	Yes	Payment due 90 days after invoice
COD	Cash on Delivery	Yes	Pay at time of delivery
ADV	Advance Payment	Yes	100% upfront
PART	Partial Advance	Yes	50% advance, 50% on delivery

3.6 Role-Based Access — Supplier Management

Screen / action	Admin	Proc. Mgr	Purchase Off.	Inv. Mgr	Requester
View supplier list	**Full**	**Full**	**View**	**View**	**None**
Add new supplier	**Full**	**Full**	**None**	**None**	**None**
Edit supplier details	**Full**	**Full**	**None**	**None**	**None**
Approve supplier	**Full**	**Full**	**None**	**None**	**None**
Blacklist supplier	**Full**	**Full**	**None**	**None**	**None**
View bank details	**Full**	**View**	**None**	**None**	**None**
Delete supplier record	**Full**	**None**	**None**	**None**	**None**
Export supplier data	**Full**	**Full**	**View**	**None**	**None**
Manage supplier contracts	**Full**	**Full**	**None**	**None**	**None**

4 Module — Demand Planning & Purchase Requisition

Captures internal demand through Purchase Requisitions (PRs), manages the quotation process, and converts approved demand into Purchase Orders. A PR may be converted into one or more POs (split by vendor). Multiple PRs can be consolidated onto a single PO. The module also supports an optional quotation cycle that can be initiated from either a PR or a PO, ensuring market rates are validated before commitment.

4.1 Screens & Features

- Requisition List (My PRs / All PRs — filterable grid)
- Create / Edit Purchase Requisition
- PR Line Item Management
- PR Approval Workflow
- Quotation Request — create from PR or PO
- Quotation Comparison (side-by-side vendor quotes)
- Convert PR → PO (single vendor or split by vendor)
- Consolidate multiple PRs → single PO
- Purchase Order List — manual create or PR-sourced
- PO Amendment Cycle (vendor feedback loop)
- Demand Forecast Dashboard
- Budget Check Integration

4.2 PR to PO Conversion — Process Rules

The PR-to-PO conversion is the central bridge between demand and procurement. The following rules govern how PRs become POs:

Rule	Description
Manual PO creation	A PO can be created directly from the PO screen without any PR. The user enters all details manually. No PR linkage is required.
Convert single PR to PO	From the PR detail screen, an approved PR can be converted to a PO. The system asks the user to select one vendor. All PR lines are carried to the PO. The PR status becomes Converted.
Split PR by vendor	Where different lines on a PR are sourced from different vendors, the Convert action allows the user to assign a vendor per line. The system then generates one PO per vendor, each containing only the lines assigned to that vendor.
Consolidate multiple PRs onto one PO	From the PO creation screen, the user can search and attach multiple approved PRs to a single PO, provided they share the same vendor. Lines from all attached PRs are combined on the PO.
PR-to-PO linkage tracking	Each PO line records the originating pr_id and pr_line_id. A PR is marked Partially Converted if only some lines have been converted, and Fully Converted when all lines are covered.
Quotation before PO	If the PR requester or Procurement Officer flags a PR line as 'Quotation Required', a Quotation Request (RFQ) must be completed and a winning quote selected before that line can be added to a PO.
PO without quotation	If no quotation is required, the Procurement Officer may source the rate directly (phone quote, catalogue, or framework contract) and enter the unit price manually on the PO line. The source of the rate must be documented in the line notes field.

4.3 Purchase Requisition Header — Field Definitions

Field name	Data type	Required	CRUD	Lookup / values	Notes
pr_id	Auto (UUID)	Yes	N/A	System generated	Primary key
pr_number	Text (20)	Yes	Auto	PR-YYYY-NNNNN	Human-readable number
pr_title	Text (200)	Yes	Free text	—	Short description of need
department	Lookup	Yes	Yes	Finance / IT / Operations / HR…	Requesting department
requester_id	FK → User	Yes	Auto	Current user	Who raised the PR
requested_date	Date	Yes	Date	—	When items are

Field name	Data type	Required	CRUD	Lookup / values	Notes
			picker		needed
priority	Lookup	Yes	Yes	Low / Normal / High / Urgent	See lookup below
pr_type	Lookup	Yes	Yes	Goods / Services / Asset / Recurring	PR category
requires_quotation	Boolean	No	Toggle	Yes / No	If Yes, RFQ must be done before PO
justification	Text (500)	Yes	Free text	—	Business reason
estimated_total	Decimal	Yes	Calculated	Sum of line items	Auto-calculated
budget_code	FK → Budget	No	Lookup	Active budget codes	Cost center / budget
budget_available	Decimal	No	Auto	System check	Available budget amount
status	Lookup	Yes	Auto	Draft / Submitted / Approved / Rejected / Partially Converted / Fully Converted	Workflow state
approved_by	FK → User	No	Auto	Set on approval	Approver user ID
approved_at	DateTime	No	Auto	System generated	Approval timestamp
rejection_reason	Text (300)	No	Free text	—	Required if rejected
linked_quotation_ids	FK[] → Quotation	No	Auto	Set when RFQ created	Quotations from this PR
linked_po_ids	FK[] → PO	No	Auto	Set on conversion	POs created from this PR
attachments	File[]	No	Upload	PDF/Word/Excel/ Image	Supporting documents
notes	Text (500)	No	Free text	—	Additional comments
created_at	DateTime	Yes	Auto	System generated	—
created_by	FK → User	Yes	Auto	System assigned	—

4.4 PR Line Item — Field Definitions

Field name	Data type	Required	CRUD	Lookup / values	Notes
line_id	Auto (UUID)	Yes	N/A	System generated	Primary key
pr_id	FK → PR	Yes	Auto	Parent PR	Link to header
line_no	Integer	Yes	Auto	1, 2, 3…	Line sequence
product_id	FK → Product	No	Lookup	Product catalog	Leave blank for free-text item
item_description	Text (300)	Yes	Free text	—	What is being requested
specification	Text (500)	No	Free text	—	Technical specs, size, grade
unit_of_measure	Lookup	Yes	Yes	Each / Kg / Litre / Box / Set	UOM
quantity	Decimal	Yes	Number	—	Requested quantity
estimated_unit_price	Decimal	Yes	Number	—	Expected unit cost
line_total	Decimal	Yes	Calculated	qty × unit price	Auto-calculated
preferred_supplier	FK → Supplier	No	Lookup	Active suppliers	Optional vendor preference
requires_quotation	Boolean	No	Toggle	Yes / No	Override PR-level flag per line
quotation_status	Lookup	No	Auto	Not Required / Pending / Received / Awarded	Quotation state for this line
awarded_quotation_line_id	FK → QuotationLine	No	Auto	Set when quote awarded	Winning quote line
line_status	Lookup	Yes	Auto	Open / Partially Converted / Fully Converted / Cancelled	Conversion state
required_date	Date	No	Date picker	—	Per-line delivery date if

Field name	Data type	Required	CRUD	Lookup / values	Notes
					different

4.5 Quotation (RFQ) — Full Specification

A Quotation (also called a Request for Quotation, RFQ) can be initiated from two starting points: (1) directly from an approved Purchase Requisition where one or more lines are flagged as 'Quotation Required', or (2) from a Purchase Order that has not yet been sent to a supplier, where the Procurement Officer wishes to validate market rates before committing. Both paths produce the same Quotation record structure.

Quotation Header — Field Definitions

Field name	Data type	Required	CRUD	Lookup / values	Notes
quotation_id	Auto (UUID)	Yes	N/A	System generated	PK
quotation_number	Text (20)	Yes	Auto	QTN-YYYY-NNNNN	Human-readable reference
quotation_title	Text (200)	Yes	Free text	—	Subject / description
source_type	Lookup	Yes	Yes	PR / PO / Standalone	Where quotation originated
source_pr_id	FK → PR	No	Lookup	Approved PRs	Populated if source = PR
source_po_id	FK → PO	No	Lookup	Draft POs	Populated if source = PO
issue_date	Date	Yes	Date picker	Default: today	Date RFQ issued
closing_date	Date	Yes	Date picker	—	Deadline for vendor submissions
status	Lookup	Yes	Auto	Draft / Sent / Responses Received / Under Evaluation / Awarded / Cancelled	Lifecycle state
invited_supplier_ids	FK[] → Supplier	Yes	Multi-select	Active suppliers	All vendors invited to quote
evaluation_criteria	Text (500)	No	Free text	Price / Delivery / Quality /	Scoring basis

Field name	Data type	Required	CRUD	Lookup / values	Notes
				Warranty	
awarded_supplier_id	FK → Supplier	No	Auto	Set on award	Winning vendor
award_date	Date	No	Auto	—	Date quotation awarded
award_notes	Text (300)	No	Free text	—	Justification for selection
terms	Text (500)	No	Free text	—	Special T&C communicated to vendors
created_by	FK → User	Yes	Auto	System assigned	—
created_at	DateTime	Yes	Auto	System generated	—

Quotation Line — Field Definitions

Field name	Data type	Required	CRUD	Lookup / values	Notes
qtn_line_id	Auto (UUID)	Yes	N/A	System generated	PK
quotation_id	FK → Quotation	Yes	Auto	Parent quotation	—
pr_line_id	FK → PR Line	No	Auto	Source PR line	If sourced from PR
po_line_id	FK → PO Line	No	Auto	Source PO line	If sourced from PO
item_description	Text (300)	Yes	Free text	—	Item being quoted
specification	Text (500)	No	Free text	—	Technical requirements
uom	Lookup	Yes	Yes	Each / Kg / Litre / Box	Unit of measure
quantity_requested	Decimal	Yes	Number	—	Quantity to be priced
line_no	Integer	Yes	Auto	1, 2, 3…	—

Vendor Quotation Response — Field Definitions

Each invited supplier submits one Vendor Quotation Response record against the Quotation. The system stores all responses for side-by-side comparison.

Field name	Data type	Required	CRUD	Lookup / values	Notes
response_id	Auto (UUID)	Yes	N/A	System generated	PK
quotation_id	FK → Quotation	Yes	Auto	Parent quotation	—
supplier_id	FK → Supplier	Yes	Lookup	Invited suppliers only	Responding vendor
response_date	Date	Yes	Date picker	—	Date vendor submitted quote
validity_days	Integer	Yes	Number	—	Days quote remains valid
currency	Lookup	Yes	Yes	PKR / USD / AED	Quote currency
total_quoted_value	Decimal	Yes	Calculated	Sum of response lines	Auto-calculated
delivery_lead_days	Integer	No	Number	—	Vendor-promised delivery time
payment_terms_offered	Lookup	No	Yes	Net 30 / COD / Advance	What vendor is offering
status	Lookup	Yes	Auto	Received / Under Review / Awarded / Rejected	Response state
rejection_reason	Text (300)	No	Free text	—	Required if rejected
attachments	File[]	No	Upload	PDF quotation scan	Vendor's formal quote doc
notes	Text (300)	No	Free text	—	Internal evaluation notes
entered_by	FK → User	Yes	Auto	Procurement Officer	Who entered the response
entered_at	DateTime	Yes	Auto	—	—

Vendor Quotation Response Line — Field Definitions

Field name	Data type	Required	CRUD	Lookup / values	Notes
resp_line_id	Auto (UUID)	Yes	N/A	System generated	PK

Field name	Data type	Required	CRUD	Lookup / values	Notes
response_id	FK → VendorResponse	Yes	Auto	Parent response	—
qtn_line_id	FK → QuotationLine	Yes	Lookup	Quotation lines	Which item being priced
unit_price_quoted	Decimal	Yes	Number	—	Price per unit offered
discount_pct	Decimal	No	Number	0–100%	Vendor discount offered
net_unit_price	Decimal	Yes	Calculated	unit_price × (1-disc%)	Effective unit price
line_total	Decimal	Yes	Calculated	qty × net_unit_price	Total for this line
can_supply	Boolean	Yes	Toggle	Yes / No	Can vendor supply this item?
partial_supply_qty	Decimal	No	Number	—	If cannot supply full qty
alternative_description	Text (300)	No	Free text	—	If vendor proposes substitute
alternative_spec	Text (300)	No	Free text	—	Substitute specification detail
notes	Text (200)	No	Free text	—	Per-line vendor remarks

4.6 Vendor Interaction Scenarios on Purchase Orders

After a PO is issued to a vendor, the procurement cycle does not end. Real-world vendor responses require the system to handle a defined set of scenarios. Each scenario has a specific workflow, field impact, and status change. These scenarios apply equally to POs created from PRs and manually created POs.

Scenario	Description	System action required	Resulting PO line status
Partial Shipment	Vendor can deliver some but not all quantity on a line (e.g. ordered 100, can supply 60 now). Vendor notifies procurement. Partial	Procurement Officer updates quantity_confirmed on the PO line. System creates a Partial Delivery flag. PO status becomes Partially Received.	Partially Delivered

Scenario	Description	System action required	Resulting PO line status
	delivery is accepted and the remaining quantity stays open on the PO.	Remaining qty tracked as quantity_pending. PO stays open until all lines are fully received or manually closed.	
Vendor Short-Closing (Refusal)	Vendor cannot supply an item at all and will not be able to in future. Vendor communicates this formally to procurement.	Procurement Officer marks the PO line as Short-Closed with mandatory reason entry. System sets line quantity_cancelled = remaining pending qty. If all lines are closed, PO status = Closed. A new PR or PO can be raised for the unfulfilled quantity from an alternative supplier.	Short-Closed / Cancelled
Specification Mismatch	Vendor cannot supply the exact specification but offers a substitute (e.g. ordered 100ml bottles, vendor offers 60ml). Vendor submits an alternative specification to procurement for approval.	Procurement Officer creates a PO Amendment record against the line, recording original_spec, proposed_spec, and reason. The amendment is routed to the original PR approver (if PR-sourced) or Procurement Manager for approval. If approved, po_line.description, specification, and unit_price are updated. A full audit record of original vs amended values is preserved.	Amended — Pending Approval → Confirmed
Price Change	Market price of an item has changed after PO issuance. Vendor requests a price revision or procurement discovers a new rate.	Procurement Officer raises a PO Price Amendment, entering original_unit_price, new_unit_price, and price_change_reason. Amendment is routed for approval per the standard PO approval tier (based on the value of the change). Upon approval, po_line.unit_price and	Price Amended — Pending Approval → Confirmed

Scenario	Description	System action required	Resulting PO line status
		line_total are updated. PO grand_total is recalculated. Original price preserved in the amendment log.	
Quotation-triggered Rate Sourcing	PR or PO line requires market validation. Procurement team needs to gather rates from vendors without a formal contract.	If requires_quotation = Yes: Procurement Officer initiates a Quotation Request from the PR or PO. Multiple vendors are invited. Responses are entered and compared. The winning response is awarded. On award, the vendor's quoted unit price flows back to the PO line as the confirmed unit_price. PO can then proceed to approval and issuance.	Quoted → Confirmed
Direct Rate Sourcing (No Quotation)	Procurement Officer obtains a verbal or catalogue price directly from vendor. No formal RFQ needed.	Procurement Officer enters the unit price directly on the PO line. The field rate_source_notes is mandatory in this case, requiring documentation of how the rate was obtained (e.g. 'Phone quote from Ahmed at Vendor Co., 12 May 2026', or 'Framework contract ref FC-2026-001'). This ensures auditability even without a formal RFQ.	Confirmed (Direct Rate)

4.7 PO Amendment — Field Definitions

A PO Amendment record is created whenever a confirmed PO line requires a change after the PO has been approved or sent. All amendments are versioned and require re-approval.

Field name	Data type	Required	CRUD	Lookup / values	Notes
amendment_id	Auto (UUID)	Yes	N/A	System generated	PK
amendment_number	Text (20)	Yes	Auto	AMD-YYYY-	Reference

Field name	Data type	Required	CRUD	Lookup / values	Notes
				NNNNN	
po_id	FK → PO	Yes	Lookup	Approved/Sent POs	Which PO
po_line_id	FK → PO Line	Yes	Lookup	PO lines	Which line (if line-level)
amendment_type	Lookup	Yes	Yes	Qty Change / Price Change / Spec Change / Short-Close / Cancellation	Category
original_quantity	Decimal	No	Auto	Snapshot	Before change
new_quantity	Decimal	No	Number	—	After change
original_unit_price	Decimal	No	Auto	Snapshot	Before change
new_unit_price	Decimal	No	Number	—	After change
original_spec	Text (500)	No	Auto	Snapshot	Before change
new_spec	Text (500)	No	Free text	—	After change
amendment_reason	Lookup	Yes	Yes	Partial Shipment / Price Change / Spec Mismatch / Short-Close / Other	Why changing
reason_detail	Text (500)	Yes	Free text	—	Full explanation
vendor_communication_ref	Text (100)	No	Free text	—	Email/letter ref from vendor
status	Lookup	Yes	Auto	Draft / Pending Approval / Approved / Rejected	Amendment state
approved_by	FK → User	No	Auto	—	Who approved amendment
approved_at	DateTime	No	Auto	—	Approval timestamp
created_by	FK → User	Yes	Auto	—	—
created_at	DateTime	Yes	Auto	—	—

4.8 Lookup Collections

Lookup: Priority (CRUD: Yes)

Code / ID	Display value	Active	Notes
LOW	Low	Yes	No urgency, plan in next cycle
NORMAL	Normal	Yes	Standard processing
HIGH	High	Yes	Needed within 1–2 weeks
URGENT	Urgent	Yes	Needed within 48 hours

Lookup: PR Type (CRUD: Yes)

Code / ID	Display value	Active	Notes
GOODS	Goods / Products	Yes	Physical inventory items
SVC	Services	Yes	Labour, consulting, maintenance
ASSET	Capital Asset	Yes	Fixed assets, equipment
RECUR	Recurring	Yes	Monthly / regular needs

Lookup: Quotation Source Type (CRUD: No)

Code / ID	Display value	Active	Notes
PR	From Purchase Requisition	N/A	Quotation raised from an approved PR
PO	From Purchase Order	N/A	Quotation raised from a draft PO
STANDALONE	Standalone / Market Survey	N/A	No source document, for benchmarking

Lookup: PO Amendment Type (CRUD: Yes)

Code / ID	Display value	Active	Notes
QTY	Quantity Change	Yes	Ordered qty revised
PRICE	Price Change	Yes	Unit price revised after PO issue
SPEC	Specification Change	Yes	Item specification or description changed
SHORT	Short-Close	Yes	Vendor cannot supply, line being closed

Code / ID	Display value	Active	Notes
CANCEL	Full Cancellation	Yes	Entire PO or line cancelled

4.9 Role-Based Access — Demand Planning

Screen / action	Admin	Proc. Mgr	Purchase Off.	Inv. Mgr	Requester
Create requisition	**Full**	**Full**	**Full**	**None**	**Full**
Edit own draft PR	**Full**	**Full**	**Full**	**None**	**Full**
Submit PR for approval	**Full**	**Full**	**Full**	**None**	**Full**
Flag PR line: quotation required	**Full**	**Full**	**Full**	**None**	**Full**
Approve / reject PR	**Full**	**Full**	**None**	**None**	**None**
View all PRs	**Full**	**Full**	**Full**	**View**	**None**
View own PRs only	**Full**	**Full**	**Full**	**None**	**Full**
Create quotation / RFQ	**Full**	**Full**	**Full**	**None**	**None**
Enter vendor quote response	**Full**	**Full**	**Full**	**None**	**None**
Award quotation	**Full**	**Full**	**None**	**None**	**None**
Convert PR to PO	**Full**	**Full**	**Full**	**None**	**None**
Create PO (manual)	**Full**	**Full**	**Full**	**None**	**None**
Consolidate PRs onto PO	**Full**	**Full**	**Full**	**None**	**None**
Create PO amendment	**Full**	**Full**	**Full**	**None**	**None**
Approve PO amendment	**Full**	**Full**	**None**	**None**	**None**
Short-close PO line	**Full**	**Full**	**None**	**None**	**None**
Delete draft PR	**Full**	**Full**	**Own only**	**None**	**Own only**

Section 4.10 below corrects the award-gate defect on the View/Edit RFQ and Score screen described against Section 4.5 (Quotation/RFQ). Section 4.11 then adds a public, unauthenticated, token-based supplier portal that extends the same Quotation/RFQ capability, allowing suppliers to submit quotation responses directly instead of requiring a Procurement Officer to transcribe them.

4.10 Group A — Quotation: 'Compare and Award' Gate (Addendum 19)

#1	**DEFECT**	**'Compare and Award' button enabled with zero submitted quotations**

4.10.1 Current Behaviour

On the View/Edit RFQ and Score screen, the 'Compare and Award' button is clickable regardless of whether any invited supplier has actually submitted a Vendor Quotation Response (FSD v3 Section 4.5). Clicking it with zero responses either opens an empty comparison view or allows an award action with nothing to compare.

4.10.2 Required Behaviour

Enable condition	'Compare and Award' is enabled only when COUNT(VendorResponses WHERE quotation_id = this RFQ AND status = 'RECEIVED' or higher) >= 1
Disabled state	Button renders visually disabled (greyed, non-interactive) when zero responses exist — not hidden, so Procurement can see the action exists and understand why it is unavailable
Tooltip on disabled state	'No vendor responses have been submitted yet. At least one response is required to compare and award.'
Live re-evaluation	The button's enabled state re-evaluates without a full page reload whenever a new VendorResponse is recorded against this RFQ while the screen is open (e.g. Procurement Officer manually records a response in another tab)

4.10.3 API Support

GET /api/quotations/{id} (existing endpoint, FSD v3 Section 4.5) is extended to include a computed field submittedResponseCount, which the React screen uses to drive the button's disabled state without an additional round-trip.

4.10.4 Acceptance Criteria

- Button is disabled on page load for any RFQ with zero submitted responses
- Button enables automatically once the first response is recorded, without requiring page refresh
- Disabled button shows the specified tooltip on hover
- Clicking the button while disabled performs no action and triggers no API call

4.11 Supplier RFQ Self-Service Portal

4.11.1 Purpose and Overview (Addendum 18)

Today, recording a supplier's quotation response requires a Procurement Officer to manually enter the data on the internal 'Record Vendor Response' screen — typically after receiving the quote by phone, email, or in person. This addendum introduces a self-service alternative: the supplier's nominated contact person receives a secure, one-time link directly on their email and WhatsApp, opens a public page showing the RFQ items, and enters the quoted rate and delivery days themselves. No supplier login or account is required.

Design principle	The public-facing page is a constrained, single-purpose view of the same underlying RFQ data model already defined in FSD v3 Section 4.5 — it is not a new data model, only a new access path with no authentication, a one-time-use token, and an expiry. Internal users continue to use 'Record Vendor Response' exactly as today; this is an additional channel, not a replacement.

4.11.1.1 End-to-End Flow

#	Stage	What happens
1	**RFQ sent**	Procurement Officer sends the Quotation/RFQ to one or more invited suppliers (existing FSD v3 Section 4.5 flow, POST /api/quotations/{id}/send).
2	**Contact person selected**	For each invited supplier, the Procurement Officer selects which Supplier Contact (from FSD v3 Section 3, SupplierContacts) should receive the response link.
3	**Mobile number verified**	System checks that the selected contact's mobile number on file matches the format and is present — this is the number WhatsApp will be sent to. If missing or invalid, the system blocks sending and prompts the Procurement Officer to update the contact record first.
4	**Secure link generated**	System generates a unique, single-use access token tied to this Quotation + Supplier + Contact combination. Token expiry is calculated as the earlier of the RFQ's closing_date or a fixed window from generation (Section 3.2).
5	**Notifications sent**	System sends an Email (to the supplier's registered email) and a WhatsApp message (to the contact person's verified mobile number), each containing the same secure link.
6	**Supplier opens link**	Contact person clicks the link on email or WhatsApp. System validates the token — if valid and unused, renders the RFQ Items page (Section 5) pre-filled with item details.
7	**Supplier submits rates**	Contact person enters unit price and delivery days per item and submits. System records the response exactly as if entered via 'Record Vendor Response', and marks the link as consumed.
8	**Repeat access blocked**	If the same link is opened again — whether already submitted or simply expired — the system shows a standard message rather than the form (Section 4.3).

4.11.2 Contact Person Selection and Mobile Number Check

Before a link can be generated for a supplier, the Procurement Officer must select which of that supplier's registered contacts will receive it. This screen is added to the existing 'Send RFQ' step (FSD v3 Section 4.5, POST /api/quotations/{id}/send).

4.11.2.1 UI Behaviour

Trigger	Procurement Officer clicks 'Send' on a Quotation/RFQ with one or more invited suppliers
Per-supplier prompt	For each invited supplier, a dropdown lists that supplier's SupplierContacts (FSD v3 Section 3, table SupplierContacts) — the Procurement Officer must select exactly one contact per supplier before the send action is enabled
Mobile number display	Once a contact is selected, their phone field is displayed read-only beside the dropdown so the Procurement Officer can visually confirm it before sending
Validation gate	The Send button remains disabled for any supplier whose selected contact

	has a missing, malformed, or non-mobile phone number — see validation rules below
Inline correction	If the contact's number fails validation, an inline 'Edit Contact' link opens the SupplierContacts edit form (FSD v3 Section 3) without leaving the Send RFQ screen

4.11.2.2 Mobile Number Validation Rules

- The contact's phone field must not be null or empty.
- The number must conform to E.164 international format after normalisation (e.g. +923001234567). The system attempts auto-normalisation for common local formats (e.g. 0300-1234567 to +923001234567 using the supplier's country code from the Supplier Master) before rejecting.
- The normalised number must pass a basic mobile-line-type check (length and prefix plausibility for the supplier's country) — this is a format check, not a live carrier lookup.
- A supplier contact flagged is_primary = true on the Supplier Master is suggested as the default selection but the Procurement Officer may choose any active contact.
- If the supplier has zero active contacts with a usable mobile number, the system blocks RFQ sending to that supplier entirely and surfaces a clear message directing the Procurement Officer to add a contact first.

4.11.3 Secure Link Generation

Each link is single-use and scoped to exactly one Quotation + Supplier + Contact combination. The link itself carries an opaque, cryptographically random token — never the database ID of the quotation or supplier — so the URL reveals nothing about the underlying record.

4.11.3.1 Token Design

Token format	256-bit cryptographically random value (RandomNumberGenerator.GetBytes), Base64Url-encoded for safe URL embedding
Link format	https://{app-domain}/supplier-portal/rfq/{token}
Token storage	Only the SHA-256 hash of the token is stored in RfqAccessLinks.token_hash — the raw token is never persisted, mirroring the refresh-token pattern already used in auth_schema.UserSessions
Token scope	One token = one Quotation + one Supplier + one VendorResponse record. A supplier with multiple invited contacts across multiple RFQs receives a distinct token per RFQ per supplier.
Single use	Token is marked consumed_at on successful submission. A consumed token can never be reused, regardless of expiry status.

4.11.3.2 Expiry Logic

Expiry rule	The link expires at whichever comes first: the RFQ's closing_date (from Quotations.closing_date, FSD v3 Section 4.5) or a fixed window from the moment the link was generated (default 7 days, configurable). expires_at = MIN(quotation.closing_date, generated_at + fixed_window_days).

Worked example: if an RFQ has closing_date 10 days from today and the fixed window is configured at 7 days, the link expires in 7 days (the fixed window is reached first). If the same RFQ has closing_date 3 days from today, the link expires in 3 days (the closing date is reached first).

4.11.3.3 RfqAccessLinks — Field Definitions

Field name	Data type	Required	Notes
link_id	UUID	Yes	Primary key. System generated.
token_hash	NVARCHAR(64)	Yes	SHA-256 hash of the raw token. Unique. Never store the raw token.
quotation_id	FK to Quotations	Yes	Which RFQ this link grants access to.
supplier_id	FK to Suppliers	Yes	Which supplier this link was issued to.
contact_id	FK to SupplierContacts	Yes	Which named contact this link was issued to.
response_id	FK to VendorResponses	No	Populated once the response record is created on first valid open.
generated_at	DateTime	Yes	When the link was created and sent.
generated_by	FK to Users	Yes	Procurement Officer who triggered the send.
expires_at	DateTime	Yes	MIN(quotation.closing_date, generated_at + fixed_window).
first_opened_at	DateTime	No	Timestamp of the first successful open. NULL until opened.
consumed_at	DateTime	No	Timestamp when the supplier successfully submitted the response. NULL until submitted.
consumed_ip	NVARCHAR(45)	No	IP address at submission, for audit.
status	Lookup	Yes (auto)	ACTIVE / EXPIRED / CONSUMED — see Section 4.2 state machine.
email_sent_at	DateTime	No	When the email notification was dispatched.
whatsapp_sent_at	DateTime	No	When the WhatsApp notification was dispatched.
created_at	DateTime	Yes	Record creation timestamp.

4.11.4 Link Validation and Repeat-Access Handling

Every time the public link is opened, the system runs a validation check before deciding whether to render the response form or a standard blocking message. This is the mechanism that satisfies the requirement: if the supplier enters data and again opens the link, show the standard message — you already entered the data or link is expired.

4.11.4.1 Validation Sequence on Link Open

Step	Check	Outcome
1	**Token exists**	Hash the incoming token and look up RfqAccessLinks WHERE token_hash = hash. If no match, show the generic 'Link not found'

Step	Check	Outcome
		message (does not reveal whether a token ever existed, for security).
2	**Already consumed**	If consumed_at IS NOT NULL, show the standard message: You have already submitted your response for this RFQ. Do not render the form under any circumstance.
3	**Expired**	If consumed_at IS NULL AND NOW() > expires_at, show the standard message: This link has expired. Please contact the procurement team for a new link. Set status = EXPIRED if not already set.
4	**Valid and unconsumed**	If neither consumed nor expired, set first_opened_at if this is the first open, and render the RFQ Items page (Section 5) pre-filled and ready for submission.

4.11.4.2 Link Status State Machine

Status	Triggered by	Meaning
ACTIVE	Link generated and sent	Default state. Token is valid and unconsumed. Form renders normally.
EXPIRED	NOW greater than expires_at evaluated at open time, OR a nightly Hangfire job sweeps and marks expired links proactively	Token is no longer usable. Standard expiry message shown on any future open attempt.
CONSUMED	Supplier successfully submits the response form	Token is permanently spent. Standard already-submitted message shown on any future open attempt, even if expires_at has not yet passed.

4.11.4.3 Standard Message Specification

The exact wording shown depends on which of the two blocking conditions applies. Both follow the same visual template — a simple, branded, centred message page with no form, no navigation, and no way to retry.

Condition	Message shown to supplier
consumed_at IS NOT NULL	You have already submitted your response for this RFQ. If you need to make changes, please contact our procurement team.
NOW greater than expires_at (and not consumed)	This link has expired. Please contact our procurement team if you would like to submit a quotation.
Token not found / malformed	This link is invalid. Please check the link or contact our procurement team.

Important	All three blocking conditions return HTTP 200 with the message page — never a 404 or 410 — so that the supplier's browser does not show a generic broken-link

error. The page itself communicates the reason. This also avoids leaking information about which HTTP status differentiates valid vs invalid tokens.

4.11.5 RFQ Items Page — Public Supplier View

This is the page the supplier's contact person lands on after a valid link is opened. It shows the same RFQ item list and basic information already present on the internal 'Record Vendor Response' screen (FSD v3 Section 4.5), but rendered as a public, unauthenticated, single-purpose page scoped to exactly one supplier's response.

4.11.5.1 Page Header — Basic Information

Matches the header information shown on the internal Record Vendor Response screen so the experience is familiar to whoever reviews it internally afterward.

Field	Source	Notes
RFQ Number	Quotations.quotation_number	Read-only display
RFQ Title / Subject	Quotations.quotation_title	Read-only display
Issue Date	Quotations.issue_date	Read-only display
Closing Date	Quotations.closing_date	Read-only display — also drives the expiry banner if close to expiring
Supplier Name	Suppliers.supplier_name	Read-only display, confirms identity to the contact
Contact Person	SupplierContacts.contact_name	Read-only — confirms who the link was issued to
Terms / Special Instructions	Quotations.terms	Read-only display if populated

4.11.5.2 RFQ Item List — Response Entry Grid

Every line item from the RFQ (QuotationLines, FSD v3 Section 4.5) is listed with the same item information shown internally, plus the two fields the supplier is being asked to provide.

Column	Source / behaviour	Editable?	Validation	Notes
Item Description	QuotationLines.item_description	**No**	—	Read-only
Specification	QuotationLines.specification	**No**	—	Read-only, shown if populated
UOM	QuotationLines.uom	**No**	—	Read-only
Quantity Requested	QuotationLines.quantity_requested	**No**	—	Read-only
Unit Price (Rate)	New input — maps to VendorResponseLines.unit_price_quoted	**Yes**	None — per requirement, no validation on this page	Numeric text input
Delivery Days	New input — maps to VendorResponses.delivery_lead_da	**Yes**	None — no validation	Numeric text input

Column	Source / behaviour	Editable?	Validation	Notes
	ys			
Can Supply?	New input — maps to VendorResponseLines.can_supply	**Yes**	None	Yes/No toggle, defaults to Yes
Remarks	New input — maps to VendorResponseLines.notes	**Yes**	None	Free text, optional

No validation by design	Per requirement, the public RFQ Items page applies no validation on the rate or delivery-days fields — the supplier can submit any value including blank, zero, or non-numeric text where applicable. Internal review and any data-quality correction happens after submission, when the Procurement Officer reviews the response on the existing internal Quotation Comparison screen (FSD v3 Section 4.5) exactly as they would for a phone-in quote.

4.11.5.3 Submission Behaviour

- A single Submit Response button at the bottom of the page submits all line entries in one request.
- On submit, the system creates one VendorResponses record and one VendorResponseLines record per RFQ line, using the same entities already defined in FSD v3 Section 4.5 — no new response data model, only a new entry path.
- On successful submission, RfqAccessLinks.consumed_at is set, RfqAccessLinks.response_id is populated with the newly created VendorResponses.response_id, and the page redirects to a simple thank-you confirmation screen.
- If the supplier closes the browser before submitting, no record is created and the link remains ACTIVE and reusable until it expires or is successfully submitted.
- There is no partial-save / draft capability on this page — submission is all-or-nothing in a single action, consistent with there being no validation to support incremental correction.

4.11.6 Notification Module — Email and WhatsApp

When the link is generated (Section 3), two notifications are dispatched: an email to the supplier's registered email address, and a WhatsApp message to the contact person's verified mobile number. Both carry the identical link URL. This extends the existing Notification Sidecar with a new WhatsApp channel.

4.11.6.1 Email Notification

Uses the existing SendGrid-based email path already defined in the architecture document (no new infrastructure required).

Trigger	Link generation completes (Section 3) — enqueued as a Hangfire job, same pattern as existing PO/GRN notifications
Recipient	Suppliers.email (the supplier's primary registered email, not the contact's personal email)
Template	New Razor template: RfqResponseRequest.cshtml — includes RFQ number, closing date, and the secure link as a button/CTA

Subject line	Request for Quotation — {quotation_number} — Response Required by {closing_date}

4.11.6.2 WhatsApp Notification Service — New Capability

A new provider-agnostic WhatsApp sending service is added alongside the existing email path. The interface is deliberately abstracted from any specific WhatsApp Business API provider so the underlying provider (Twilio, Meta Cloud API, Gupshup, or others) can be selected or swapped at configuration time without changing calling code.

4.11.6.2.1 Service Interface

Element	Specification
Interface	IWhatsAppNotificationService — defined in the Notification Sidecar project, provider implementation injected via DI based on configuration (WhatsApp:Provider setting)
Primary method	SendMessageAsync(recipients, templateCode, templateData) — accepts a list of recipients so the same call can notify multiple contacts in one invocation, per requirement to pass a user list to whom it should send the message
WhatsAppRecipient	Record type: MobileNumber, DisplayName, ReferenceId — ReferenceId allows correlating the send back to the RfqAccessLinks row for tracking
Message content	Rendered from a configurable text template containing the clickable link — for the RFQ use case, the same URL sent by email
Delivery tracking	Each send attempt is logged with provider message ID (if returned), status (QUEUED/SENT/DELIVERED/FAILED), and timestamp — stored in a new WhatsAppMessageLog table
Failure handling	Hangfire automatic retry on transient failures, same pattern as existing email retry logic. A WhatsApp failure never blocks the email from sending — both channels are dispatched independently.

4.11.6.2.2 Method Signature (illustrative)

```
public interface IWhatsAppNotificationService
{
Task<WhatsAppSendResult> SendMessageAsync(
IEnumerable<WhatsAppRecipient> recipients,
string templateCode,
object templateData,
CancellationToken cancellationToken = default);
}
```

4.11.6.2.3 RFQ-Specific Usage

For this addendum specifically, the calling code resolves a single recipient (the selected contact's verified mobile number) per supplier and calls SendMessageAsync with a one-item list, but the same service supports multi-recipient broadcast for future use cases (e.g. notifying an entire approver group) without modification.

4.11.7 API Endpoints

4.11.7.1 Internal Endpoints (Authenticated — Procurement Officer)

Method	Endpoint	Description
GET	/api/suppliers/{id}/contacts/eligible	Returns active contacts for a supplier with mobile validation status per contact, used to populate the contact-selection dropdown
POST	/api/quotations/{id}/send-with-link	Extends existing send action — accepts supplierId/contactId pairs; generates one RfqAccessLink per pair; triggers email + WhatsApp dispatch
GET	/api/quotations/{id}/access-links	Lists all generated links for this RFQ with status (ACTIVE/EXPIRED/CONSUMED), useful for Procurement Officer to track who has responded
POST	/api/quotations/{id}/access-links/{linkId}/resend	Re-sends the same active (non-expired, non-consumed) link's notifications — does not generate a new token

4.11.7.2 Public Endpoints (Unauthenticated — Supplier Portal)

Method	Endpoint	Description
GET	/api/public/rfq-portal/{token}	Validates token per Section 4.1 sequence; returns either the RFQ + items payload (valid) or a status indicating which standard message to show (consumed / expired / invalid)
POST	/api/public/rfq-portal/{token}/submit	Accepts the line-by-line response payload; creates VendorResponses + VendorResponseLines; marks link consumed_at; returns confirmation

Security note	Public endpoints are exempt from JWT authentication but are NOT exempt from standard protections: rate limiting per IP, the token-hash lookup pattern (Section 3.1) so raw tokens are never stored, and HTTPS-only access. No supplier or internal data beyond the single scoped RFQ is ever exposed through these endpoints.

4.11.8 Role-Based Access

Screen / action	Admin	Proc. Mgr	Purch. Off.	Supplier
Select contact + send RFQ link	Full	Full	Full	—
View access link status / history	Full	Full	Full	—
Resend link	Full	Full	Full	—
Open public RFQ Items page (own RFQ only, via link)	—	—	—	Full*
Submit RFQ response (own RFQ only, via link)	—	—	—	Full*

* Supplier access is not role-based in the conventional sense — there is no supplier user account or login. Access is governed entirely by possession of a valid, unconsumed, unexpired token as described in Section 4. Full here means the contact person can view and submit exactly once for the single RFQ their link was scoped to, and nothing else in the system.

— End of Addendum —

5 Module — Procurement

Handles the formal purchase process: RFQ issuance, quote comparison, purchase order creation, multi-level approval, and supplier acknowledgement. This is the core transactional engine of the supply system.

5.1 Screens & Features

- RFQ Management (Create, Send, Track)
- Quotation Comparison (side-by-side vendor quotes)
- Purchase Order List
- Create / Edit Purchase Order
- PO Approval Workflow (multi-level)
- PO Acknowledgement Tracking
- PO Amendment / Cancellation
- Blanket / Framework Orders

5.2 Purchase Order Header — Field Definitions

Field name	Data type	Required	CRUD	Lookup / values	Notes
po_id	Auto (UUID)	Yes	N/A	System generated	Primary key
po_number	Text	Yes	Auto	PO-YYYY-NNNNN	Unique PO

Field name	Data type	Required	CRUD	Lookup / values	Notes
	(20)				number
po_type	Lookup	Yes	Yes	Standard / Blanket / Emergency / Service	PO category
supplier_id	FK → Supplier	Yes	Lookup	Active suppliers only	Must be approved
pr_id	FK → PR	No	Auto	Approved PRs	Source requisition
contract_id	FK → Contract	No	Lookup	Active contracts	If against a contract
po_date	Date	Yes	Date picker	Default: today	Date issued
delivery_date	Date	Yes	Date picker	—	Expected delivery
delivery_address	FK → Warehouse	Yes	Lookup	Active warehouses	Delivery location
currency	Lookup	Yes	Yes	PKR / USD / AED / EUR	Order currency
exchange_rate	Decimal	No	Number	1.00 for local	FX rate to base currency
payment_terms	Lookup	Yes	Yes	Inherited from supplier	Can override
subtotal	Decimal	Yes	Calculated	Sum of lines	Pre-tax total
discount_pct	Decimal	No	Number	0–100%	Overall discount
discount_amount	Decimal	No	Calculated	subtotal × disc%	Auto-calculated
tax_amount	Decimal	Yes	Calculated	Per tax rules	GST / sales tax
shipping_cost	Decimal	No	Number	—	Freight / delivery charge
grand_total	Decimal	Yes	Calculated	subtotal - disc + tax + ship	Auto-calculated
status	Lookup	Yes	Auto	Draft/Pending/Approved/Sent/Partial/Received/Closed/Cancelled	Lifecycle state
approval_level	Integer	No	Auto	1, 2, 3	Current approval step

Field name	Data type	Required	CRUD	Lookup / values	Notes
approved_by_l1	FK → User	No	Auto	—	First approver
approved_at_l1	DateTime	No	Auto	—	L1 approval time
approved_by_l2	FK → User	No	Auto	—	Second approver
approved_at_l2	DateTime	No	Auto	—	L2 approval time
sent_to_supplier_at	DateTime	No	Auto	—	When PO was emailed/sent
supplier_ack_at	DateTime	No	Auto	—	Supplier confirmation time
terms_conditions	Text (1000)	No	Free text	Default from settings	T&C text
internal_notes	Text (500)	No	Free text	—	Internal only notes
attachments	File[]	No	Upload	PDF/Excel	Supporting files
created_at	DateTime	Yes	Auto	System generated	—
created_by	FK → User	Yes	Auto	System assigned	—

5.3 Purchase Order Line — Field Definitions

Field name	Data type	Required	CRUD	Lookup / values	Notes
po_line_id	Auto (UUID)	Yes	N/A	System generated	Primary key
po_id	FK → PO	Yes	Auto	Parent PO	Link to header
line_no	Integer	Yes	Auto	1, 2, 3...	Line sequence
product_id	FK → Product	No	Lookup	Product catalog	Optional
description	Text (300)	Yes	Free text	—	Item description
uom	Lookup	Yes	Yes	Each / Kg / Litre / Box /	Unit of measure

Field name	Data type	Required	CRUD	Lookup / values	Notes
				Set	
quantity_ordered	Decimal	Yes	Number	—	Ordered quantity
quantity_received	Decimal	No	Auto	Updated by GRN	Received so far
quantity_pending	Decimal	No	Calculated	ordered - received	Outstanding qty
unit_price	Decimal	Yes	Number	—	Agreed unit price
line_discount_pct	Decimal	No	Number	0–100%	Per-line discount
line_total	Decimal	Yes	Calculated	qty × price × (1-disc%)	Net line value
tax_code	Lookup	No	Yes	GST 17% / Exempt / Zero-rated	Tax treatment
tax_amount	Decimal	No	Calculated	Auto per tax code	Line tax
delivery_date	Date	No	Date picker	—	Per-line if different
warehouse_id	FK → Warehouse	No	Lookup	Active warehouses	Specific location
gl_account	FK → GL	No	Lookup	Chart of accounts	Cost posting code
notes	Text (200)	No	Free text	—	Per-line remarks

5.4 RFQ — Field Definitions

Field name	Data type	Required	CRUD	Lookup / values	Notes
rfq_id	Auto (UUID)	Yes	N/A	System generated	PK
rfq_number	Text (20)	Yes	Auto	RFQ-YYYY-NNNNN	Human-readable
rfq_title	Text (200)	Yes	Free text	—	Subject
pr_id	FK → PR	No	Lookup	Approved PRs	Source PR if any
issue_date	Date	Yes	Date picker	Default: today	—
closing_date	Date	Yes	Date picker	—	Quote submission deadline

Field name	Data type	Required	CRUD	Lookup / values	Notes
status	Lookup	Yes	Auto	Draft / Sent / Closed / Cancelled	—
invited_suppliers	FK[] → Supplier	Yes	Multi-select	Active suppliers	Suppliers invited
evaluation_criteria	Text (500)	No	Free text	Price / Quality / Delivery	Scoring basis
terms	Text (500)	No	Free text	—	Special T&C
created_by	FK → User	Yes	Auto	System assigned	—
created_at	DateTime	Yes	Auto	System generated	—

5.5 Lookup Collections

Lookup: PO Type (CRUD: Yes)

Code / ID	Display value	Active	Notes
STD	Standard PO	Yes	One-time purchase
BLANK	Blanket Order	Yes	Open order with call-offs
EMRG	Emergency PO	Yes	Urgent, bypasses normal flow
SVC	Service Order	Yes	Services, no physical delivery

Lookup: PO Status (CRUD: No — system controlled)

Code / ID	Display value	Active	Notes
DRAFT	Draft	N/A	Being prepared, not submitted
PEND	Pending Approval	N/A	Submitted for approval
APPR	Approved	N/A	Fully approved, ready to send
SENT	Sent to Supplier	N/A	PO transmitted to vendor
PART	Partially Received	N/A	Some lines received
RECV	Fully Received	N/A	All lines received
CLOSED	Closed	N/A	Completed and closed
CANCEL	Cancelled	N/A	Voided before

Code / ID	Display value	Active	Notes
			delivery

5.6 Role-Based Access — Procurement

Screen / action	Admin	Proc. Mgr	Purchase Off.	Inv. Mgr	Requester
Create PO	**Full**	**Full**	**Full**	**None**	**None**
Edit draft PO	**Full**	**Full**	**Full**	**None**	**None**
Approve PO (L1)	**Full**	**Full**	**None**	**None**	**None**
Approve PO (L2)	**Full**	**Full**	**None**	**None**	**None**
Send PO to supplier	**Full**	**Full**	**Full**	**None**	**None**
Cancel / Amend PO	**Full**	**Full**	**None**	**None**	**None**
Create RFQ	**Full**	**Full**	**Full**	**None**	**None**
View PO list	**Full**	**Full**	**Full**	**View**	**View own**
Export PO report	**Full**	**Full**	**Full**	**None**	**None**

Sections 5.7 through 5.9 below add PO Creation UX corrections and enhancements raised against the Create Purchase Order screen.

5.7 Group B — PO Creation: Import Lines Placement and Labelling (Addendum 19)

#2	**DEFECT**	**Import Lines control poorly positioned and unclearly labelled**

5.7.1 Current Behaviour

On the Create Purchase Order screen, the control for importing line items (e.g. from a PR, a spreadsheet, or a quotation) is positioned inconveniently within the screen flow, and its label does not clearly communicate what it does.

5.7.2 Required Behaviour

New position	Moved to the very top of the Create Purchase Order screen, above the PO header fields (supplier, dates, currency) — so a user importing lines does so first, before manually filling header details that may be partially auto-populated by the import
New label	'Import Items from Requisition / Quotation / File' replacing the previous unclear label — communicates the three supported sources explicitly rather than a generic 'Import Lines'
Visual treatment	Rendered as a prominent action bar (icon + label) spanning the screen width, distinct from the form fields below it, so it reads as a starting action rather than a buried form control
Behaviour unchanged	The underlying import logic (selecting a source PR/Quotation, mapping lines, or uploading a file) is unchanged — this item addresses placement

and labelling only

5.7.3 Acceptance Criteria

- Import action bar renders as the first visible element on the Create Purchase Order screen, above all header fields
- Label reads 'Import Items from Requisition / Quotation / File' exactly
- Existing import functionality (source selection, line mapping, file upload) behaves identically to before — only placement and copy changed

5.8 Group B — PO Creation: Auto-Select Supplier from Awarded Quotation (Addendum 19)

#3	**NEW FEATURE**	**Auto-select and lock supplier when converting a PO From Quotation with an award already made**

5.8.1 Current Behaviour

When creating a PO 'From Requisition' or 'From Quotation', the Supplier field on the PO header is always manually selectable, even when the source PR already has an awarded Quotation pointing to a specific winning supplier (FSD v3 Section 4.5, Quotations.awarded_supplier_id). This allows a user to accidentally select a different supplier than the one the quotation process determined.

5.8.2 Required Behaviour

Source selection	Resulting supplier field behaviour
From Requisition — PR has no linked Quotation	Supplier field remains manually selectable, as today. No automatic behaviour.
From Requisition — PR has a linked Quotation that is NOT yet Awarded (status: Draft/Sent/Responses Received/Under Evaluation)	Supplier field remains manually selectable. The quotation has not concluded, so no winning supplier exists yet to auto-select.
From Requisition — PR has a linked Quotation that IS Awarded	System auto-selects Quotations.awarded_supplier_id into the PO header Supplier field and disables (read-only, greyed) the field. A small inline note: 'Supplier set automatically from awarded quotation QTN-{number}.'
From Quotation — selecting an Awarded quotation directly	Same as above: supplier auto-selected and field disabled, with the inline note.
From Quotation — selecting a quotation that is NOT yet Awarded	Supplier field remains manually selectable (matches the 'not yet awarded' row above) — the user may proceed to manually pick a supplier or wait for the award to complete.

5.8.3 Override Path

If a Procurement Manager genuinely needs to override an auto-selected supplier (e.g. the awarded supplier became unavailable after award), the disabled field carries an 'Unlock' icon visible only to Procurement Manager role, which re-enables manual selection after a confirmation dialog warning that this deviates from the awarded quotation. This deviation is recorded as a note on the PO's internal_notes field automatically.

5.8.4 Acceptance Criteria

- Supplier field auto-populates and disables only when the linked quotation status is Awarded
- Inline note correctly references the awarded quotation number
- Supplier field remains fully manual for any quotation status other than Awarded, and for PRs/manual creation with no linked quotation at all
- Procurement Manager can unlock and override with a recorded justification; other roles cannot

5.9 Group B — PO Creation: Per-Item and Global Warehouse Selection (Addendum 19)

#6	**NEW FEATURE**	**Per-line warehouse override with a single global warehouse as PO-level default**

5.9.1 Required Behaviour

Design decision	A single 'Delivery Warehouse' field remains at the PO header level and acts as the default for every line. Each PO line additionally gets its own optional warehouse dropdown. If a line-level warehouse is selected, it overrides the header default for that line only. Lines with no override use the header default. This allows a single PO with mixed delivery destinations (e.g. cold-chain items to the Cold Storage warehouse, everything else to Main) without forcing every line to be set individually.

PO header field (existing)	delivery_warehouse_id on PurchaseOrders (FSD v3 Section 5.2) — unchanged, remains required, now explicitly documented as the default
New PO line field	warehouse_id on POLines (already present per FSD v3 Section 5.3 as an optional field, but previously unused/unsurfaced in UI) — now exposed as a per-line dropdown defaulting to blank
Effective warehouse logic	EffectiveWarehouseId for a line = POLine.warehouse_id IF NOT NULL, ELSE PurchaseOrders.delivery_warehouse_id
UI placement	New 'Warehouse' column added to the PO line grid, positioned after Quantity. Shows the header default warehouse name as placeholder/ghost text when no override is selected, with a dropdown to override.
GRN and downstream impact	All downstream logic that currently reads PurchaseOrders.delivery_warehouse_id directly (GRN creation, FSD Addendum 5A receipt tracking) is updated to read the line's EffectiveWarehouseId instead, so multi-warehouse POs receive correctly per line

5.9.2 Worked Example

PO field	Value	Line override	Effective warehouse
Header	delivery_warehouse_id = Main Warehouse	—	Main Warehouse (default)
Line 1 — Laptops	—	None selected	Main Warehouse
Line 2 —	—	Cold Storage WH	Cold Storage WH

PO field	Value	Line override	Effective warehouse
Vaccines			

5.9.3 Acceptance Criteria

- PO header retains a single required Delivery Warehouse field functioning as before
- Each PO line can independently select a warehouse that overrides the header default for that line only
- Lines without an override correctly inherit the header default — verified after header warehouse is changed post-line-entry (override lines unaffected, default lines follow the new header value)
- GRN creation against any PO line uses the line's effective warehouse, not blindly the header value

5A Purchase Order Delivery Lifecycle (Addendum 5A)

The FSD v3 Procurement module (Section 5) covers Purchase Order creation, approval workflow, and amendment cycle, but does not fully address what happens after a PO is issued to a supplier — deliveries are rarely complete in one shipment, invoices may not match receipts one-for-one, and POs often close through partial or short-close scenarios rather than full receipt. This section defines the complete PO delivery lifecycle and must be read together with the GRN Approval Workflow (Section 13) and the Finance module (Section 10).

5A.1 Purpose of This Addendum

The FSD v3 Procurement module (Section 5) covers Purchase Order creation, approval workflow, and amendment cycle. What it does not fully address is what happens after a PO is issued to a supplier — specifically the operational reality that deliveries are rarely complete, invoices may not match receipts, and POs often close through partial or short-close scenarios rather than full receipt.

This addendum defines the complete PO delivery lifecycle as a dedicated specification section. It should be treated as Section 5A of the FSD — inserted between Section 5 (Procurement) and Section 6 (Inventory Management) — and must be read together with the GRN Approval Workflow (FSD Section 13) and the Finance module (FSD Section 10).

What was missing	The FSD did not consolidate: (1) line-level quantity tracking fields and their lifecycle, (2) the logic for multiple GRNs and multiple invoices per PO, (3) the exact conditions and steps for Short-Close, (4) how PO status moves through all states including Partially Invoiced / Fully Invoiced, and (5) the precise 3-way match rules when receipts are partial. This addendum fills all five gaps.

5A.2 End-to-End PO Delivery Flow

Every Purchase Order in the SMS moves through the following high-level flow. This flow applies regardless of whether the PO was raised from a PR, a quotation award, or entered manually.

#	Stage	Owner	What happens
1	PR → PO	Procurement	Purchase Requisition converted or PO created manually. PO approved through

#	Stage	Owner	What happens
			tier-based workflow. PO status = Approved.
2	**PO sent to supplier**	Procurement Officer	PO PDF emailed to supplier. PO status = Sent. Supplier acknowledgement recorded. Delivery date confirmed.
3	**Supplier delivers goods**	Warehouse	Goods arrive at warehouse. Warehouse Operator physically checks delivery against supplier's delivery note and the PO.
4	**GRN created**	Warehouse Operator	Goods Receipt Note created against the PO. Line-level quantities entered (received, accepted, rejected). QC check performed. GRN submitted for approval.
5	**GRN approved**	Inventory Manager	GRN approved. Accepted quantities posted to inventory. PO line received quantities updated. PO status recalculated (Partially Received or Fully Received).
6	**Supplier sends invoice**	Finance Officer	Supplier invoice received. Invoice entered into SMS referencing the PO and the relevant GRN(s).
7	**3-way match run**	System (auto)	System compares PO value, GRN accepted quantities, and invoice amount. Match result: Matched, Variance, or Mismatch.
8	**Invoice approved**	Finance Officer	Matched or variance-approved invoice approved for payment. Invoice status = Approved.
9	**Payment processed**	Finance Officer	Payment scheduled and processed. Invoice status = Paid.
10	**PO closure**	System / Procurement	PO automatically closes when all lines are Fully Received + Fully Invoiced + Fully Paid. Or manually closed via Short-Close for lines that will never be delivered.

5A.3 Line-Level Quantity Tracking

Quantity tracking in the SMS operates at the PO line level, not just at the PO header level. This is critical because different items on the same PO may be in different delivery states simultaneously — one line may be fully received while another is partially delivered and a third is short-closed.

Design principle	Every quantity on a PO line is derived from transactions — GRN receipts, short-close actions, and invoice postings. No quantity field on a PO line is manually editable by a user after the PO is approved. All updates happen through system-generated transactions that maintain a full audit trail.

5A.3.1 PO Line Quantity Fields

Field name	Type	Definition and update rule
qty_ordered	Decimal	The original ordered quantity on the PO line. Set at PO

Field name	Type	Definition and update rule
		creation. Never changed after PO approval. Any change to ordered quantity requires a formal PO Amendment (qty_change type) with approval.
qty_received	Decimal	Running total of quantity received to date across all GRNs for this line. Updated automatically when a GRN line for this PO line is approved. qty_received = SUM(grn_lines.qty_accepted) for this po_line_id.
qty_rejected	Decimal	Running total of quantity rejected across all GRNs. Updated on GRN approval. qty_rejected = SUM(grn_lines.qty_rejected) for this po_line_id. Rejected quantities do not count toward receipt.
qty_pending	Decimal	Remaining quantity not yet received and not short-closed. Computed field: qty_pending = qty_ordered - qty_received - qty_short_closed. A line with qty_pending = 0 is fully resolved.
qty_short_closed	Decimal	Quantity formally cancelled via Short-Close action. Set when Procurement Officer marks a line as Short-Closed with reason and approval. qty_short_closed + qty_received must not exceed qty_ordered.
qty_invoiced	Decimal	Running total of quantity covered by approved invoices against this line. Updated when an invoice line for this PO line is approved by Finance. qty_invoiced = SUM(invoice_lines.qty_invoiced) for this po_line_id.
qty_invoice_pending	Decimal	Quantity received but not yet invoiced. Computed: qty_invoice_pending = qty_received - qty_invoiced. This is the 'invoiceable' balance. Finance uses this to validate invoice quantities.
line_status	Lookup	Current delivery and invoice state of this line. See Section 3.2 for full state machine.

5A.3.2 PO Line Status State Machine

Line status	Meaning and conditions
Open	No GRN has been approved for this line yet. qty_received = 0. qty_pending = qty_ordered.
Partially Received	At least one GRN has been approved but qty_received < qty_ordered - qty_short_closed. The line is still awaiting further deliveries.
Fully Received	qty_received = qty_ordered - qty_short_closed. All deliverable quantity has been received. The line may still be awaiting invoicing.
Partially Invoiced	qty_invoiced > 0 but qty_invoiced < qty_received. Some received quantity has been invoiced, some has not yet been covered by an approved invoice.
Fully Invoiced	qty_invoiced = qty_received. All received quantity is covered by approved invoices. This is the financially settled state for received goods.
Short-Closed	qty_short_closed > 0. The remaining undelivered quantity (or all remaining quantity) has been formally cancelled with reason and approval. The line

Line status	Meaning and conditions
	will never receive further goods against qty_short_closed.
Cancelled	The entire PO line was cancelled before any delivery occurred (qty_received = 0 and qty_short_closed = qty_ordered). Typically used when the entire line is removed from scope.

5A.3.3 PO Header Status — Derived from Line Statuses

The PO header status is automatically derived by the system from the aggregate state of all its lines. It is never set manually. The derivation rules are:

PO header status	Derivation condition
Draft	PO is being composed. No approval submitted.
Pending Approval	Submitted for approval, awaiting approver action.
Approved	Fully approved, not yet sent to supplier.
Sent to Supplier	PO transmitted to supplier. Awaiting delivery.
Partially Received	At least one line has qty_received > 0 but NOT all lines are Fully Received / Short-Closed / Cancelled.
Fully Received	ALL lines are in status Fully Received OR Short-Closed OR Cancelled. No further deliveries expected.
Partially Invoiced	At least one line has qty_invoiced > 0 but NOT all lines are Fully Invoiced.
Fully Invoiced	ALL deliverable lines are Fully Invoiced. All received quantity is covered by approved invoices.
Closed	PO is Fully Invoiced AND all payments are Processed/Cleared. The PO is complete. No further actions possible.
Short Closed	PO is closed but with at least one line in Short-Closed status. Some ordered quantity was never delivered and was formally cancelled.
Cancelled	Entire PO was cancelled before any delivery occurred.

5A.4 Multiple GRNs Against One PO

The SMS must support an unlimited number of GRNs against a single PO. This is the standard scenario for any PO where a supplier delivers in batches, where goods arrive on different days, or where a delivery is rejected and a replacement shipment is made.

5A.4.1 Multi-GRN Business Rules

- A new GRN can be created against a PO with status Sent to Supplier, Partially Received, or Partially Invoiced. It cannot be created against a Fully Received, Closed, Short Closed, or Cancelled PO.
- **Each GRN references specific PO lines and records the quantity received for each line in that delivery.**
- A GRN line quantity cannot cause the cumulative qty_received for a PO line to exceed qty_ordered by more than the configured over-receipt tolerance (default 3%).
- Each GRN is independently approved through the GRN approval workflow before its quantities are posted to inventory.

- Multiple GRNs can be open simultaneously (e.g. GRN-2 created before GRN-1 is approved) but each must complete its own approval cycle independently.
- A GRN does not need to cover all PO lines. A supplier may deliver only the items available and leave others for a later delivery.
- Every GRN is assigned a unique GRN number (GRN-YYYY-NNNNN) and maintains a permanent link to its source PO.

5A.4.2 Multi-GRN Worked Example

Purchase Order PO-2026-00123 contains two lines:

PO Line	Item	Qty ordered	Unit price	Line total
L1	Laptop (Dell Inspiron 15)	100	PKR 75,000	PKR 7,500,000
L2	Wireless Mouse	200	PKR 1,500	PKR 300,000
PO Total				**PKR 7,800,000**

Delivery 1 — Supplier delivers partial Laptops + all Mice

GRN-001	Item	Qty ordered	Qty received	Qty accepted	Qty rejected
L1	Laptop	100	60	60	0
L2	Wireless Mouse	200	200	200	0

After GRN-001 is approved:

PO Line	Item	Qty ordered	Qty received	Qty pending	Qty invoiced	Line status
L1	Laptop	100	60	40	0	Partially Received
L2	Wireless Mouse	200	200	0	0	Fully Received
PO Status	**Partially Received**					

Delivery 2 — Supplier delivers remaining 40 Laptops

GRN-002	Item	Qty ordered	Qty received	Qty accepted	Qty rejected
L1	Laptop	100	40	40	0

After GRN-002 is approved:

PO Line	Item	Qty ordered	Qty received	Qty pending	Qty invoiced	Line status
L1	Laptop	100	100	0	0	Fully Received
L2	Wireless Mouse	200	200	0	0	Fully Received

PO Line	Item	Qty ordered	Qty received	Qty pending	Qty invoiced	Line status
PO Status	**Fully Received**					

5A.5 Short-Close — Formal Cancellation of Undelivered Quantity

A Short-Close is the formal business action of cancelling the remaining undelivered quantity on a PO line when the supplier confirms it cannot or will not deliver the outstanding goods. It is not an automatic system action — it requires explicit user action, a reason, and Procurement Manager approval.

5A.5.1 When to Use Short-Close vs Cancel

Scenario	Use Short-Close	Use Cancel
Supplier has delivered some goods	Yes — close the remaining undelivered qty	No — cannot cancel if any qty received
Supplier cannot deliver any goods at all	No — use Line Cancellation	Yes — cancel entire line (qty_received = 0)
Entire PO before any delivery	No	Yes — full PO cancellation
One line undeliverable, others fine	Yes — short-close that specific line only	No — do not cancel lines that are being delivered
Business no longer needs the item	Either — depends on whether any qty received	—

5A.5.2 Short-Close Workflow — Step by Step

Step	Actor	Action	System response
1	**Supplier / vendor**	Communicates formally that remaining quantity cannot be delivered. Provides written confirmation (email, letter).	—
2	**Purchase Officer**	Opens the PO line in the system. Clicks 'Short-Close Line'. Enters: (a) reason from lookup, (b) reason detail free text, (c) vendor communication reference (email/letter number).	System creates a POAmendment record with amendment_type = SHORT_CLOSE. Records qty_short_closed = qty_pending at time of action. Status = Pending Approval.
3	**Procurement Manager**	Receives notification. Reviews the short-close request, vendor communication, and business justification.	System displays the PO line detail with original qty, qty received, qty pending, and proposed qty_short_closed.
4a	**Procurement**	Approves the short-close.	System sets po_line.qty_short_closed =

Step	Actor	Action	System response
	Manager		requested amount. Recalculates qty_pending = 0 for that line. Updates po_line.line_status = Short-Closed. Recalculates PO header status. Publishes po_line.short_closed event (triggers audit log and notification to Finance).
4b	**Procurement Manager**	Rejects the short-close (disagreement on quantity or insufficient evidence).	System sets amendment status = Rejected. PO line remains Partially Received. Purchase Officer notified.
5	**System**	Recalculates PO header status based on all line statuses.	If all lines are now Fully Received or Short-Closed or Cancelled → PO status = Short Closed. Finance module notified that invoiceable quantity has been reduced.
6	**Finance Officer**	Reviews updated invoiceable quantities. Ensures any outstanding supplier invoice does not exceed qty_received (short-closed qty is NOT invoiceable).	System flags any invoice in PENDING status whose quantity exceeds the updated qty_invoiceable. Finance Officer must approve or reject those invoices.

5A.5.3 Short-Close Worked Example

Continuing from Section 4 example — suppose after GRN-001 (60 Laptops received), the supplier confirms they cannot deliver the remaining 40 Laptops.

Item	Qty ordered	Qty received	Qty short-closed	Qty pending	Line status
Laptop	100	60	40	0	Short-Closed
Wireless Mouse	200	200	0	0	Fully Received
PO Status	**Short Closed**				

Key result: Only 60 Laptops + 200 Mice are invoiceable. The supplier cannot invoice for the 40 short-closed Laptops. Any invoice claiming 100 Laptops will fail the 3-way match and be put on hold.

5A.5.4 Short-Close Business Rules

- Short-close requires Procurement Manager approval. Purchase Officer cannot self-approve a short-close.
- **A mandatory vendor communication reference (email, letter, or call log reference) must be entered. This is the documented evidence that the supplier confirmed non-delivery.**
- The short-closed quantity is permanently locked. It cannot be un-short-closed. If the supplier later offers to deliver, a new PO must be raised.

- Short-closed quantity is not invoiceable. The Finance module uses (qty_received - qty_invoiced) as the invoiceable balance, which excludes short-closed quantities automatically.
- Short-close is recorded in the audit trail with: user, timestamp, original qty_pending, qty_short_closed, reason, and vendor communication reference.
- If a short-close is approved on a line that is Partially Invoiced, the system checks whether any pending invoice covers the short-closed quantity and flags it for Finance review.
- A partial short-close is allowed: if qty_pending = 40, the Purchase Officer can short-close 30 and leave 10 still pending (expecting partial future delivery).

5A.6 Multiple Invoices Against One PO

Just as multiple GRNs are supported per PO, the SMS must support multiple invoices per PO. This is the standard in ERP-grade procurement — suppliers invoice per delivery, not per PO. Each invoice references the specific GRN(s) it covers.

5A.6.1 Invoice-to-GRN Relationship

Relationship	Description
One invoice → one GRN	Most common case. Supplier delivers a batch, raises an invoice for exactly that delivery.
One invoice → multiple GRNs	Supplier consolidates multiple deliveries into one invoice. Allowed. The invoice must be linked to all relevant GRNs. The 3-way match validates against the sum of all linked GRNs.
Multiple invoices → one GRN	Rare but allowed. Supplier splits one delivery's invoice (e.g. goods and freight separately). Both invoices linked to the same GRN. The sum of linked invoices must not exceed GRN accepted value.
Multiple invoices → one PO	Standard scenario for partial deliveries. Each invoice covers what has been delivered so far. Total invoiced across all invoices must never exceed total qty_received across all GRNs.

5A.6.2 Multi-Invoice Business Rules

- Each invoice references a PO (mandatory) and one or more GRNs (mandatory). An invoice cannot be created without a corresponding approved GRN.
- **The system validates that the total invoiced quantity per PO line across all invoices does not exceed qty_received for that line.**
- An invoice covering multiple GRNs is valid provided the sum of the invoice amounts does not exceed the combined accepted value of all referenced GRNs.
- Each invoice is independently processed through the 3-way match engine. The match compares: this invoice's amount vs the GRN(s) it references vs the relevant PO lines.
- Invoices cannot be created for short-closed quantities. The system enforces: invoice qty per PO line ≤ qty_received (not qty_ordered).
- Invoice sequence is tracked. The system records which GRNs are 'fully invoiced' (all accepted quantity covered by approved invoices) to support PO closure logic.

- Overpayment protection: if the total amount across all invoices for a PO line would exceed the PO line total, the system flags the invoice as a variance requiring Finance Officer approval before it can proceed.

5A.6.3 Multi-Invoice Worked Example

Continuing the Section 4 example after both GRN-001 and GRN-002 are approved:

Invoice 1 — covers GRN-001 (60 Laptops + 200 Mice)

Inv line	Item	Qty invoiced	Unit price	Line total	Match status
1	Laptop	60	PKR 75,000	PKR 4,500,000	Matched ✓
2	Wireless Mouse	200	PKR 1,500	PKR 300,000	Matched ✓
Invoice-1 total				**PKR 4,800,000**	**Matched**

Invoice 2 — covers GRN-002 (40 Laptops)

Inv line	Item	Qty invoiced	Unit price	Line total	Match status
1	Laptop	40	PKR 75,000	PKR 3,000,000	Matched ✓
Invoice-2 total				**PKR 3,000,000**	**Matched**

PO line invoice tracking after both invoices approved:

Item	Qty ordered	Qty received	Qty invoiced	Qty inv. pending	Line status
Laptop	100	100	100	0	Fully Invoiced
Wireless Mouse	200	200	200	0	Fully Invoiced
PO Status					**Fully Invoiced**

5A.7 Three-Way Match — Detailed Rules

The 3-way match (3WM) is the financial control that ensures the SMS only pays for goods that were ordered (PO), actually received (GRN), and correctly invoiced (Invoice). It runs automatically when an invoice is saved. The result determines whether the invoice can proceed to payment or must be reviewed.

5A.7.1 What the 3-Way Match Compares

Document	Field compared	What it represents
Purchase Order (PO)	po_line.unit_price × qty_ordered	The committed purchase price per unit, the maximum payable per line.
Goods Receipt Note (GRN)	SUM(grn_line.qty_accepted) per po_line_id	The actual quantity physically received and accepted — the basis for the invoice.
Supplier Invoice	invoice_line.qty_invoiced × invoice_line.unit_price	What the supplier is claiming for payment.

5A.7.2 Match Scenarios and System Responses

	Scenario	Match result	System action
✓	Invoice qty = GRN qty AND invoice amount within 5% of PO value	**MATCHED**	Invoice proceeds to approval workflow. Finance Officer notified for final approval.
⚠️	Invoice qty = GRN qty BUT invoice amount > PO unit price by more than 5%	**VARIANCE**	Invoice put on hold. Finance Officer notified with variance detail. Must approve-with-reason or reject before invoice proceeds.
✗	Invoice qty > GRN accepted qty (supplier invoicing for more than was received)	**MISMATCH — HOLD**	Invoice put on hold automatically. Finance Officer notified. Procurement Manager also notified. Invoice cannot proceed until supplier issues a credit note or the invoice quantity is corrected.
✗	Invoice references a PO line that is Short-Closed (invoicing for short-closed qty)	**MISMATCH — REJECT**	Invoice automatically rejected. Notification sent to Finance Officer and supplier contact. Supplier must issue a revised invoice.
⚠️	Invoice qty < GRN qty (supplier under-invoicing — partial invoice)	**PARTIAL MATCH**	Invoice approved for the invoiced amount. The uninvoiced GRN quantity remains as qty_invoice_pending. A subsequent invoice is expected to cover the balance.
✓	Multiple invoices covering one PO — combined quantities = total GRN qty AND combined value within 5% of PO	**MATCHED (cumulative)**	All invoices matched cumulatively. PO line moves to Fully Invoiced when combined qty_invoiced = qty_received.

5A.7.3 3-Way Match — Additional Field: Invoice Line

The existing Invoice field definitions in FSD Section 10 should be extended with the following per-line fields to support multi-GRN invoicing and line-level 3WM:

Field name	Data type	Required	Notes
invoice_line_id	UUID	Yes	Primary key for invoice line
invoice_id	FK → Invoice	Yes	Parent invoice
po_line_id	FK → POLine	Yes	Which PO line this invoice line covers
grn_line_id	FK → GRNLine	Yes	Which specific GRN line this invoice line matches to
qty_invoiced	Decimal	Yes	Quantity being invoiced on this line
unit_price_invoiced	Decimal	Yes	Unit price on this invoice line

Field name	Data type	Required	Notes
line_total	Decimal	Yes	Calculated: qty_invoiced × unit_price_invoiced
po_unit_price	Decimal	Yes (auto)	Snapshot from PO line at time of invoice creation — for comparison
grn_qty_accepted	Decimal	Yes (auto)	Snapshot from GRN line accepted qty — for comparison
variance_pct	Decimal	No (auto)	Computed: ABS(unit_price_invoiced - po_unit_price) / po_unit_price × 100
line_match_status	Lookup	Yes (auto)	Matched / Variance / Mismatch per line
line_match_notes	Text (300)	No	System-generated explanation of variance or mismatch

5A.8 PO Closure Logic

A PO closes automatically when all financial obligations are settled. Manual closure is also available to Procurement Manager for special circumstances. A closed PO is immutable — no further GRNs, invoices, or amendments can be raised against it.

5A.8.1 Automatic Closure Conditions

Automatic closure rule	The system evaluates PO closure eligibility automatically after every GRN approval, invoice approval, payment clearance, and short-close action. When ALL conditions in the relevant closure type are met, the PO status is updated and all parties are notified.

Closure type	All conditions must be true	Resulting PO status
Standard closure (full delivery)	• ALL PO lines: line_status = Fully Invoiced • ALL invoices: payment_status = Paid or Cleared • No open amendments or pending approvals	**Closed**
Short-close closure (partial delivery)	• ALL PO lines: line_status = Fully Invoiced OR Short-Closed OR Cancelled • ALL invoices for received goods: payment_status = Paid or Cleared • No open amendments	**Short Closed**
Cancelled (no delivery)	• ALL PO lines: qty_received = 0 • PO Amendment of type CANCEL approved	**Cancelled**

Closure type	All conditions must be true	Resulting PO status
	• Procurement Manager approval on record	

5A.8.2 Manual Closure

- Procurement Manager can manually close a PO that meets financial settlement conditions but has not yet auto-closed due to system timing.
- **Manual closure requires: entering a closure reason, confirming no outstanding deliveries or invoices are expected, and Procurement Manager role authentication.**
- Manual closure is recorded in the audit trail with user, timestamp, reason, and PO state at time of closure.
- Manual closure cannot be performed on a PO with: pending GRN approvals, pending invoice approvals, open PO amendments, or unpaid invoices.
- A manually closed PO can be re-opened by System Admin only, with a re-open reason and audit entry.

5A.8.3 Post-Closure Rules

Action	Permitted after PO closure
Create new GRN against this PO	Not permitted. GRN creation validates PO status ≠ Closed / Short Closed / Cancelled.
Create new invoice against this PO	Not permitted. Invoice creation validates PO status ≠ Closed / Short Closed / Cancelled.
Raise a PO amendment	Not permitted. Amendment creation validates PO is not in a closed state.
View the PO and all related documents	Permitted for all roles with PO view access. Historical record remains fully accessible.
Export PO and its GRNs/invoices	Permitted for Admin, Procurement Manager, Finance Officer.
Generate audit trail report for this PO	Permitted for Admin and Auditor roles.
Raise a new PO for undelivered items	Permitted. A new PO/PR must be raised against a new or different supplier for any quantity that was short-closed and still needed.

5A.9 Updated Field Definitions — PO Line Additions

The following fields must be added to the PO Line field definition table in FSD Section 5.3. They extend the existing po_lines table and are required to support the delivery lifecycle described in this addendum.

Field name	Data type	Required	Notes
qty_rejected	Decimal	No (auto)	Running total of rejected qty across all GRNs for this line. Updated on GRN approval. System managed — not manually editable.
qty_short_closed	Decimal	No	Quantity formally short-closed via

Field name	Data type	Required	Notes
			approved POAmendment of type SHORT_CLOSE. Defaults to 0. Requires Procurement Manager approval to set.
qty_invoiced	Decimal	No (auto)	Running total of quantity covered by approved invoices. Updated when invoice line for this po_line_id is approved. System managed.
qty_invoice_pending	Decimal	No (computed)	Computed: qty_received - qty_invoiced. Read-only. Represents the invoiceable balance not yet covered by any invoice.
line_receipt_status	Lookup	Yes (auto)	Delivery state: Open / Partially Received / Fully Received / Short-Closed / Cancelled. Updated by system on GRN approval or short-close.
line_invoice_status	Lookup	Yes (auto)	Invoice state: Not Invoiced / Partially Invoiced / Fully Invoiced. Updated by system on invoice line approval.
last_grn_date	Date	No (auto)	Date of the most recent approved GRN against this PO line. Updated on each GRN approval.
last_invoice_date	Date	No (auto)	Date of the most recent approved invoice line against this PO line.
short_close_reason	Lookup	No	Required if qty_short_closed > 0. From PO Amendment reason lookup: Partial Shipment / Price Change / Spec Mismatch / Short-Close / Other.
short_close_ref	Text (100)	No	Vendor communication reference for the short-close evidence (email ID, letter reference, call log number).
short_closed_at	DateTime	No (auto)	Timestamp when short-close was approved.
short_closed_by	FK → User	No (auto)	User who approved the short-close.

5A.10 Updated Lookup — PO Status (Full List)

The PO Status lookup in FSD Section 5.5 must be updated to include Partially Invoiced and Fully Invoiced statuses, which were missing. The complete updated list is:

Code	Display value	CRUD	Notes
DRAFT	Draft	N/A	Being prepared, not submitted
PEND	Pending Approval	N/A	Submitted for approval chain
APPR	Approved	N/A	Fully approved, not yet sent
SENT	Sent to Supplier	N/A	PO transmitted, awaiting delivery
PART_RCV	Partially Received	N/A	Some GRN lines received — updated

Code	Display value	CRUD	Notes
			from SENT
FULL_RCV	Fully Received	N/A	All deliverable lines received — no qty_pending on any line
PART_INV	**Partially Invoiced**	N/A	NEW — At least one line has qty_invoiced > 0, not all fully invoiced
FULL_INV	**Fully Invoiced**	N/A	NEW — All received quantities are covered by approved invoices
CLOSED	Closed	N/A	All invoices paid. PO complete. Immutable.
SHORT_CLOSED	Short Closed	N/A	Closed with at least one Short-Closed line. Some qty was never delivered.
CANCEL	Cancelled	N/A	Voided — no delivery occurred

Rows highlighted in amber are new additions to the FSD v3 lookup. The development team must add these to the PO status lookup seed data and update the status derivation logic in the ProcurementService.

— End of Addendum —

6 Module — Inventory Management

Tracks stock levels across all warehouses and locations in real time. Manages product catalog, stock adjustments, reorder rules, stock transfers, and cycle counts.

6.1 Screens & Features

- Product Catalog (master list of all SKUs)
- Stock Levels (by warehouse, zone, bin)
- Stock Adjustment Entry
- Reorder Rule Configuration
- Inter-Warehouse Transfer
- Cycle Count / Stocktake
- Inventory Valuation Report
- Expiry / Batch / Serial Tracking (optional)

6.2 Product Master — Field Definitions

Field name	Data type	Required	CRUD	Lookup / values	Notes
product_id	Auto (UUID)	Yes	N/A	System generated	Primary key
sku	Text (30)	Yes	Manual / Auto	Unique e.g. PRD-00001	Stock keeping unit

Field name	Data type	Required	CRUD	Lookup / values	Notes
product_name	Text (200)	Yes	Free text	—	Full product name
short_name	Text (50)	No	Free text	—	Abbreviated name
category_id	FK → Category	Yes	Lookup + CRUD	Product categories	See lookup below
sub_category_id	FK → SubCategory	No	Lookup + CRUD	Sub-categories	Child of category
brand	Text (100)	No	Free text	—	Brand / manufacturer
unit_of_measure	Lookup	Yes	Yes	Each / Kg / Litre / Box / Set	Base UOM
weight_kg	Decimal	No	Number	—	Weight per unit
dimensions	Text (50)	No	Free text	L×W×H cm	Physical size
shelf_life_days	Integer	No	Number	—	0 = no expiry
is_batch_tracked	Boolean	No	Toggle	Yes / No	Batch/lot tracking
is_serial_tracked	Boolean	No	Toggle	Yes / No	Serial number tracking
barcode	Text (50)	No	Free text	EAN/UPC/QR	Scan code
standard_cost	Decimal	Yes	Number	—	Standard valuation cost
last_purchase_price	Decimal	No	Auto	Updated by PO receipt	—
sales_price	Decimal	No	Number	—	If also sold
reorder_point	Decimal	Yes	Number	—	Trigger reorder below this
reorder_qty	Decimal	Yes	Number	—	How much to reorder
min_stock	Decimal	No	Number	—	Absolute minimum
max_stock	Decimal	No	Number	—	Warehouse capacity limit
lead_time_days	Integer	No	Number	—	Days to receive from supplier
preferred_supplier_id	FK → Supplier	No	Lookup	Active suppliers	Default vendor
status	Lookup	Yes	Yes	Active / Inactive / Discontinued	Product status
notes	Text (300)	No	Free text	—	Internal notes
image_url	URL	No	Upload	JPEG/PNG	Product image

Field name	Data type	Required	CRUD	Lookup / values	Notes
created_at	DateTime	Yes	Auto	—	—
created_by	FK → User	Yes	Auto	—	—

6.3 Inventory Item (Stock Record) — Field Definitions

Field name	Data type	Required	CRUD	Lookup / values	Notes
inv_id	Auto (UUID)	Yes	N/A	System generated	PK — one row per product/warehouse/bin
product_id	FK → Product	Yes	Auto	Product catalog	Which product
warehouse_id	FK → Warehouse	Yes	Lookup	Active warehouses	Which warehouse
zone_id	FK → Zone	No	Lookup	Warehouse zones	Storage zone
bin_id	FK → Bin	No	Lookup	Bin locations	Exact shelf location
qty_on_hand	Decimal	Yes	Calculated	Updated by transactions	Current physical stock
qty_reserved	Decimal	No	Auto	Reserved by open orders	Cannot pick
qty_available	Decimal	Yes	Calculated	on_hand - reserved	Available to promise
qty_on_order	Decimal	No	Auto	Open PO quantities	Incoming stock
batch_number	Text (50)	No	Free text	—	Batch / lot number
serial_number	Text (50)	No	Free text	—	Serial if tracked
expiry_date	Date	No	Date picker	—	For perishables
last_count_date	Date	No	Auto	Updated by cycle count	—
valuation_method	Lookup	No	Yes	FIFO / LIFO / Weighted Avg	Costing method
unit_cost	Decimal	Yes	Auto	From last receipt / weighted avg	Current cost
total_value	Decimal	Yes	Calculated	qty × unit_cost	Stock value

6.4 Stock Adjustment — Field Definitions

Field name	Data type	Required	CRUD	Lookup / values	Notes
adj_id	Auto (UUID)	Yes	N/A	System generated	PK
adj_number	Text (20)	Yes	Auto	ADJ-YYYY-NNNNN	Reference number
adj_type	Lookup	Yes	Yes	Write-off / Damage / Count / Transfer	Reason type
product_id	FK → Product	Yes	Lookup	Product catalog	Which item
warehouse_id	FK → Warehouse	Yes	Lookup	Active warehouses	Location
qty_before	Decimal	Yes	Auto	System reads current stock	Pre-adjustment qty
qty_adjusted	Decimal	Yes	Number	Positive = add / Negative = remove	Change amount
qty_after	Decimal	Yes	Calculated	before + adjusted	Post-adjustment qty
reason	Lookup	Yes	Yes	Damage / Expiry / Theft / Count Variance / Other	—
reference_doc	Text (100)	No	Free text	—	e.g. count sheet number
approved_by	FK → User	No	Auto	Required if >100 units	Approval for large adj
notes	Text (300)	No	Free text	—	Explanation
created_at	DateTime	Yes	Auto	—	—
created_by	FK → User	Yes	Auto	—	—

6.5 Lookup Collections

Lookup: Product Category (CRUD: Yes)

Code / ID	Display value	Active	Notes
ELEC	Electronics	Yes	Computers, cables, devices
FURN	Furniture	Yes	Office furniture and fixtures
STAT	Stationery	Yes	Paper, pens, consumables

Code / ID	Display value	Active	Notes
MED	Medical Supplies	Yes	Healthcare equipment and consumables
RAW	Raw Materials	Yes	Production inputs
PKG	Packaging	Yes	Boxes, labels, wrap
MAINT	Maintenance	Yes	Tools, spare parts

Lookup: Unit of Measure (CRUD: Yes)

Code / ID	Display value	Active	Notes
EA	Each	Yes	Individual unit
KG	Kilogram	Yes	Weight measure
LTR	Litre	Yes	Volume measure
BOX	Box	Yes	Packaged box
SET	Set	Yes	Group of items
MTR	Meter	Yes	Length measure
PKT	Packet	Yes	Small packet
PAIR	Pair	Yes	Two items together

6.6 Role-Based Access — Inventory Management

Screen / action	Admin	Proc. Mgr	Purchase Off.	Inv. Mgr	Requester
View stock levels	**Full**	**Full**	**Full**	**Full**	**View**
Add / edit product	**Full**	**View**	**None**	**Full**	**None**
Stock adjustment	**Full**	**None**	**None**	**Full**	**None**
Approve large adjustment	**Full**	**Full**	**None**	**None**	**None**
Reorder rule config	**Full**	**Full**	**None**	**Full**	**None**
Inter-warehouse transfer	**Full**	**None**	**None**	**Full**	**None**
Run cycle count	**Full**	**None**	**None**	**Full**	**Full**
View valuation report	**Full**	**Full**	**None**	**Full**	**None**

Sections 6.7 through 6.10 below add Product Category and Sub-Category navigation, view, edit, and delete corrections raised against the Inventory Management screens.

6.7 Item 1 — Dedicated Sub-Category Navigation Link (Addendum 17)

#1	**DEFECT**	**Sub-Category has no separate, dedicated navigation link**

6.7.1 Current Behaviour

Sub-Category is only reachable by opening a specific Product Category record and looking for its sub-categories inside that record's detail view. There is no top-level menu item for Sub-Category. A user wanting to browse, search, or manage sub-categories independently of a specific parent category has no way to do so.

6.7.2 Required Behaviour

Add a new, independent navigation menu item: Inventory → Product Categories → Sub-Categories, positioned as a sibling link directly beneath or alongside the existing Product Categories link — not nested inside an individual category's detail page.

Navigation element	Specification
Menu path	Inventory › Product Categories › Sub-Categories
Route	/inventory/sub-categories
Landing behaviour	Opens directly to the full Sub-Category list (Section 3) — not filtered to any single parent category by default
Visibility / RBAC	Visible to Admin and Inventory Manager (same access as Product Categories per FSD v3 Section 6.6)
Breadcrumb	Inventory › Product Categories › Sub-Categories (always shown, regardless of entry point)

The existing entry point — viewing sub-categories from within a specific Category's detail page — remains available as a convenience filter (a deep link into the same Sub-Category list, pre-filtered by parent category). It is not removed; it is now a filtered view of the same independent screen rather than the only way to reach sub-categories.

6.8 Item 2 — 'View Sub Categories' Page Repair (Addendum 17)

#2	**DEFECT**	**'View Sub Categories' page does not load or function**

6.8.1 Current Behaviour

Clicking 'View Sub Categories' from the Product Categories screen either fails to load, loads an empty/broken view, or does not correctly display the sub-categories belonging to the selected category. This is a functional defect against the screen that should already exist per FSD v3 Section 6.1 (Screens & Features).

6.8.2 Required Behaviour — Sub-Category List Screen

This is now the same screen introduced in Section 2 (route /inventory/sub-categories), but specified in full here as the fix target.

Column	Source field	Behaviour
Sub-Category Code	sub_category_code	Sortable, searchable
Sub-Category Name	sub_category_name	Sortable, searchable, primary display field
Parent Category	category_id (joined to ProductCategories.category_name)	Filterable dropdown above the grid

Column	Source field	Behaviour
Active Status	is_active	Toggle filter: All / Active / Inactive
Product Count	Computed — COUNT(Products WHERE sub_category_id = this row)	Read-only, shown for reference, links to filtered Product list
Actions	—	Edit and Delete icons per row (see Sections 5 and 6 for behaviour)

6.8.3 API Endpoint

Method	Endpoint	Description
GET	/api/product-categories/sub-categories	Paginated list. Query params: categoryId (optional filter), search, isActive, page, pageSize. Returns sub-category rows joined with parent category name and live product count.
GET	/api/product-categories/{categoryId}/sub-categories	Same list, pre-filtered to one parent category. This is the endpoint the 'View Sub Categories' link from within a Category detail page calls.

6.8.4 Acceptance Criteria

- Clicking 'View Sub Categories' from any Product Category record opens the Sub-Category list filtered to that category, with no error and no blank state
- Sub-categories with zero products still display correctly (Product Count = 0, not a broken row)
- Navigating to /inventory/sub-categories directly (Item 1) shows all sub-categories across all parent categories, unfiltered
- Search, sort, and the Parent Category filter dropdown all function correctly on first load

6.9 Item 4 — Product Category Edit and Delete Repair (Addendum 17)

#4	**DEFECT**	**Edit and Delete actions on Product Categories are non-functional**

6.9.1 Current Behaviour

On the Product Categories list screen, clicking Edit does not open a working edit form (or the form does not save changes), and clicking Delete either does nothing, throws an error, or does not correctly remove/deactivate the category.

6.9.2 Required Behaviour — Edit

Trigger	Edit icon on Product Categories list row
Form fields	category_name (required), description (optional), is_active (toggle)
Immutable field	category_code is read-only on edit — same protection pattern as role_code in Section 4.5
Save action	PUT /api/product-categories/{id} — updates the record; returns updated

	row for grid refresh
Validation	category_name required, max 100 characters; duplicate name check (case-insensitive) against other active categories

6.9.3 Required Behaviour — Delete

Design decision	Categories are never hard-deleted if they have any product or sub-category references — this would silently orphan data across Inventory, Procurement, and Demand Planning. 'Delete' on a referenced category performs a soft deactivation. 'Delete' on a category with zero references performs a true delete.

Condition	Action taken	System response
Category has zero Products and zero Sub-Categories referencing it	**Hard delete**	DELETE /api/product-categories/{id} removes the row. Confirmation dialog required before action.
Category has one or more Products or Sub-Categories referencing it	**Soft deactivate**	DELETE /api/product-categories/{id} is rejected with 409 Conflict; UI instead offers 'Deactivate' which sets is_active = false. Category remains visible in historical records but excluded from new-product category pickers.

6.9.4 Acceptance Criteria

- Edit form opens pre-populated with current category values and saves changes correctly on submit
- category_code field is visible but disabled/read-only in the edit form
- Delete on an unreferenced category removes it permanently after confirmation
- Delete on a referenced category is blocked with a clear message and a Deactivate alternative is offered
- Grid refreshes automatically after a successful edit, delete, or deactivate action — no manual page reload required

6.10 Item 5 — Sub-Category Edit and Delete (New Capability) (Addendum 17)

#5	**DEFECT**	**Sub-Category has no Edit or Delete action at all**

6.10.1 Current Behaviour

The Sub-Category list (now specified independently in Item 2) does not currently expose Edit or Delete actions per row at all — not broken, but entirely absent. Once a sub-category is created there is no way to change its name, reassign its parent category, or remove it.

6.10.2 Required Behaviour — Edit

Trigger	Edit icon on Sub-Category list row (Section 3 grid)
Form fields	sub_category_name (required), parent category_id (dropdown — re-assignable), description (optional), is_active (toggle)
Immutable field	sub_category_code is read-only on edit

Save action	PUT /api/product-categories/sub-categories/{id}
Validation	sub_category_name required, max 100 characters; duplicate name check scoped within the same parent category (the same sub-category name can exist under two different parent categories)
Re-parenting effect	If parent category_id is changed, all Products currently assigned to this sub-category retain the sub-category assignment — only the sub-category's parent link changes, not the products' direct links

6.10.3 Required Behaviour — Delete

Follows the same referenced-vs-unreferenced pattern established for Product Category in Section 5.3, scoped to products only (a sub-category has no further child level).

Condition	Action taken	System response
Sub-category has zero Products referencing it	**Hard delete**	DELETE /api/product-categories/sub-categories/{id} removes the row after confirmation
Sub-category has one or more Products referencing it	**Soft deactivate**	Request rejected with 409 Conflict; UI offers Deactivate (is_active = false) instead. Affected product count shown in the confirmation dialog.

6.10.4 Role-Based Access — Items 1, 2, 4, 5

Screen / action	Admin	Proc. Mgr	Inv. Mgr	All others
View Sub-Categories list / link	Full	View	Full	None
Create Category / Sub-Category	Full	None	Full	None
Edit Category / Sub-Category	Full	None	Full	None
Delete / Deactivate Category / Sub-Category	Full	None	Full	None

— End of Addendum —

7 Module — Order Management

Manages both inbound (from suppliers) and outbound (to customers or internal departments) order flows, tracking status from creation to fulfillment.

7.1 Order Header — Field Definitions

Field name	Data type	Required	CRUD	Lookup / values	Notes
order_id	Auto (UUID)	Yes	N/A	System generated	PK
order_number	Text (20)	Yes	Auto	ORD-YYYY-NNNNN	Reference

Field name	Data type	Required	CRUD	Lookup / values	Notes
					number
order_type	Lookup	Yes	Yes	Purchase / Sales / Internal Transfer	Direction
order_date	Date	Yes	Date picker	Default: today	—
expected_date	Date	Yes	Date picker	—	Requested delivery date
source_id	FK → Supplier / Dept	Yes	Lookup	Depends on order type	From whom
destination_id	FK → Warehouse / Dept	Yes	Lookup	Active warehouses	To where
status	Lookup	Yes	Auto	New/Confirmed/Processing/Shipped/Delivered/Cancelled	Order state
priority	Lookup	Yes	Yes	Low / Normal / High / Urgent	Fulfillment urgency
total_lines	Integer	No	Auto	Count of lines	—
total_qty	Decimal	No	Calculated	Sum of line qtys	—
total_value	Decimal	No	Calculated	Sum of line totals	—
linked_po_id	FK → PO	No	Auto	If sourced from PO	—
linked_shipment_id	FK → Shipment	No	Auto	When dispatched	—
notes	Text (300)	No	Free text	—	—
created_at	DateTime	Yes	Auto	—	—
created_by	FK → User	Yes	Auto	—	—

7.2 Role-Based Access — Order Management

Screen / action	Admin	Proc. Mgr	Purchase Off.	Inv. Mgr	Requester
Create order	**Full**	**Full**	**Full**	**Full**	**None**

Screen / action	Admin	Proc. Mgr	Purchase Off.	Inv. Mgr	Requester
Edit open order	**Full**	**Full**	**Full**	**Full**	**None**
Confirm / approve order	**Full**	**Full**	**Full**	**None**	**None**
Cancel order	**Full**	**Full**	**None**	**None**	**None**
View all orders	**Full**	**Full**	**Full**	**Full**	**View own**
Export order list	**Full**	**Full**	**Full**	**Full**	**None**

8 Module — Warehouse Management

Controls all physical warehouse operations: goods receiving, put-away, bin management, picking, packing, dispatch, and stock transfers.

8.1 Screens & Features

- Warehouse Master Setup
- Zone and Bin Configuration
- Goods Receipt Note (GRN) Entry
- Put-Away Task Management
- Pick List Generation
- Dispatch / Shipment Preparation
- Internal Transfer Order
- Bin-Level Stock Inquiry

8.2 Warehouse Master — Field Definitions

Field name	Data type	Required	CRUD	Lookup / values	Notes
warehouse_id	Auto (UUID)	Yes	N/A	System generated	PK
warehouse_code	Text (10)	Yes	Manual	WH-01	Short code
warehouse_name	Text (100)	Yes	Free text	—	Full name
warehouse_type	Lookup	Yes	Yes	Main / Transit / Cold Storage / Bonded	See lookup
country	Lookup	Yes	Yes	Country list	Location
city	Text (100)	Yes	Free text	—	City
address	Text (300)	Yes	Free text	—	Full address
manager_id	FK → User	Yes	Lookup	Users with WH role	Warehouse manager
phone	Text (20)	No	Free text	—	—

Field name	Data type	Required	CRUD	Lookup / values	Notes
capacity_sqm	Decimal	No	Number	—	Floor area m^2
is_default	Boolean	No	Toggle	Yes / No	Default receiving WH
status	Lookup	Yes	Yes	Active / Inactive / Under Maintenance	—
notes	Text (200)	No	Free text	—	—

8.3 Goods Receipt Note (GRN) — Field Definitions

Field name	Data type	Required	CRUD	Lookup / values	Notes
grn_id	Auto (UUID)	Yes	N/A	System generated	PK
grn_number	Text (20)	Yes	Auto	GRN-YYYY-NNNNN	Reference number
po_id	FK → PO	Yes	Lookup	Approved, sent POs	Source PO
supplier_id	FK → Supplier	Yes	Auto	Inherited from PO	—
warehouse_id	FK → Warehouse	Yes	Lookup	Active warehouses	Receiving location
received_date	Date	Yes	Date picker	Default: today	Physical receipt date
delivery_note_no	Text (50)	No	Free text	—	Supplier's delivery note
vehicle_no	Text (30)	No	Free text	—	Delivery vehicle
driver_name	Text (100)	No	Free text	—	—
status	Lookup	Yes	Auto	Draft / Pending QC / Accepted / Rejected / Partial	GRN state
qc_passed	Boolean	No	Toggle	Yes / No	Quality check result
qc_notes	Text (300)	No	Free text	—	QC findings
qc_done_by	FK → User	No	Auto	—	QC inspector
received_by	FK → User	Yes	Auto	Current user	Receiving clerk
approved_by	FK → User	No	Auto	—	Supervisor approval
invoice_no	Text (50)	No	Free text	—	Supplier invoice

Field name	Data type	Required	CRUD	Lookup / values	Notes
					number
notes	Text (300)	No	Free text	—	—
created_at	DateTime	Yes	Auto	—	—

8.4 GRN Line — Field Definitions

Field name	Data type	Required	CRUD	Lookup / values	Notes
grn_line_id	Auto (UUID)	Yes	N/A	System generated	PK
grn_id	FK → GRN	Yes	Auto	Parent GRN	—
po_line_id	FK → POLine	Yes	Lookup	PO lines	Source PO line
product_id	FK → Product	Yes	Auto	Inherited from PO line	—
qty_ordered	Decimal	Yes	Auto	From PO line	Expected qty
qty_received	Decimal	Yes	Number	—	Actual qty received
qty_accepted	Decimal	Yes	Number	—	QC accepted qty
qty_rejected	Decimal	No	Number	—	QC rejected qty
rejection_reason	Lookup	No	Yes	Damaged / Wrong item / Short expiry / Over qty	—
bin_id	FK → Bin	No	Lookup	Available bins	Put-away location
batch_number	Text (50)	No	Free text	—	If batch tracked
expiry_date	Date	No	Date picker	—	If expiry tracked
unit_cost	Decimal	No	Auto	From PO line	For valuation

8.5 Lookup Collections

Lookup: Warehouse Type (CRUD: Yes)

Code / ID	Display value	Active	Notes
MAIN	Main Warehouse	Yes	Primary storage facility
TRANSIT	Transit / Cross-dock	Yes	Short-term storage in transit
COLD	Cold Storage	Yes	Temperature-controlled
BOND	Bonded Warehouse	Yes	Customs-bonded storage

Code / ID	Display value	Active	Notes
VAN	Van / Mobile	Yes	Field sales vehicle stock

8.6 Role-Based Access — Warehouse Management

Screen / action	Admin	Proc. Mgr	Purchase Off.	Inv. Mgr	Requester
View GRN list	**Full**	**Full**	**Full**	**Full**	**View**
Create / edit GRN	**Full**	**None**	**Full**	**Full**	**None**
Approve GRN	**Full**	**Full**	**None**	**Full**	**None**
Reject GRN line	**Full**	**Full**	**Full**	**Full**	**None**
Warehouse master config	**Full**	**None**	**None**	**Full**	**None**
Run pick list	**Full**	**None**	**None**	**Full**	**Full**
Update bin location	**Full**	**None**	**None**	**Full**	**Full**

Sections 8.7 and 8.8 below add a Google Maps location field to the Warehouse Master and formalise the rack/shelf/bin storage hierarchy.

8.7 Group C — Warehouse Master: Google Maps Location Field (Addendum 19)

#4	**NEW FEATURE**	**Add Google Maps location to the Warehouse Master**

8.7.1 Required Behaviour

Field	Type	Required	Notes
google_maps_url	Text (500)	No	Direct link to the Google Maps location, e.g. https://maps.google.com/?q=lat,lng. Stored as-is; the system does not need to parse coordinates separately unless lat/lng are also captured (see next two rows).
latitude	Decimal(9,6)	No	Optional structured coordinate if the system wants to render an embedded map pin rather than only a clickable link.
longitude	Decimal(9,6)	No	Paired with latitude.

8.7.2 UI Behaviour

- On the Add/Edit Warehouse form, a new 'Location' field group is added beneath the existing Address fields: a text input for pasting a Google Maps link, plus an inline 'Pick on Map' button that opens an embedded map picker (Google Maps JavaScript API) to set latitude/longitude visually and auto-generate the google_maps_url.
- On the Warehouse detail/view screen, if google_maps_url or lat/lng is populated, render a small embedded map preview and an 'Open in Google Maps' link.

- The field is fully optional — existing warehouses with no location data continue to function unaffected; the map section simply does not render if no location data is present.

8.7.3 Acceptance Criteria

- Warehouse create/edit form accepts a Google Maps URL and/or coordinates picked from an embedded map
- Warehouse detail screen renders an embedded map preview when location data exists
- Saving a warehouse with no location data succeeds with no validation error — field is optional

8.8 Group C — Warehouse Structure: Rack / Shelf / Bin Management (Addendum 19)

#5	**NEW FEATURE**	**Full structural CRUD for Rack/Shelf/Bin within a warehouse**

8.8.1 Background and Gap

FSD v3 Section 8 already defines Zones and Bins as part of the Warehouse module, but provides no screen or API for a warehouse-level user (Inventory Manager / Warehouse Operator) to actually create and manage that structure themselves. This item adds the missing management capability and introduces an explicit Rack/Shelf level between Zone and Bin to match real-world warehouse layout (Warehouse → Zone → Rack → Shelf → Bin).

8.8.2 Revised Structural Hierarchy

Level	Entity	Notes
1	**Warehouse (existing)**	Top-level facility, FSD v3 Section 8.2
2	**Zone (existing)**	Area within a warehouse, FSD v3 Section 8 — e.g. 'Receiving', 'Cold Storage Zone'
3	**Rack (NEW)**	Physical racking unit within a zone — e.g. 'Rack A', 'Rack B'
4	**Shelf (NEW)**	A level within a rack — e.g. 'Shelf 1', 'Shelf 2'
5	**Bin (existing, now nested under Shelf)**	Exact pickable location, FSD v3 Section 8 — previously nested directly under Zone, now nested under Shelf for finer granularity

Backward compatibility	Existing Bins created directly under a Zone (pre-this-addendum) remain valid with rack_id and shelf_id left NULL — Bin.zone_id is retained alongside the new optional Bin.rack_id/shelf_id so historical data and any code path referencing Zone-level bins directly continues to work without a breaking migration.

8.8.3 New Entities — Field Definitions

8.8.3.1 Racks

Field	Type	Required	Notes
rack_id	UUID	Yes	Primary key
zone_id	FK to Zones	Yes	Parent zone
rack_code	Text(20)	Yes	e.g. RACK-A. Unique within zone.
rack_name	Text(100)	No	Optional descriptive name
is_active	Boolean	Yes	Default true
created_at / created_by	DateTime / FK	Yes	Standard audit fields

8.8.3.2 Shelves

Field	Type	Required	Notes
shelf_id	UUID	Yes	Primary key
rack_id	FK to Racks	Yes	Parent rack
shelf_code	Text(20)	Yes	e.g. SHELF-01. Unique within rack.
shelf_level	Integer	No	Numeric level for sort order, e.g. 1 = bottom
is_active	Boolean	Yes	Default true
created_at / created_by	DateTime / FK	Yes	Standard audit fields

8.8.3.3 Bins — Updated Field Definitions

Field	Type	Required	Notes
bin_id	UUID	Yes	Primary key — unchanged
zone_id	FK to Zones	Yes	Retained for backward compatibility
rack_id	FK to Racks	No (new)	NULL for legacy bins created directly under a zone
shelf_id	FK to Shelves	No (new)	NULL for legacy bins; required going forward when created via the new structured flow
bin_code	Text(20)	Yes	Unchanged
is_active	Boolean	Yes	Unchanged

8.8.4 Management Screen

- New screen: Warehouse Detail → 'Manage Structure' tab, showing an expandable tree: Zone → Rack → Shelf → Bin.
- Inventory Manager and Warehouse Operator (the same roles already permitted to manage Zones/Bins per FSD v3 Section 8.6) can add, edit, and deactivate Racks, Shelves, and Bins inline from the tree view — no separate full-page forms required for simple add actions.

- Adding a Rack requires selecting the parent Zone; adding a Shelf requires selecting the parent Rack; adding a Bin requires selecting the parent Shelf (or, for legacy-style bins, the parent Zone directly, preserving the old flow as an option).
- Deactivating a Rack or Shelf that still has active child Shelves/Bins is blocked with a reference count shown, consistent with the reference-aware deactivation pattern already established for Categories (Addendum 17).
- Bin-level stock inquiry (FSD v3 Section 8.1) is updated to display the full path: Zone › Rack › Shelf › Bin, for any bin that has the new structure populated, or Zone › Bin for legacy bins.

8.8.5 API Endpoints

Method	Endpoint	Description
GET	/api/warehouses/{id}/structure	Returns the full Zone → Rack → Shelf → Bin tree for a warehouse
POST	/api/zones/{zoneId}/racks	Create rack under a zone
PUT	/api/racks/{id}	Edit rack
PATCH	/api/racks/{id}/deactivate	Deactivate; blocked if active shelves/bins reference it
POST	/api/racks/{rackId}/shelves	Create shelf under a rack
PUT	/api/shelves/{id}	Edit shelf
PATCH	/api/shelves/{id}/deactivate	Deactivate; blocked if active bins reference it
POST	/api/shelves/{shelfId}/bins	Create bin under a shelf (new structured flow)

8A Supplier Returns Management (Addendum 8A)

FSD v3 and Section 5A (PO Delivery Lifecycle) do not specify how goods are returned to suppliers. The GRN approval workflow (Section 13) references an auto-created Supplier Return Order for rejected lines but, until this section, no specification existed for what that entity contains, who approves it, how goods are physically dispatched, or how the return is financially resolved. Returns are handled within the existing Warehouse and Finance modules; no separate module is required. New entities: SupplierReturnOrders, SupplierReturnOrderLines (warehouse_schema); CreditNotes, DebitNotes (finance_schema).

8A.1 Purpose of This Addendum

The SMS FSD v3 and Addendum 5A (PO Delivery Lifecycle) do not specify how goods are returned to suppliers. The GRN approval workflow references 'auto-create a Supplier Return Order for rejected lines' but provides no specification of what that entity contains, who approves it, how goods are physically dispatched, or how the return is financially resolved.

This addendum defines the complete supplier returns lifecycle as Section 8A, inserted after Section 8 (Warehouse Management). It covers all three scenarios in which a return arises, the Supplier Return Order (SRO) entity that manages the return, the dispatch process, and the three resolution paths: credit note, replacement delivery, and debit note.

Design principle	Returns do not require a separate module. They are handled within the existing Warehouse module (SRO creation, approval, dispatch) and Finance module (credit

note, debit note). The new entities add four database tables to warehouse_schema and finance_schema respectively. No new ASP.NET Core project is needed.

8A.2 Return Scenarios

There are three distinct scenarios that trigger a supplier return in the SMS. Each has a different origin point, a different trigger mechanism, and slightly different data requirements. All three converge into the same Supplier Return Order workflow.

#	Scenario	Origin / trigger	How it enters the system
1	**GRN Rejection**	Warehouse Operator enters qty_rejected > 0 on a GRN line during receiving. QC Officer confirms rejection. GRN approved with rejected lines.	System auto-creates a draft SRO for each GRN line where qty_rejected > 0 immediately after GRN approval. Procurement Manager notified to review. The SRO is pre-populated from the GRN line data (product, supplier, qty, rejection reason).
2	**Post-Receipt Defect**	Goods passed QC at receiving (GRN approved, stock posted). Defect discovered later — damage found during put-away, expiry date issue identified, quality failure discovered in use.	Inventory Manager creates SRO manually from the Warehouse module. References the original approved GRN and the affected InventoryItem. A stock adjustment (type = RETURN_PENDING) is created to reserve the defective quantity, preventing it from being issued or sold while the return is in progress.
3	**Wrong Item / Overdelivery**	Supplier delivered a different product than ordered, or delivered more units than the PO quantity. May be caught at GRN entry or discovered during put-away.	If caught at GRN: Warehouse Operator marks the line as wrong item (rejection_reason = WRONG_ITEM) — follows Scenario 1 path. If caught post-receipt: Inventory Manager creates SRO manually — follows Scenario 2 path. Over-delivery against the PO tolerance generates an automatic system alert for Procurement Manager review.

8A.3 Supplier Return Order Workflow

The Supplier Return Order (SRO) is the central document for all returns. It passes through a defined lifecycle from creation through approval, dispatch, and resolution.

8A.3.1 Step-by-Step Workflow

Step	Actor	Action	System response
1	**System (auto) or Inventory Manager**	SRO created — either auto-triggered from rejected GRN lines or manually created for	System assigns SRO number (SRO-YYYY-NNNNN). Status = DRAFT. Notifies Procurement Manager via Hangfire job. For post-receipt returns, creates a stock

Step	Actor	Action	System response
		post-receipt defect. SRO populated with: supplier, original GRN reference, original PO reference, product, qty to return, return reason.	reservation (StockAdjustment type = RETURN_PENDING, negative qty) to quarantine defective stock.
2	**Procurement Manager**	Reviews SRO — verifies return reason, confirms return is warranted, checks supplier contact details for return authorisation.	System displays SRO with original GRN and PO details for context. Shows supplier's current status and contact information.
3a	**Procurement Manager**	Approves SRO. Optionally records supplier's Return Merchandise Authorisation (RMA) number if supplier has issued one.	Status = APPROVED. Warehouse Operator notified to prepare goods for dispatch. Finance module notified that a return is in progress against this PO/invoice.
3b	**Procurement Manager**	Rejects SRO — goods to be kept (e.g. supplier offers discount instead of return).	Status = REJECTED with reason. For post-receipt returns: RETURN_PENDING stock reservation cancelled; stock returned to available. Audit record created.
4	**Warehouse Operator**	Physically prepares goods for return — segregates from good stock, packs, attaches return label. Records outbound vehicle and dispatch details.	System records dispatch date, carrier, and tracking reference on SRO. Status = DISPATCHED. InventoryItem.QtyOnHand decremented by return qty via stock adjustment (type = RETURNED). POLine.QtyReceived decremented by return qty. PO header status recalculated.
5	**Procurement Officer**	Confirms supplier has received the returned goods. Records supplier's acknowledgement reference.	Status = SUPPLIER_RECEIVED. System creates a resolution task for Finance Officer — to process credit note, debit note, or track replacement delivery.
6a	**Finance Officer**	Processes supplier credit note — enters credit note number and value issued by supplier.	Credit Note record created in finance_schema. Invoice balance updated. If invoice already paid, credit carried forward to next payment. SRO status = RESOLVED_CREDIT. POLine.QtyReceived finalised at net received qty.
6b	**Finance Officer / Procurement**	Tracks replacement delivery — supplier	SRO status = AWAITING_REPLACEMENT. When replacement arrives, a new GRN is

Step	Actor	Action	System response
	Officer	agrees to re-supply the returned goods.	raised against the original PO (multiple GRNs per PO supported per Addendum 5A). On new GRN approval, SRO status = RESOLVED_REPLACEMENT.
6c	**Finance Officer**	Raises debit note — supplier refuses return or does not respond within SLA.	Debit Note record created in finance_schema. Raised against supplier as a formal financial claim for the return value. SRO status = RESOLVED_DEBIT.

8A.3.2 SRO Status State Machine

Status	Triggered by	Meaning
DRAFT	Auto (GRN rejection) or manual (Inventory Manager)	SRO created. Awaiting Procurement Manager review.
APPROVED	Procurement Manager approval action	Return authorised. Warehouse Operator to prepare dispatch.
REJECTED	Procurement Manager rejection action	Return not proceeding. Stock reservation released. SRO closed.
DISPATCHED	Warehouse Operator records dispatch	Goods physically sent back to supplier. Inventory decremented. PO qty decremented.
SUPPLIER_RECEIVED	Procurement Officer confirms receipt	Supplier has acknowledged receipt of returned goods. Awaiting financial resolution.
AWAITING_REPLACEMENT	Finance Officer selects replacement outcome	Supplier to re-supply. System watching for new GRN against original PO.
RESOLVED_CREDIT	Finance Officer creates credit note	Credit note received and applied to invoice. SRO financially closed.
RESOLVED_REPLACEMENT	New GRN approved for replacement qty	Replacement delivery received and accepted. SRO closed.
RESOLVED_DEBIT	Finance Officer raises debit note	Debit note raised against supplier. SRO closed regardless of supplier response.
ESCALATED	System — SLA breach (configurable, default 14 days from DISPATCHED)	Supplier has not responded within SLA. Procurement Manager and Finance Officer alerted.

8A.4 Field Definitions

8A.4.1 Supplier Return Order Header — Field Definitions

Entity: SupplierReturnOrders | Schema: warehouse_schema | EF Core context: WarehouseDbContext

Field name	Data type	Required	Notes
sro_id	UUID	Yes	Primary key. System generated.
sro_number	Text (20)	Yes (auto)	SRO-YYYY-NNNNN. Unique reference number.
sro_type	Lookup	Yes	GRN_REJECTION / POST_RECEIPT_DEFECT / WRONG_ITEM / OVERDELIVERY. Determines origin scenario.
supplier_id	FK → Supplier	Yes	Supplier to whom goods are being returned. Inherited from source GRN or selected manually.
original_grn_id	FK → GRN	Yes	The GRN that received the goods being returned. Mandatory for all scenarios — post-receipt returns reference the original accepted GRN.
original_po_id	FK → PO	Yes	The PO under which the goods were ordered. Inherited from the GRN.
return_date	Date	Yes	Date SRO was created. Default: today.
return_reason	Lookup	Yes	See Return Reason lookup. Mandatory.
return_reason_detail	Text (500)	Yes	Free-text explanation of defect, wrong item description, or overdelivery detail.
rma_number	Text (100)	No	Return Merchandise Authorisation number issued by supplier. Entered by Procurement Manager on approval.
dispatch_date	Date	No (auto)	Date goods physically dispatched to supplier. Set when Warehouse Operator records dispatch.
dispatch_carrier	Text (100)	No	Carrier used for return shipment.
dispatch_tracking_ref	Text (100)	No	Outbound tracking reference.
supplier_received_date	Date	No	Date supplier confirmed receipt of returned goods.
supplier_received_ref	Text (100)	No	Supplier's acknowledgement reference number.
resolution_type	Lookup	No (auto)	CREDIT_NOTE / REPLACEMENT / DEBIT_NOTE. Set when resolution path is chosen.
resolution_date	Date	No (auto)	Date SRO was resolved (credit note

Field name	Data type	Required	Notes
			created, replacement received, or debit note raised).
sla_deadline	DateTime	Yes (auto)	Calculated: dispatch_date + configured SLA days (default 14). Escalation triggers after this.
status	Lookup	Yes (auto)	See Section 3.2 state machine.
approved_by	FK → User	No (auto)	Procurement Manager who approved the SRO.
approved_at	DateTime	No (auto)	Approval timestamp.
rejection_reason	Text (300)	No	Required if status = REJECTED.
notes	Text (500)	No	Internal remarks.
created_by	FK → User	Yes (auto)	User who created the SRO (system or Inventory Manager).
created_at	DateTime	Yes (auto)	Creation timestamp.

8A.4.2 Supplier Return Order Line — Field Definitions

Entity: SupplierReturnOrderLines | Schema: warehouse_schema | EF Core context: WarehouseDbContext

Field name	Data type	Required	Notes
sro_line_id	UUID	Yes	Primary key.
sro_id	FK → SRO	Yes	Parent SRO.
grn_line_id	FK → GRNLine	Yes	The specific GRN line from which the return originates.
po_line_id	FK → POLine	Yes	The PO line the goods were ordered on. Inherited from GRN line.
product_id	FK → Product	Yes	The product being returned.
qty_to_return	Decimal	Yes	Quantity being returned on this line. Must not exceed original GRN line qty_accepted (for post-receipt returns).
qty_dispatched	Decimal	No (auto)	Quantity actually dispatched. Confirmed by Warehouse Operator at dispatch. Initially = qty_to_return.
qty_supplier_confirmed	Decimal	No	Quantity supplier confirmed as received. May differ from qty_dispatched (transit loss).
unit_value	Decimal	Yes (auto)	Unit cost of the goods being returned. Inherited from PO line unit_price. Used to calculate return value and credit/debit note amounts.

Field name	Data type	Required	Notes
return_line_value	Decimal	Yes (computed)	qty_to_return × unit_value. Computed field.
return_reason	Lookup	Yes	See Return Reason lookup. Can differ per line.
defect_description	Text (300)	No	Detailed description of defect per line if post-receipt.
line_status	Lookup	Yes (auto)	PENDING / DISPATCHED / SUPPLIER_RECEIVED / RESOLVED / CANCELLED.
bin_id	FK → Bin	No	Bin where defective goods are quarantined pending dispatch.
notes	Text (200)	No	Per-line remarks.

8A.4.3 Credit Note — Field Definitions

Entity: CreditNotes | Schema: finance_schema | EF Core context: FinanceDbContext

Field name	Data type	Required	Notes
credit_note_id	UUID	Yes	Primary key.
credit_note_number	Text (20)	Yes (auto)	CN-YYYY-NNNNN. System-assigned reference.
supplier_credit_note_no	Text (50)	Yes	The credit note reference number issued by the supplier.
sro_id	FK → SRO	Yes	The Supplier Return Order this credit note resolves.
supplier_id	FK → Supplier	Yes	Supplier issuing the credit.
original_invoice_id	FK → Invoice	No	The invoice this credit note offsets. Null if no invoice exists yet (goods returned before invoicing).
credit_date	Date	Yes	Date on the supplier's credit note.
received_date	Date	Yes	Date SMS received the credit note. Default: today.
currency	Lookup	Yes	Credit note currency. Should match original invoice currency.
credit_amount	Decimal	Yes	Total credit value on this note.
tax_amount	Decimal	No	Any tax credit included.
total_credit_amount	Decimal	Yes	credit_amount + tax_amount.
application_status	Lookup	Yes (auto)	PENDING / APPLIED_TO_INVOICE / CARRIED_FORWARD. Set when Finance applies the credit.
applied_to_invoice_id	FK →	No	Invoice to which this credit was

Field name	Data type	Required	Notes
	Invoice		applied. Null if carried forward.
applied_amount	Decimal	No	Amount actually applied (may differ if partial application).
carried_forward_amount	Decimal	No	Balance carried to next payment cycle if invoice already fully paid.
approved_by	FK → User	Yes (auto)	Finance Officer who processed the credit note.
approved_at	DateTime	Yes (auto)	Processing timestamp.
notes	Text (300)	No	Internal remarks.
attachment	File URL	No	Scanned copy of supplier's credit note document.
created_at	DateTime	Yes (auto)	Record creation timestamp.
created_by	FK → User	Yes (auto)	Finance Officer who entered the credit note.

8A.4.4 Debit Note — Field Definitions

Entity: DebitNotes | Schema: finance_schema | EF Core context: FinanceDbContext

Field name	Data type	Required	Notes
debit_note_id	UUID	Yes	Primary key.
debit_note_number	Text (20)	Yes (auto)	DN-YYYY-NNNNN. System-assigned reference.
sro_id	FK → SRO	Yes	The Supplier Return Order this debit note relates to.
supplier_id	FK → Supplier	Yes	Supplier being charged.
original_invoice_id	FK → Invoice	No	The invoice against which this debit is raised, if any.
issue_date	Date	Yes	Date the debit note is issued to the supplier. Default: today.
debit_reason	Lookup	Yes	RETURN_NOT_ACCEPTED / NO_RESPONSE_SLA / DAMAGED_GOODS / WRONG_ITEM_UNRESOLVED / OTHER.
debit_reason_detail	Text (500)	Yes	Mandatory explanation of the debit.
currency	Lookup	Yes	Debit note currency.
debit_amount	Decimal	Yes	Total amount claimed.
tax_amount	Decimal	No	Tax component if applicable.
total_debit_amount	Decimal	Yes	debit_amount + tax_amount.
status	Lookup	Yes (auto)	DRAFT / ISSUED / ACKNOWLEDGED / DISPUTED / SETTLED / WRITTEN_OFF.
supplier_response	Text (500)	No	Supplier's response to the debit note

Field name	Data type	Required	Notes
			(dispute reason or acceptance reference).
settlement_date	Date	No	Date debit was settled (deducted from future payment or paid separately).
approved_by	FK → User	Yes (auto)	Finance Officer who approved the debit note.
approved_at	DateTime	Yes (auto)	Approval timestamp.
notes	Text (300)	No	Internal remarks.
attachment	File URL	No	Supporting evidence document (return confirmation, photos).
created_at	DateTime	Yes (auto)	Creation timestamp.
created_by	FK → User	Yes (auto)	Finance Officer who created the debit note.

8A.5 Lookup Collections

8A.5.1 SRO Type (CRUD: No — system controlled)

Code	Display value	Notes
GRN_REJECTION	GRN Rejection	Goods rejected at receiving during GRN entry. Auto-triggered.
POST_RECEIPT_DEFECT	Post-Receipt Defect	Defect found after GRN was approved and stock posted to inventory.
WRONG_ITEM	Wrong Item Delivered	Supplier delivered a different product than ordered.
OVERDELIVERY	Overdelivery	Supplier delivered more quantity than ordered, excess to be returned.

8A.5.2 Return Reason (CRUD: Yes — Admin + Procurement Manager)

Code	Display value	Notes
DAMAGED	Damaged in transit	Goods physically damaged before or during delivery.
DEFECTIVE	Manufacturing defect	Product defect unrelated to transit — factory or production fault.
WRONG_ITEM	Wrong item delivered	Different SKU or product delivered than ordered.
WRONG_QTY	Wrong quantity	Different quantity delivered than ordered (over or under — return of excess).
SHORT_EXPIRY	Short / expired shelf life	Expiry date does not meet minimum shelf life requirement.

Code	Display value	Notes
SPEC_MISMATCH	Specification mismatch	Product does not match technical specification on the PO.
DUPLICATE	Duplicate delivery	Goods already received on a previous GRN — duplicate delivery.
QUALITY_FAIL	Quality test failure	Failed internal or third-party quality testing post-receipt.
OTHER	Other	Other reason — detail mandatory in return_reason_detail.

8A.5.3 Debit Note Reason (CRUD: Yes — Admin + Finance Officer)

Code	Display value
RETURN_NOT_ACCEPTED	Supplier refused return without valid reason
NO_RESPONSE_SLA	No response from supplier within SLA period
DAMAGED_GOODS	Supplier responsible for goods arriving damaged
WRONG_ITEM_UNRESOLVED	Wrong item delivered — no replacement or credit provided
QUALITY_CLAIM	Formal quality claim against supplier
OTHER	Other — detail mandatory

8A.6 Inventory and PO Impact

Returns affect inventory levels, PO line quantities, and invoice balances. The impacts are applied at specific points in the SRO lifecycle and must be atomic with the triggering action.

8A.6.1 Inventory Impact by SRO Stage

SRO stage	Inventory action	Detail
SRO DRAFT created (post-receipt)	StockAdjustment created: type = RETURN_PENDING, qty = −qty_to_return	Quarantines defective stock. InventoryItem.QtyOnHand is NOT decremented yet — the adjustment is in pending state. The quantity is effectively reserved/locked from use.
SRO REJECTED (post-receipt)	RETURN_PENDING StockAdjustment cancelled	Defective quantity released back to available stock. QtyOnHand unchanged.
SRO DISPATCHED (all scenarios)	StockAdjustment finalised: type = RETURNED, qty = −qty_dispatched	InventoryItem.QtyOnHand decremented by qty_dispatched. For GRN-rejection returns, this is the first inventory impact (no prior reservation). The goods are now physically gone from the warehouse.
Replacement delivery received	New GRN approved against original PO	InventoryItem.QtyOnHand incremented by replacement qty_accepted via the normal GRN approval stock posting. SRO.status set to RESOLVED_REPLACEMENT.

8A.6.2 PO Line Impact

Key rule	When goods are dispatched back to a supplier, the POLine.QtyReceived is decremented by the dispatched quantity. This may reopen a PO line that was previously Fully Received. The PO header status is recalculated automatically via RecalculatePoStatus() — the same engine defined in Addendum 5A Section 3.3.

Scenario	POLine and PO header impact
Goods dispatched back, credit note received	POLine.QtyReceived -= qty_dispatched. POLine.LineReceiptStatus recalculated. If now QtyReceived < QtyOrdered, status reverts to PARTIALLY_RECEIVED. PO header status recalculated. Invoice.MatchedGRNValue reduced. Credit note applied to invoice balance.
Goods dispatched back, replacement received	POLine.QtyReceived first decremented on dispatch. Then re-incremented when replacement GRN approved. Net effect: QtyReceived unchanged if full replacement. PO may temporarily show PARTIALLY_RECEIVED during the gap.
Goods dispatched back, debit note raised	POLine.QtyReceived -= qty_dispatched. Same as credit note scenario for inventory and PO. The financial recovery is via debit note rather than supplier credit.
GRN rejection return (goods never entered inventory)	POLine.QtyReceived was already reduced by the qty_rejected at GRN approval (qty_received only counts qty_accepted). POLine.QtyPending increases. PO status stays PARTIALLY_RECEIVED until: replacement delivery, short-close, or PO amendment cancels the line.

8A.6.3 Invoice and Payment Impact

Resolution outcome	Invoice and payment impact
Credit note — invoice not yet paid	Credit note linked to Invoice. Invoice.MatchedGRNValue reduced. 3-way match re-run. If credit brings invoice into tolerance, MatchStatus may upgrade to MATCHED. Payment can be scheduled for the net amount (Invoice.TotalAmount − CreditNote.TotalCreditAmount).
Credit note — invoice already paid	CreditNote.ApplicationStatus = CARRIED_FORWARD. CreditNote.CarriedForwardAmount recorded. Finance Officer allocates the carried-forward credit against the next invoice from this supplier. Supplier ledger balance updated.
Replacement delivery	No direct invoice impact until new GRN is approved and supplier issues a new invoice for the replacement. The original invoice stands for the originally received (non-returned) quantity.
Debit note raised	DebitNote created against supplier. Finance Officer deducts debit note value from next payment to supplier, or raises a formal claim. DebitNote.Status transitions through ISSUED → ACKNOWLEDGED / DISPUTED → SETTLED / WRITTEN_OFF.
Return before any invoice	If goods are returned before supplier has invoiced (SRO raised immediately on GRN rejection), no invoice adjustment is needed. Finance is notified not to accept or process an invoice for the returned quantity.

8A.7 Business Rules

8A.7.1 SRO Creation Rules

- An SRO must always reference an original approved GRN. Returns cannot be processed without a GRN — there must be a receipt record to return against.
- **For GRN rejection returns, the SRO qty_to_return must equal the GRN line qty_rejected. It cannot be greater.**
- **For post-receipt defect returns, qty_to_return must not exceed the original GRN line qty_accepted.**
- **An SRO cannot be raised against a GRN that belongs to a CANCELLED PO.**
- **Multiple SROs can exist against the same GRN if multiple defects are discovered at different times.**
- **A single SRO can cover multiple lines from the same GRN (e.g. two different products rejected in the same delivery).**
- **The SRO return_reason_detail field is mandatory regardless of whether a lookup reason is selected — every return must have a written explanation.**

8A.7.2 Approval and Dispatch Rules

- All SROs require Procurement Manager approval before goods can be dispatched. Warehouse Operators cannot self-approve a return.
- **Goods cannot be dispatched to the supplier before SRO approval. Dispatch on a DRAFT SRO returns a 422 validation error.**
- **The Procurement Manager should obtain a supplier RMA (Return Merchandise Authorisation) number before approving where the supplier requires one. The RMA number is recorded on the SRO and accompanies the return shipment.**
- **Once dispatched, an SRO cannot be reversed. A new SRO must be created if additional goods need to be returned for the same reason.**
- **If goods are physically dispatched but the SRO has not been approved (emergency situation), Procurement Manager must retrospectively approve within 24 hours. The system flags these as out-of-sequence dispatches in the audit trail.**

8A.7.3 Resolution Rules

- Every dispatched SRO must be resolved — it cannot remain permanently in DISPATCHED or SUPPLIER_RECEIVED status. The system enforces a resolution SLA (default 14 days from dispatch date). Overdue SROs trigger escalation notifications.
- **A credit note must reference a valid supplier credit note number. The Finance Officer cannot create a credit note with only an internal reference — the supplier's document reference is mandatory.**
- **A credit note amount must not exceed the SRO's total return value (qty_to_return × unit_value). If the supplier issues a credit for a different amount, the difference must be explained in the credit note notes field.**
- **A debit note can only be raised by a Finance Officer. Procurement Manager or Warehouse Operators cannot raise debit notes.**

- **When a debit note is raised, a copy is sent to the supplier via the Notification Service (Hangfire job). The supplier contact email on the Supplier Master is used.**
- **A replacement delivery is tracked by linking the original SRO to the new GRN via SRO.resolution_type = REPLACEMENT. When the new GRN is approved, the system auto-updates SRO.status = RESOLVED_REPLACEMENT.**
- **Supplier performance rating is updated after SRO resolution: each resolved return deducts from the supplier's quality score. The deduction weight depends on return reason (DAMAGED and DEFECTIVE have higher weights than WRONG_QTY).**

8A.7.4 Notification Templates

Trigger event	Recipient	Subject line
SRO created (GRN rejection — auto)	Procurement Manager	Action required: Supplier return SRO-{number} pending approval
SRO created (post-receipt — manual)	Procurement Manager	Action required: Supplier return SRO-{number} pending approval
SRO approved	Warehouse Operator	Prepare return dispatch: SRO-{number} approved
SRO approved	Supplier contact	Return authorised: Please expect return shipment for PO {po_number}
SRO dispatched	Procurement Officer + Finance Officer	Return dispatched: SRO-{number} sent to supplier
Supplier received confirmed	Finance Officer	Action required: Resolve SRO-{number} — credit note, replacement, or debit note
SLA breach (14 days from dispatch)	Procurement Manager + Finance Officer	Overdue: SRO-{number} unresolved for {days} days
Credit note created	Finance Officer (confirmation)	Credit note CN-{number} recorded for SRO-{number}
Debit note issued	Supplier contact	Debit note DN-{number} raised against your account
Replacement GRN approved	Procurement Officer + Finance	Replacement delivery received: SRO-{number} resolved

8A.7.5 Role-Based Access — Returns

Screen / action	Admin	Proc. Mgr	Purch. Off.	Inv. Mgr	Fin. Off.	W/H Op
View SRO list	Full	Full	Full	Full	Full	View
Create SRO (manual)	Full	Full	None	Full	None	None
Approve / reject SRO	Full	Full	None	None	None	None
Record dispatch	Full	None	None	None	None	Full
Confirm supplier receipt	Full	None	Full	None	None	None
Create credit note	Full	None	None	None	Full	None
Create debit note	Full	None	None	None	Full	None
Approve debit note	Full	Full	None	None	Full	None

Screen / action	Admin	Proc. Mgr	Purch. Off.	Inv. Mgr	Fin. Off.	W/H Op
Apply credit to invoice	Full	None	None	None	Full	None
View credit / debit notes	Full	Full	None	None	Full	None
Export SRO report	Full	Full	Full	None	Full	None

— End of Addendum —

9 Module — Logistics & Shipping

Manages outbound dispatches, carrier assignments, shipment tracking, delivery confirmation, and returns processing.

9.1 Shipment — Field Definitions

Field name	Data type	Required	CRUD	Lookup / values	Notes
shipment_id	Auto (UUID)	Yes	N/A	System generated	PK
shipment_number	Text (20)	Yes	Auto	SHP-YYYY-NNNNN	Reference
order_id	FK → Order	Yes	Lookup	Confirmed orders	Source order
carrier_id	FK → Carrier	Yes	Lookup	Active carriers	Logistics company
shipment_type	Lookup	Yes	Yes	Courier / LTL / FTL / Air / Sea	Transport mode
dispatch_date	Date	Yes	Date picker	—	Actual dispatch date
estimated_arrival	Date	Yes	Date picker	—	ETA at destination
actual_arrival	Date	No	Date picker	—	Actual delivery date
tracking_number	Text (100)	No	Free text	—	Carrier tracking ref
tracking_url	URL (300)	No	Free text	—	Carrier tracking link
origin_warehouse	FK → Warehouse	Yes	Lookup	Active warehouses	Dispatch from
destination_address	Text (300)	Yes	Free text	—	Deliver to
weight_kg	Decimal	No	Number	—	Total

Field name	Data type	Required	CRUD	Lookup / values	Notes
					shipment weight
volume_cbm	Decimal	No	Number	—	Cubic meters
freight_cost	Decimal	No	Number	—	Shipping charge
status	Lookup	Yes	Auto	Preparing/Dispatched/In Transit/Delivered/Returned	Status
proof_of_delivery	File	No	Upload	PDF/Image	Signed delivery note
notes	Text (300)	No	Free text	—	—
created_at	DateTime	Yes	Auto	—	—
created_by	FK → User	Yes	Auto	—	—

9.2 Carrier Master — Field Definitions

Field name	Data type	Required	CRUD	Lookup / values	Notes
carrier_id	Auto (UUID)	Yes	N/A	System generated	PK
carrier_name	Text (100)	Yes	Free text	—	Logistics company name
carrier_code	Text (20)	Yes	Manual	DHL / TCS / LEOPARD	Short code
service_type	Lookup	Yes	Yes	Express / Economy / Overnight / Freight	Service level
tracking_url_template	URL (300)	No	Free text	https://track.dhl.com/{{no}}	Parameterised URL
contact_name	Text (100)	No	Free text	—	Account manager
contact_phone	Text (20)	No	Free text	—	—
contact_email	Email (100)	No	Free text	—	—
rate_per_kg	Decimal	No	Number	—	Pricing reference
status	Lookup	Yes	Yes	Active / Inactive	Carrier status

9.3 Role-Based Access — Logistics & Shipping

Screen / action	Admin	Proc. Mgr	Purchase Off.	Inv. Mgr	Requester
Create shipment	**Full**	**None**	**Full**	**Full**	**None**
Update tracking info	**Full**	**None**	**Full**	**Full**	**None**
Upload proof of delivery	**Full**	**None**	**Full**	**Full**	**None**
View shipment list	**Full**	**Full**	**Full**	**Full**	**View own**
Manage carriers	**Full**	**Full**	**None**	**None**	**None**
Process return	**Full**	**None**	**Full**	**Full**	**None**

10 Module — Finance & Payments

Manages the financial settlement layer of the supply chain: supplier invoice processing, three-way matching, payment scheduling, accounts payable, and budget monitoring.

10.1 Invoice — Field Definitions

Field name	Data type	Required	CRUD	Lookup / values	Notes
invoice_id	Auto (UUID)	Yes	N/A	System generated	PK
invoice_number	Text (20)	Yes	Auto	INV-YYYY-NNNNN	System number
supplier_invoice_no	Text (50)	Yes	Free text	—	Supplier's own ref
supplier_id	FK → Supplier	Yes	Lookup	Active suppliers	Vendor
po_id	FK → PO	Yes	Lookup	Approved POs	Matched PO
grn_id	FK → GRN	Yes	Lookup	Accepted GRNs	Matched GRN
invoice_date	Date	Yes	Date picker	—	Date on invoice
received_date	Date	Yes	Date picker	Default: today	When SMS received it
due_date	Date	Yes	Calculated	received + payment terms	Payment deadline
currency	Looku	Yes	Yes	PKR / USD / AED	Invoice

Field name	Data type	Required	CRUD	Lookup / values	Notes
	p				currency
subtotal	Decimal	Yes	Number	—	Pre-tax amount
tax_amount	Decimal	No	Number	—	Tax charged
total_amount	Decimal	Yes	Number	—	Total invoice value
matched_po_value	Decimal	No	Auto	Sum of matched PO lines	For 3WM
matched_grn_value	Decimal	No	Auto	Sum of accepted GRN lines	For 3WM
variance_amount	Decimal	No	Calculated	invoice - PO/GRN	3WM difference
match_status	Lookup	Yes	Auto	Pending/Matched/Variance/ Approved/Rejected	3WM result
payment_status	Lookup	Yes	Auto	Unpaid / Scheduled / Partial / Paid / Overdue	Payment state
payment_method	Lookup	No	Yes	Bank Transfer / Cheque / Online / Cash	How paid
bank_account_id	FK → BankAcc	No	Lookup	Supplier bank accounts	Which account to pay
approved_by	FK → User	No	Auto	Finance Officer / Admin	—
approved_at	DateTime	No	Auto	—	—
notes	Text (300)	No	Free text	—	—
attachment	File	No	Upload	PDF scan of invoice	Original invoice
created_at	DateTime	Yes	Auto	—	—
created_by	FK → User	Yes	Auto	—	—

10.2 Payment — Field Definitions

Field name	Data type	Required	CRUD	Lookup / values	Notes
payment_id	Auto (UUID)	Yes	N/A	System generated	PK

Field name	Data type	Required	CRUD	Lookup / values	Notes
payment_number	Text (20)	Yes	Auto	PAY-YYYY-NNNNN	Reference
invoice_id	FK → Invoice	Yes	Lookup	Approved invoices	Which invoice
supplier_id	FK → Supplier	Yes	Auto	Inherited from invoice	—
payment_date	Date	Yes	Date picker	—	Date payment made
amount_paid	Decimal	Yes	Number	—	This payment amount
payment_method	Lookup	Yes	Yes	Bank Transfer / Cheque / Online / Cash	—
bank_reference	Text (100)	No	Free text	—	Bank transaction ref
cheque_number	Text (30)	No	Free text	—	If cheque payment
account_debited	FK → GL	No	Lookup	Chart of accounts	Company bank account
status	Lookup	Yes	Auto	Pending / Processed / Cleared / Reversed	—
notes	Text (200)	No	Free text	—	—
processed_by	FK → User	Yes	Auto	—	—
processed_at	DateTime	Yes	Auto	—	—

10.3 Lookup Collections

Lookup: Invoice Match Status (CRUD: No — system controlled)

Code / ID	Display value	Active	Notes
PEND	Pending Match	N/A	Awaiting 3-way match check
MATCH	Fully Matched	N/A	PO, GRN and invoice align
VAR	Variance Detected	N/A	Amounts do not reconcile
APPR	Approved for Payment	N/A	Variance accepted or cleared
REJECT	Rejected	N/A	Invoice sent back to supplier

Lookup: Payment Method (CRUD: Yes)

Code / ID	Display value	Active	Notes
TT	Bank Transfer / TT	Yes	Telegraphic transfer
CHQ	Cheque	Yes	Physical cheque
ONLINE	Online / IBFT	Yes	Internet banking
CASH	Cash	Yes	Physical cash payment
DD	Demand Draft	Yes	Bank demand draft

10.4 Role-Based Access — Finance & Payments

Screen / action	Admin	Proc. Mgr	Purchase Off.	Inv. Mgr	Requester
Create / enter invoice	**Full**	**None**	**None**	**None**	**None**
Run 3-way match	**Full**	**None**	**None**	**None**	**None**
Approve invoice	**Full**	**Full**	**None**	**None**	**None**
Schedule payment	**Full**	**None**	**None**	**None**	**None**
Process payment	**Full**	**None**	**None**	**None**	**None**
View invoice list	**Full**	**Full**	**Full**	**View**	**None**
View payment records	**Full**	**Full**	**None**	**None**	**None**
Export finance reports	**Full**	**Full**	**None**	**None**	**None**

11 Module — Reports & Analytics

Provides operational dashboards, standard management reports, ad-hoc query tools, and a full audit trail across all modules.

11.1 Standard Reports

Report name	Module	Key metrics	Access
Supplier Performance	Supplier Mgmt	On-time %, quality score, PO count	Admin, Proc Mgr
Purchase Order Summary	Procurement	PO count, value, status breakdown	Admin, Proc Mgr, PO
Spend by Supplier	Procurement	Total spend per vendor by period	Admin, Proc Mgr
Pending Approvals	All Modules	PRs, POs, GRNs awaiting action	Admin, Proc Mgr
Stock Level Report	Inventory	On-hand, reserved, available, value	Admin, Inv Mgr
Reorder Alert Report	Inventory	Items at or below reorder	Admin, Inv Mgr, PO

Report name	Module	Key metrics	Access
		point	
Inventory Valuation	Inventory	Stock value by category/warehouse	Admin, Inv Mgr
GRN vs PO Variance	Warehouse	Qty ordered vs received differences	Admin, Proc Mgr
Shipment Tracker	Logistics	In-transit orders, ETAs, overdue	Admin, PO, WH Op
Invoice Aging	Finance	Outstanding invoices by due date bucket	Admin, Finance
Payment Summary	Finance	Payments made, pending, overdue	Admin, Finance
Budget Utilisation	Finance	Budget vs actual spend by dept	Admin, Finance
Audit Trail Log	All Modules	All create/update/delete actions	Admin only
User Activity Report	System	Login history, actions per user	Admin only

11.2 KPI Dashboard — Key Metrics

- PO Cycle Time (days from PR to PO approval)
- Supplier On-Time Delivery Rate (%)
- Purchase Order Fill Rate (%)
- Stock Turnover Ratio
- Inventory Accuracy (% from cycle counts)
- Invoice Processing Time (days)
- 3-Way Match Rate (% invoices auto-matched)
- Budget Variance (% over/under spend)
- GRN Rejection Rate (% by supplier)
- Reorder Trigger Frequency (items hitting reorder point)

11.3 Audit Trail — Field Definitions

Field name	Data type	Required	CRUD	Lookup / values	Notes
audit_id	Auto (UUID)	Yes	N/A	System generated	PK
timestamp	DateTim e	Yes	Auto	System generated	When it happene d
user_id	FK →	Yes	Auto	Logged-in user	Who did

Field name	Data type	Required	CRUD	Lookup / values	Notes
	User				it
module	Lookup	Yes	Auto	Supplier/Procurement/Inventory…	Which module
action	Lookup	Yes	Auto	CREATE/UPDATE/DELETE/ APPROVE/LOGIN	Action type
entity_type	Text (50)	Yes	Auto	PO / GRN / Invoice…	Which entity
entity_id	UUID	Yes	Auto	Record ID	Which record
field_changed	Text (100)	No	Auto	e.g. status	For updates
old_value	Text (500)	No	Auto	Previous value	Before change
new_value	Text (500)	No	Auto	New value	After change
ip_address	Text (45)	Yes	Auto	IPv4 or IPv6	User's IP
notes	Text (300)	No	Auto	System or user note	—

12 Purchase Order Approval Workflow

The PO Approval Workflow enforces a configurable multi-level approval chain before any Purchase Order can be transmitted to a supplier. The workflow is value-based: lower-value POs require fewer approval levels, while high-value or emergency POs require additional sign-off. No PO may be sent to a supplier without completing its full approval chain.

12.1 Approval Tiers — Value Thresholds

Tier	PO Value (PKR)	Approver Level 1	Approver Level 2	Approver Level 3
1	Up to 50,000	Purchase Officer	—	—
2	50,001 – 250,000	Purchase Officer	Procurement Manager	—
3	250,001 – 1,000,000	Purchase Officer	Procurement Manager	Finance Officer
4	Above 1,000,000	Purchase Officer	Procurement Manager	System Admin / CEO
E	Emergency PO (any value)	Procurement Manager	System Admin	—

Threshold values are configurable by System Admin. Currency thresholds auto-convert using the system exchange rate for foreign-currency POs.

12.2 PO Approval — Step-by-Step Workflow

Step	Actor	Action	System Response	Status Change
1	**Purchase Officer**	Creates PO, fills all fields, attaches documents	System validates mandatory fields, calculates totals, checks budget	Draft
2	**Purchase Officer**	Clicks 'Submit for Approval'	System determines tier, assigns L1 approver, sends email/in-app notification	Pending L1
3	**L1 Approver**	Reviews PO, line items, pricing, supplier validity	System displays PO detail with approve / reject / request revision buttons	Pending L1
4a	**L1 Approver**	Approves PO	System records timestamp + user, advances to L2 (if required) or moves to Approved	L2 Pending or Approved
4b	**L1 Approver**	Rejects PO with reason	System notifies creator, PO status set to Rejected, reason logged	Rejected
4c	**L1 Approver**	Requests Revision	System notifies creator with comment, PO returned to Draft	Draft (Revision)
5	**Creator**	Revises PO and re-submits	Workflow restarts from Step 2	Pending L1
6	**L2 Approver**	Reviews and approves (if tier requires)	System records approval, advances to L3 or Approved	L3 Pending or Approved
7	**L3 Approver**	Final approval (if tier requires)	System sets status Approved, notifies creator and procurement team	Approved
8	**Purchase Officer**	Sends approved PO to supplier	System records sent timestamp, generates PDF, emails supplier	Sent to Supplier
9	**Supplier**	Acknowledges PO (manual entry or portal)	System records acknowledgement date and reference	Acknowledged

12.3 PO Approval — Business Rules

- A PO cannot be edited once submitted for approval. The approver must first 'Request Revision' to unlock it.
- If an approver does not act within the configured SLA (default: 48 hours), an escalation notification is sent to the approver's manager.
- If no action is taken within 72 hours after escalation, the System Admin receives a critical alert.
- The creator of a PO cannot approve their own PO at any level.
- Procurement Manager can self-approve Tier 1 POs they create only if the System Admin has explicitly granted this exception.
- A rejected PO can be revised and resubmitted up to 3 times. On the 4th rejection, the PO is automatically cancelled.
- PO amendments (changes after approval) restart the full approval workflow.
- Emergency POs bypass standard creation workflow but still require 2-level approval before being sent.
- All approval actions are permanently recorded in the Audit Trail with user ID, timestamp, IP address, and comments.
- Approval notification channels: in-app notification, email alert, and optional SMS (configurable per user).

12.4 PO Approval Workflow — Field Definitions

Field name	Data type	Required	CRUD	Lookup / values	Notes
approval_id	Auto (UUID)	Yes	N/A	System generated	PK
po_id	FK → PO	Yes	Auto	Parent PO	Which PO
approval_level	Integer	Yes	Auto	1, 2, 3	Step number
approver_id	FK → User	Yes	Auto	System assigned by tier rule	Required approver
delegate_id	FK → User	No	Lookup	Active users	Delegated approver
status	Lookup	Yes	Auto	Pending / Approved / Rejected / Escalated / Delegated	Step status
action_taken_at	DateTime	No	Auto	Set on action	When decision made
sla_deadline	DateTime	Yes	Auto	Submitted + SLA hours	Escalation trigger
decision_notes	Text (500)	No	Free text	—	Approver's

Field name	Data type	Required	CRUD	Lookup / values	Notes
					comments
escalated_to	FK → User	No	Auto	Manager of approver	Set on escalation
escalated_at	DateTime	No	Auto	—	Escalation timestamp
created_at	DateTime	Yes	Auto	—	Step creation time

12.5 PO Approval — Notification Templates

Trigger event	Recipient	Subject line	Channel
PO submitted	L1 Approver	Action required: PO {po_number} awaiting your approval	Email + In-app
L1 approved (Tier 2+)	L2 Approver	Action required: PO {po_number} requires your approval	Email + In-app
PO fully approved	PO Creator	PO {po_number} has been approved	Email + In-app
PO rejected	PO Creator	PO {po_number} was rejected — action required	Email + In-app
Revision requested	PO Creator	PO {po_number} returned for revision	Email + In-app
SLA breached	Approver + Manager	Overdue: PO {po_number} awaiting approval for 48h	Email
PO sent to supplier	Supplier contact	Purchase Order {po_number} from {company}	Email (PDF attached)
Supplier acknowledgement	PO Creator	Supplier confirmed PO {po_number}	In-app

13 Goods Receipt Note (GRN) Approval Workflow

The GRN Approval Workflow governs the acceptance of physical goods received from suppliers. It enforces quality inspection, quantity verification, and financial sign-off before stock is committed to inventory. A GRN that is not fully approved does not update inventory quantities or trigger invoice matching.

13.1 GRN Approval — Trigger Conditions

Condition	Approval required from	Reason
All GRNs (standard)	Inventory Manager	Mandatory minimum for all receipts
Partial receipt (qty < PO qty)	Inventory Manager + Proc. Manager	Shortfall must be acknowledged and PO kept open
Quantity variance > 5%	Inventory Manager + Proc. Manager	Discrepancy requires procurement review
QC failure on any line	Inventory Manager + QC Officer	Rejected goods must be formally documented
GRN value > 500,000 PKR	Inventory Manager + Finance Officer	High-value receipts need financial sign-off
First receipt from new supplier	Inventory Manager + Proc. Manager	New vendor quality gate
Cold-chain / perishable goods	Inventory Manager + QC Officer	Temperature and condition check required
Asset / capital goods	Inventory Manager + Finance Officer	Fixed asset registration trigger

13.2 GRN Approval — Step-by-Step Workflow

Step	Actor	Action	System Response	GRN Status
1	**Warehouse Operator**	Physically receives delivery, checks supplier delivery note	—	—
2	**Warehouse Operator**	Creates GRN against the PO, enters received quantities per line	System pre-fills from PO. Flags lines where qty received differs from qty ordered.	Draft
3	**Warehouse Operator**	Records QC result per line (Accepted / Rejected / Partially accepted) with reason	System calculates accepted qty, rejected qty, and variance	Draft
4	**Warehouse Operator**	Attaches delivery note photo/scan and submits GRN	System checks if additional approvers required (see 13.1), sends notifications	Pending QC / Pending Approval
5	**QC Officer (if required)**	Reviews QC findings, may physically re-inspect goods	System shows GRN detail with line-level QC data	Pending QC
6a	**QC Officer**	Confirms or overrides QC result per line	System updates accepted/rejected quantities	Pending Approval
6b	**QC Officer**	Rejects entire GRN (all goods refused)	System notifies Warehouse,	Rejected

Step	Actor	Action	System Response	GRN Status
			Procurement Manager, and supplier. GRN status set to Rejected.	
7	**Inventory Manager**	Reviews GRN — quantities, QC results, supplier note	System shows variance report vs PO	Pending Approval
8a	**Inventory Manager**	Approves GRN	System commits accepted quantities to inventory. Updates PO received qty. Triggers 3-way match check.	Approved
8b	**Inventory Manager**	Rejects GRN with reason	Goods not accepted. Return process initiated. PO remains open.	Rejected
9	**Finance Officer (if triggered)**	Reviews and approves high-value or asset receipt	System completes financial posting	Approved (Finance)
10	**System**	Stock updated, PO status recalculated, invoice match triggered	Inventory levels updated. PO becomes Partial or Received. Finance module notified.	Closed / Partial

13.3 GRN Approval — Business Rules

- A GRN must always reference an approved, sent Purchase Order. GRNs cannot be created against cancelled or draft POs.
- Received quantity on any line cannot exceed ordered quantity by more than the configured over-receipt tolerance (default: 3%).
- Rejected goods must be assigned a rejection reason from the lookup list. A mandatory Supplier Return Order is auto-created for all rejected lines.
- Partial GRNs are allowed. The PO remains open with status 'Partially Received' until all lines are fully received or the PO is manually closed.
- Once a GRN is approved, its quantities are locked. Corrections require a Stock Adjustment with a second-level approval.
- A GRN cannot be deleted after submission. It can only be rejected. The rejection and reason are permanently logged.
- QC inspection is mandatory for: food/pharma products, items with shelf life, any supplier with a QC failure in the past 6 months.
- The system automatically flags a 'first delivery from new supplier' to prompt extra scrutiny.

- GRN approval SLA: Inventory Manager must act within 24 hours of submission. After 24 hours, an escalation alert is sent.
- All approved GRN quantities feed directly into the 3-way matching engine in the Finance module.
- Goods with expiry dates must have expiry date entered before GRN can be submitted.

13.4 GRN Approval — Status State Machine

From status	Action / trigger	To status	Who can trigger
—	Warehouse Operator creates GRN	**Draft**	Warehouse Operator
Draft	Submit for approval	**Pending QC**	Warehouse Operator
Draft	Submit (no QC required)	**Pending Approval**	Warehouse Operator
Pending QC	QC Officer confirms	**Pending Approval**	QC Officer
Pending QC	QC Officer rejects all	**Rejected**	QC Officer
Pending Approval	Inventory Mgr approves	**Approved**	Inventory Manager
Pending Approval	Inventory Mgr rejects	**Rejected**	Inventory Manager
Approved	High-value flag — Finance review	**Pending Finance**	System (auto)
Pending Finance	Finance Officer approves	**Approved**	Finance Officer
Approved	System posts to inventory	**Closed**	System (auto)
Rejected	Return Order created	**Return in Progress**	System (auto)

13.5 GRN Approval — Notification Templates

Trigger event	Recipient	Subject line	Channel
GRN submitted	Inventory Manager	Action required: GRN {grn_number} pending your approval	Email + In-app
QC check required	QC Officer	QC inspection needed: GRN {grn_number}	Email + In-app
GRN approved	Warehouse Operator + Finance	GRN {grn_number} approved — stock updated	In-app
GRN rejected	Warehouse Operator + Proc. Mgr	GRN {grn_number} rejected: {reason}	Email + In-app
Partial receipt detected	Procurement Manager	Partial receipt: GRN {grn_number}, PO {po_number} remains open	Email + In-app
SLA breached (24h)	Inventory Manager + Admin	Overdue GRN approval: {grn_number}	Email

Trigger event	Recipient	Subject line	Channel
Return order created	Supplier + Proc. Mgr	Return initiated for GRN {grn_number}	Email
High-value GRN	Finance Officer	Finance approval required: GRN {grn_number} value {amount}	Email + In-app

Sections 13.6 through 13.10 below add GRN intake and inspection corrections and new capabilities raised against the Add New GRN screen.

13.6 Group D — GRN: Search Purchase Order Field (Addendum 19)

#7	**NEW FEATURE**	**Smart-filtered PO search on Add New GRN with free-text fallback**

13.6.1 Required Behaviour

Default dropdown content	On focus/click (before typing), the dropdown pre-populates with POs matching: status IN ('SENT','PART_RCV') — i.e. awarded/sent to supplier AND not yet fully received, per Addendum 5A PO status lookup. This is exactly the set of POs a Warehouse Operator would realistically be receiving against.
Sort order	Most recently sent POs first (sent_to_supplier_at DESC), so the most active/likely candidates surface without typing
Type-ahead search	As the user types, the list filters by po_number, supplier name, or partial match — not restricted to only the pre-filtered status set; the user can type any PO number (any status) and it will appear if it matches, satisfying 'other than dropdown user can type any other purchase order also'
Selecting an out-of-scope PO	If a user types and selects a PO that is NOT in (SENT, PART_RCV) — e.g. a DRAFT, CLOSED, or CANCELLED PO — the system allows the selection but immediately shows an inline warning banner: 'This PO is in {status} status and cannot currently receive a GRN.' The GRN cannot be saved in this state (existing validation already blocks GRN creation against non-receivable PO statuses per Addendum 5A Section 4.1).
Display format per option	{po_number} — {supplier_name} — {grand_total} {currency} — {qty_pending} units pending

13.6.2 API Endpoint

GET /api/purchase-orders/search?status=receivable&q={query} — the existing PO list endpoint extended with a status=receivable shorthand that the GRN screen uses for the default pre-filtered view, and the existing free-text q parameter handles the type-ahead-any-PO behaviour.

13.7 Group D — GRN: Receiving Date and Time (Addendum 19)

#8	**DEFECT**	**GRN only captures receiving date, not time**

13.7.1 Required Behaviour

Field change	GRNs.received_date (FSD v3 Section 8.3, currently Date type) is changed to received_at (DateTime type) — capturing both date and time of physical receipt
Default value	Defaults to the current date/time when the Warehouse Operator opens the Add New GRN screen, editable if the physical receipt happened earlier (e.g. recorded the next morning for a late-evening delivery)
Display format	Shown as 'DD-MMM-YYYY, HH:mm' throughout the UI (GRN list, GRN detail, reports) rather than date-only
Downstream impact	Any report or screen currently sorting/filtering by GRN received_date (e.g. GRN vs PO Variance report, FSD v3 Section 11.1) continues to function — date-only displays simply truncate the time portion where appropriate (e.g. a daily summary report).

13.8 Group D — GRN: Optional Inspection — Per-Receipt Decision (Addendum 19)

#9	**NEW FEATURE**	**Make inspection optional, decided per-GRN at the point of receiving**

#10	**NEW FEATURE**	**Radio button on GRN screen to flag inspection requirement**

13.8.1 Design Decision

Confirmed scope	Whether inspection is required is decided per-GRN, at the moment of receiving, by the Warehouse Operator — not as a global system setting and not tied to product category. The same product received twice on two different GRNs could be inspected once and not the other, reflecting real-world variability (e.g. a trusted repeat delivery from a long-standing supplier vs a first-time or post-complaint delivery).

13.8.2 Required Behaviour

New field	GRNs.requires_inspection (Boolean, NOT NULL, default true) — replaces the previous implicit assumption that all GRNs are inspected
UI control	A radio button group on the Add New GRN screen: 'Inspection Required for this Receipt?' — options Yes / No. Defaults to Yes.
Placement	Positioned directly above the item list, so the Warehouse Operator decides before entering item-level data, and the form below adapts based on the choice (Section 11)
Effect of 'No'	If No is selected, the Inspection Results section (Section 11) is hidden/not required. GRN lines still capture qty_received and qty_accepted (qty_accepted = qty_received automatically when no inspection occurs, qty_rejected = 0 by default but remains editable in case of an obvious visual defect noted without formal inspection).
Effect of 'Yes'	Inspection Results section (Section 11) becomes mandatory before the GRN

	can proceed to final approval (Section 12).
Changing the choice	The Yes/No choice can be changed any time before the GRN is submitted for approval. Once submitted, it locks — consistent with other FSD v3 submitted-document immutability patterns (e.g. PO cannot be edited once submitted, Section 12.3).

13.9 Group D — GRN: Inspection Results Section (Addendum 19)

#11	**NEW FEATURE**	**Dedicated inspection-results section for all items against the PO**

13.9.1 Required Behaviour

When requires_inspection = Yes (Section 10), a new section appears on the GRN screen, below the basic receiving information and above the final submit action, listing every line item on the GRN with inspection-specific fields.

13.9.2 Inspection Results Grid — Per-Line Fields

Field	Type	Required	Notes
Item / Product	Read-only display	—	From GRN line, same item shown in the basic receiving grid
Qty Received	Read-only display	—	Carried from the basic receiving entry, shown for reference
Inspection Result	Dropdown: Pass / Fail / Partial Pass	Yes (if inspection required)	Maps to existing GRNLines.qc_result-equivalent. 'Partial Pass' triggers the Qty Accepted / Qty Rejected split below.
Qty Accepted	Number	Yes (if Pass or Partial Pass)	Defaults to Qty Received when result = Pass
Qty Rejected	Number	Yes (if Fail or Partial Pass)	Defaults to Qty Received when result = Fail; must equal Qty Received minus Qty Accepted when Partial Pass
Rejection Reason	Dropdown (existing lookup, FSD v3 Section 8.4 rejection_reason)	Yes (if Qty Rejected > 0)	DAMAGED / WRONG_ITEM / SHORT_EXPIRY / OVER_QTY etc.
Inspector Remarks	Text (300)	No	Free-text notes from the inspector
Inspected By	Auto — current user	—	Captured automatically, not user-editable
Inspected At	Auto — timestamp	—	Captured automatically on save of this line's inspection entry

13.9.3 Completeness Rule

Inspection is considered 'completed' for the GRN only when every single line on the GRN has an Inspection Result recorded (none left at a blank/unset state). This completeness check is the gate referenced in Section 12.

13.10 Group D — GRN: Final Approval Gate and Stock Posting (Addendum 19)

#12	**NEW FEATURE**	**Approval blocked when inspection is required but incomplete; approved GRN posts quantities to stock**

13.10.1 Confirmed Scope

Gate applies at final approval only	Per confirmed scope, the inspection-completeness gate blocks only the final Inventory Manager approval step. Earlier steps — saving a GRN as Draft, submitting for QC (FSD v3 Section 13.2) — are unaffected and behave as already specified; a GRN can be saved or move through earlier states with inspection results still incomplete. The block is enforced specifically at the POST /api/grns/{id}/approve action.

13.10.2 Approval Validation Logic

Condition at approval attempt	System behaviour
requires_inspection = No	Approval proceeds normally — no inspection-related check applies. Existing approval rules (FSD v3 Section 13, Addendum 5A) apply unchanged.
requires_inspection = Yes AND all lines have an Inspection Result recorded	Approval proceeds normally. Section 11.3 completeness rule satisfied.
requires_inspection = Yes AND one or more lines have NO Inspection Result recorded	Approval is rejected with HTTP 422 and message: 'Inspection is required for this GRN but is not complete. {N} of {Total} items have not been inspected.' The approval action does not execute; GRN status remains unchanged.

13.10.3 Stock Posting on Approval

This formalises the existing behaviour from FSD v3 Section 13.2 (Step 10) and Addendum 5A, now explicitly gated by the rule above:

- On successful approval (gate passed or not applicable), the system posts accepted quantities to inventory exactly as already specified: InventoryItem.qty_on_hand incremented by qty_accepted per line, within the same atomic transaction as the approval action.
- POLine.qty_received is updated by the accepted quantity, and PO header status is recalculated per the existing RecalculatePoStatus() engine (Addendum 5A Section 3.3) — unchanged by this addendum.
- If requires_inspection = Yes, the posted qty_accepted per line is taken from the Inspection Results grid (Section 11.2) rather than the basic receiving entry, since the inspection result is the authoritative accepted quantity once inspection has occurred.

- If requires_inspection = No, the posted qty_accepted is taken directly from the basic GRN line entry (qty_received, with any manually-noted qty_rejected subtracted) since no formal inspection step occurred.

13.10.4 Updated GRN Field Definitions Summary

Field	Type	Required	Notes
received_at	DateTime	Yes	Replaces received_date (Section 9). Date AND time.
requires_inspection	Boolean	Yes	Default true. Locks once submitted for approval (Section 10).
inspection_completed_at	DateTime	No (auto)	Set automatically when the last required line inspection result is recorded.

— End of Addendum —

14 System Logging Requirements

The SMS must maintain comprehensive, tamper-proof logs across four distinct logging layers: Application Logs, Audit Logs, Security Logs, and Error Logs. Each layer has a defined purpose, retention policy, storage strategy, and access control.

14.1 Logging Architecture Overview

Log type	Purpose	Storage	Retention	Access
Audit Log	Business actions (CRUD, approvals)	Database (dedicated table)	7 years (regulatory)	Admin + Auditor only
Application Log	System events, performance, info	File system + Log aggregator	90 days	Admin + DevOps
Security Log	Auth events, access violations	Separate secure store	2 years	Admin only
Error Log	Exceptions, stack traces, failures	File system + Alerting tool	180 days	Admin + DevOps
API Access Log	All API calls, response codes, latency	Log aggregator	30 days	Admin + DevOps
Job / Scheduler Log	Background tasks, batch jobs	Database + File	60 days	Admin + DevOps

14.2 Audit Log — Detailed Specification

The Audit Log is the primary compliance and accountability record. Every business action that creates, modifies, approves, or deletes data must generate an audit entry. Audit records are write-once and cannot be updated or deleted by any user including System Admin.

Auditable Events by Module

Module	Events logged
Supplier Mgmt	Create supplier, edit supplier, approve supplier, blacklist supplier, add/edit contract, delete contact
Demand Planning	Create PR, edit PR, submit PR, approve PR, reject PR, convert PR to PO
Procurement	Create PO, edit PO, submit PO, L1 approve, L2 approve, L3 approve, reject PO, send to supplier, receive acknowledgement, cancel PO, amend PO
Inventory	Create product, edit product, stock adjustment (all), reorder rule change, transfer order, cycle count submission
Warehouse / GRN	Create GRN, submit GRN, QC decision, approve GRN, reject GRN, GRN stock posted, create return order
Order Mgmt	Create order, confirm order, cancel order, status change
Logistics	Create shipment, update tracking, upload POD, delivery confirmation, return processed
Finance	Create invoice, 3WM match result, approve invoice, reject invoice, schedule payment, process payment, payment reversal
System / Auth	User login, logout, failed login, password change, role change, user created, user deactivated, settings change

14.3 Audit Log — Field Definitions (Extended)

Field name	Data type	Required	CRUD	Lookup / values	Notes
audit_id	UUID	Yes	Auto	System generated	PK — immutable
log_timestamp	DateTime (UTC)	Yes	Auto	Microsecond precision	Time of event
user_id	FK → User	Yes	Auto	Logged-in user	Who performed action
user_name	Text (100)	Yes	Auto	Snapshot at time of event	Preserved even if user renamed
user_role	Text (50)	Yes	Auto	Role at time of action	Snapshot
session_id	UUID	Yes	Auto	Active session token	Trace across a session
ip_address	Text (45)	Yes	Auto	IPv4 or IPv6	User's IP address
device_info	Text (200)	No	Auto	Browser / OS / user-agent	Client metadata
module	Looku	Yes	Auto	All modules	Which

Field name	Data type	Required	CRUD	Lookup / values	Notes
	p				system module
action_type	Lookup	Yes	Auto	CREATE/READ/UPDATE/DELETE/APPROVE/REJECT/LOGIN/EXPORT	Category of action
entity_type	Text (50)	Yes	Auto	PO / GRN / Invoice / Supplier…	Object type
entity_id	UUID	Yes	Auto	Record ID	Which specific record
entity_ref	Text (50)	No	Auto	e.g. PO-2026-00123	Human-readable reference
field_name	Text (100)	No	Auto	Which field changed	Blank for create/delete
old_value	Text (1000)	No	Auto	Value before change	Null on create
new_value	Text (1000)	No	Auto	Value after change	Null on delete
action_status	Lookup	Yes	Auto	Success / Failed / Blocked	Result of action
failure_reason	Text (300)	No	Auto	e.g. Validation error / Unauthorised	If status = Failed
request_id	UUID	No	Auto	API request correlation ID	For API-sourced actions
notes	Text (500)	No	Auto or manual	Contextual detail	—

14.4 Application Log — Log Levels and Events

Level	When to use	Example events	Alert triggered
DEBUG	Development / diagnostics only	Query execution, variable states, loop iterations	No — Dev only
INFO	Normal application events	User logged in, PO created, batch job started, email sent	No
WARN	Unexpected but recoverable	Slow query (>1s), retry attempt, cache miss rate high, deprecated API call	No (monitored)
ERROR	Recoverable system error	Email delivery failure, external API timeout, validation exception	Yes — DevOps alert

Level	When to use	Example events	Alert triggered
CRITICAL	System-threatening failure	Database unreachable, payment gateway down, out-of-memory	Yes — PagerDuty / SMS

14.5 Security Log — Events Specification

Security event	Severity	Immediate alert	Data captured
Successful login	INFO	No	User, IP, timestamp, device
Failed login (single attempt)	WARN	No	User, IP, timestamp, reason
Failed login (5+ consecutive)	HIGH	Yes — Admin	User, IP, all timestamps, lockout triggered
Account locked out	HIGH	Yes — Admin	User, IP, lockout time, trigger count
Password reset request	INFO	No	User, IP, timestamp, method
Unauthorised resource access	HIGH	Yes — Admin	User, IP, resource requested, role vs required
Role / permission change	CRITICAL	Yes — Admin	Changed by, target user, old role, new role
Admin privilege granted	CRITICAL	Yes — Admin + CEO	All details above
Bulk data export	WARN	Yes — Admin	User, export type, record count, filters, timestamp
Login from new country / IP	WARN	Yes — User (email)	User, IP, geo-location, timestamp
Session token reuse after logout	CRITICAL	Yes — Admin	Session ID, user, IP, timestamps
API key created or revoked	HIGH	Yes — Admin	Key ID, user, scope, timestamp

14.6 Log Retention, Archival, and Compliance

- Audit Logs: 7-year retention (regulatory minimum for financial records). After 2 years, logs are archived to cold storage but remain queryable.
- Security Logs: 2-year online retention, then archived for an additional 3 years.
- Application / Error Logs: 90–180 days online, then purged unless an open incident references them.
- All log storage must be separate from application data to prevent tampering.
- Log integrity: each audit record includes a SHA-256 hash of its own content. A log verification job runs nightly to detect any tampering.

- Logs must not store raw passwords, full credit card numbers, or full bank account numbers.
- Log export: Admin can export audit logs as CSV/PDF for regulatory inspection. All export actions are themselves audit-logged.
- Log ingestion must not degrade application response time. Logging is asynchronous (fire-and-forget) via a message queue.

15 Caching Requirements

Caching is required across multiple layers of the SMS to ensure performance targets are met under concurrent multi-user load. This section defines what data should be cached, at which layer, with what expiry policy, and how cache invalidation is handled.

15.1 Caching Architecture — Layers

Cache layer	Technology	Scope	Location	Use case
Browser cache	HTTP Cache-Control headers	Static assets only	Client browser	JS, CSS, images — 30-day max-age
CDN cache	CDN (e.g. CloudFront)	Public static assets	Edge nodes	Report PDFs, product images
Application cache	Redis (recommended) / Memcached	Shared server-side data	App server / Redis cluster	Lookups, user sessions, computed data
Database query cache	DB-level / ORM cache	Expensive read queries	DB server	Stock totals, approval state
In-memory cache	Local process memory	Per-instance, volatile	Application process	Config, constants, role maps

15.2 Cache Objects — Detailed Specification

Cache object	Cache key pattern	TTL	Layer	Invalidation trigger
All lookup lists	lookup:{type}	24 hours	Redis	Admin edits any lookup value
Country / currency lists	ref:countries, ref:currencies	7 days	Redis	Admin updates reference data
User session + permissions	session:{user_id}:{session_id}	30 min (sliding)	Redis	Logout, role change, password change
User role permission map	role:{role_id}:permissions	1 hour	Redis	Any role permission updated

Cache object	Cache key pattern	TTL	Layer	Invalidation trigger
Product catalog (full list)	products:catalog	1 hour	Redis	Product created, edited, or deactivated
Single product detail	product:{product_id}	2 hours	Redis	Product record updated
Stock levels per warehouse	stock:{warehouse_id}:{product_id}	5 minutes	Redis	GRN approved, stock adjustment, PO receipt
Reorder alert list	reorder:alerts	15 minutes	Redis	Any stock transaction that affects qty
PO approval state	po:approval:{po_id}	Until resolved	Redis	Any approval action taken on PO
Supplier list (active only)	suppliers:active	2 hours	Redis	Supplier status change
Single supplier detail	supplier:{supplier_id}	4 hours	Redis	Supplier record updated
Dashboard KPI totals	dashboard:kpi:{user_id}	10 minutes	Redis	Any significant transaction
Report results (generated)	report:{report_type}:{params_hash}	30 minutes	Redis	Underlying data changes or TTL expires
Approval threshold config	config:approval:thresholds	1 hour	In-memory	Admin updates threshold settings
System-wide settings	config:system	1 hour	In-memory	Admin updates system settings
Static UI assets (JS/CSS)	—	30 days	Browser + CDN	New deployment (cache-busted by hash)

15.3 Cache Invalidation Strategy

Event-Driven Invalidation (primary strategy)

Cache entries are invalidated immediately when the underlying data changes. The application publishes domain events (e.g. PO.Approved, Stock.Adjusted) to an internal event bus. Cache invalidation handlers subscribe to these events and delete or refresh the relevant keys.

Domain event	Cache keys invalidated	Scope
Supplier.StatusChanged	suppliers:active, supplier:{id}	Supplier module

Domain event	Cache keys invalidated	Scope
Product.Updated	product:{id}, products:catalog, stock:{wh}:{id}	Inventory module
Stock.Adjusted	stock:{warehouse_id}:{product_id}, reorder:alerts	Inventory module
GRN.Approved	stock:{warehouse_id}:{product_id}, reorder:alerts, dashboard:kpi:*	Warehouse + Dashboard
PO.ApprovalStateChanged	po:approval:{po_id}, dashboard:kpi:*	Procurement + Dashboard
PO.Approved	po:approval:{po_id}	Procurement module
Lookup.Updated	lookup:{type}, ref:*	System-wide
User.RoleChanged	session:{user_id}:*, role:{role_id}:permissions	Auth / Security
Config.ThresholdsUpdated	config:approval:thresholds	Approval engine
Report.Requested	report:{type}:{hash} — force refresh	Reporting module

TTL-Based Expiry (fallback strategy)

All cache entries have a maximum TTL as defined in section 15.2. Even if an invalidation event is missed (e.g. direct DB update by admin), the cache entry will expire and be refreshed on the next access. TTL is the safety net, not the primary invalidation mechanism.

15.4 Cache Consistency Rules

- Stock quantity caches (stock:{wh}:{product}) have a maximum TTL of 5 minutes. For financial transactions (invoice matching, payment processing), the system always reads directly from the database, bypassing cache.
- User session caches use a sliding TTL — each authenticated API request resets the 30-minute clock. An explicit logout immediately deletes the session cache key.
- If a role permission cache (role:{role_id}:permissions) is stale when a permission check fails, the system falls back to a database read and refreshes the cache. It does not deny access based on a cold cache.
- Cache warm-up: on application startup, the system pre-loads the following keys into Redis: all lookup lists, active supplier list, active product catalog, approval threshold config, system settings.
- Cache stampede prevention: use a mutex lock (Redis SETNX) when a high-traffic cache key expires, ensuring only one process rebuilds it while others wait.
- No sensitive data in cache: passwords, bank account numbers, and payment instrument details must never be stored in Redis or any shared cache layer.
- Redis persistence: configure Redis with AOF (Append-Only File) persistence to survive restarts. Redis must not be used as the only store for any critical data.

15.5 Caching — Monitoring and Alerting

Metric	Warning threshold	Action
Cache hit rate (Redis)	Below 85%	Investigate which keys are missing or

Metric	Warning threshold	Action
		expiring too fast
Cache memory usage	Above 80% of allocated RAM	Increase Redis memory or reduce TTLs on large objects
Redis connection pool exhausted	Above 90% utilisation	Scale Redis cluster or increase connection pool size
Cache invalidation lag	Above 2 seconds	Check event bus throughput, investigate consumer backlog
Cold cache (post-restart)	Warm-up duration > 60 seconds	Optimise pre-load batch size
Session cache miss rate	Above 5%	Check TTL config and load balancer session stickiness

15A Module — Material Issue & Consumption / MIC (Addendum 20)

FSD v3 and its addenda fully specify how stock enters the warehouse (PR -> PO -> GRN, Section 5A) and how stock returns to a supplier (Section 8A), but not how stock is consumed internally by projects, departments, or production. This section adds the Material Issue & Consumption module: request, approval, reservation, issue, consumption, return, wastage, and project/department cost allocation. Core principle: GRN is the only inbound stock document and Material Issue is the only outbound stock document — no module updates InventoryItem quantities directly; every movement is mediated by the Stock Ledger. New schema: material_schema.

15A.1 Purpose and End-to-End Flow

FSD v3 and its addendums fully specify how stock enters the warehouse (PR -> PO -> GRN, Addendum 5A) and how stock returns to a supplier (Addendum 8A). They do not specify how stock leaves the warehouse for legitimate internal consumption — issuing material to a project site, a department, a maintenance job, or an employee. This addendum closes that gap by introducing the Material Issue Request (MIR) module: the complete consumption-side counterpart to the procurement-side modules already specified.

Core principle	Inventory quantities are never updated directly by any screen or service. Every increase or decrease to InventoryItem.qty_on_hand happens exclusively through a write to the Stock Ledger (Section 15 of this addendum), inside the same transaction as the business action that caused it. This applies equally to GRN receipts (already true per Addendum 5A) and now to every consumption-side transaction introduced here.

15A.1.1 End-to-End Flow

#	Stage	What happens
1	**GRN**	Goods received and posted to warehouse stock — fully specified in Addendum 5A. Stock Ledger entry: type GRN_RECEIPT, quantity_in.
2	**Warehouse Stock**	Stock sits as available InventoryItem.qty_on_hand, ready to be

#	Stage	What happens
		requested.
3	**Material Issue Request (MIR)**	Requester (Project Manager, Department Head, Maintenance, or Employee) raises a request for specific items and quantities (Section 2).
4	**Approval**	MIR routes through a configurable approval chain, with rules escalating by value (Section 3).
5	**Stock Reservation**	On approval, the approved quantity is reserved against InventoryItem.qty_reserved — physically still in the warehouse, but earmarked (Section 5).
6	**Material Issue Voucher**	Warehouse staff physically issues the material against the approved/reserved request, recording batch/serial where applicable (Section 6).
7	**Project / Site Store**	Issued material moves to the consuming location — a project site, department, or maintenance store (Section 13).
8	**Consumption**	Material is actually used. For direct-consumption categories, issue and consumption happen in the same transaction (Section 12).
9	**Return / Wastage**	Unused issued material is returned to the warehouse, or wastage is recorded with reason and approval (Sections 11, 14).
10	**Project / Department Costing**	Every issue and consumption event posts a cost entry against the project or department, feeding cost reporting (Sections 9, 10).

15A.2 Material Issue Request (MIR)

The MIR is the formal request for material from the warehouse. It is the consumption-side equivalent of the Purchase Requisition (FSD v3 Section 4) — the same design discipline applies: a header capturing who/why/when, and lines capturing what.

15A.2.1 MIR Header — Field Definitions

Field	Type	Required	Notes
request_no	Text(20)	Yes (auto)	MIR-YYYY-NNNNN
request_date	Date	Yes	Default: today
request_type	Lookup	Yes	PROJECT / DEPARTMENT / MAINTENANCE / EMPLOYEE
requested_by	FK -> Users	Yes (auto)	Current user
project_id	FK -> Projects	Conditional	Required when request_type = PROJECT. See Section 2.2 for the new Projects master.
department	Lookup	Conditional	Required when request_type = DEPARTMENT or EMPLOYEE
maintenance_ref	Text(100)	Conditional	Required when request_type = MAINTENANCE — e.g. work order or asset reference
required_date	Date	Yes	When material is needed on site

Field	Type	Required	Notes
priority	Lookup	Yes	LOW / NORMAL / HIGH / URGENT — reuses existing Priority lookup (FSD v3 Section 4.8)
remarks	Text(500)	No	Free-text justification
estimated_value	Decimal	Yes (auto)	Sum of line (requested_qty x last known unit_cost). Drives the approval tier (Section 3).
status	Lookup	Yes (auto)	DRAFT / PENDING_APPROVAL / APPROVED / PARTIALLY_APPROVED / REJECTED / PARTIALLY_ISSUED / FULLY_ISSUED / CLOSED / CANCELLED
warehouse_id	FK -> Warehouses	Yes	Source warehouse the material is requested from
created_at / created_by	DateTime / FK	Yes	Standard audit fields

15A.2.2 New Master: Projects

request_type = PROJECT requires a project reference. No project master currently exists in FSD v3. This addendum introduces a minimal Projects table sufficient to support MIR and project costing (Section 9); a full Project Management module is out of scope here.

Field	Type	Required	Notes
project_id	UUID	Yes	Primary key
project_code	Text(20)	Yes (UQ)	e.g. PRJ-2026-001
project_name	Text(200)	Yes	
project_manager_id	FK -> Users	Yes	Drives the Project Manager approval step, Section 3
site_warehouse_id	FK -> Warehouses	No	If the project has a dedicated site store, Section 13
status	Lookup	Yes	ACTIVE / COMPLETED / ON_HOLD / CANCELLED
budget_amount	Decimal	No	Optional — for future budget-vs-actual reporting

15A.2.3 MIR Line — Field Definitions

Field	Type	Required	Notes
line_id	UUID	Yes	Primary key
request_id	FK -> MIR	Yes	Parent header
line_no	Integer	Yes	Sequence
product_id	FK -> Products	Yes	From existing Product catalog (FSD v3 Section 6.2)
uom	Lookup	Yes (auto)	Inherited from Product, can differ if a

Field	Type	Required	Notes
			conversion is configured
requested_qty	Decimal	Yes	
approved_qty	Decimal	No (auto)	Set during approval (Section 3) — may be less than requested_qty
issued_qty	Decimal	No (auto)	Running total issued so far across one or more Material Issue Vouchers (Section 6)
pending_qty	Decimal	No (computed)	approved_qty minus issued_qty
purpose	Text(300)	No	Free-text — what the item will be used for on this line
line_status	Lookup	Yes (auto)	OPEN / PARTIALLY_ISSUED / FULLY_ISSUED / CANCELLED

15A.3 Approval Workflow

Implementation note	MIR approval is implemented as a registered interface on the Dynamic Approval Workflow Engine (Addendum 16) — not as new bespoke approval code. This section defines the business rules; the engine (workflow_schema entities, resolution logic, escalation) is reused as-is from Addendum 16.

15A.3.1 Configurable Approval Levels

The standard chain has up to four sequential steps, configured as a WorkflowDefinition with interface_code = MIR:

Step	Approver	Condition
1	**Requester**	Implicit — creates and submits the MIR. Not a formal approval step, included for completeness of the chain diagram.
2	**Supervisor**	approver_type = DEPARTMENT (resolves the requester's department head) — applies to all request types
3	**Project Manager**	approver_type = USER, resolved dynamically from Projects.project_manager_id of the requested project — applies only when request_type = PROJECT. Skipped (skip_condition) for DEPARTMENT/MAINTENANCE/EMPLOYEE requests.
4	**Store Manager / Warehouse Manager**	approver_type = ROLE (Inventory Manager) — final step, confirms warehouse can fulfil before stock is reserved

15A.3.2 Value-Based Escalation Rule

Threshold rule	estimated_value < 50,000 -> Manager approval only (Supervisor + Store Manager, Project Manager step included only if request_type = PROJECT). estimated_value >= 50,000 -> Manager + Director approval — an additional Director-level step (approver_type = ROLE, Director) is inserted after Store Manager and before final approval. Threshold is configurable Admin-side via the workflow's condition_value_min/max fields (Addendum 16 Section 4.2), default 50,000.

15A.3.3 Approval Action — Partial Approval

Unlike a simple Approve/Reject, MIR approval supports partial quantity approval at the line level — an approver may reduce approved_qty below requested_qty on any line (e.g. supervisor approves 60 of 100 requested bags due to budget or availability concerns) without rejecting the line entirely.

Per-line approval action	Each approval step can set approved_qty independently per line, defaulting to requested_qty if left unchanged
Header status when any line is reduced	MIR.status = PARTIALLY_APPROVED if at least one line's approved_qty < requested_qty, else APPROVED
Final approver sees prior approvals	If Supervisor reduced a quantity, the Project Manager and Store Manager steps see the Supervisor's approved_qty as the new ceiling — later steps can reduce further but never increase beyond the previous step's approved figure
Rejection vs reduction	A line can also be explicitly rejected (approved_qty = 0) without rejecting the entire MIR — other lines proceed normally

15A.4 Stock Availability Validation

Before any approval step can approve a line (whether in full or partially), the system validates and displays live availability so the approver makes an informed decision rather than approving blind.

15A.4.1 Displayed Figures Per Line

Figure	Source	Notes
Current Stock	InventoryItem.qty_on_hand for (product_id, warehouse_id)	Physical quantity in the warehouse, regardless of any reservations
Reserved Stock	InventoryItem.qty_reserved	Already earmarked by other approved-but-not-yet-issued requests, confirmed Sales/Transfer Orders, etc.
Available Stock	InventoryItem.qty_available (computed: qty_on_hand minus qty_reserved)	What can genuinely still be promised to this request
Reorder Level	Products.reorder_point (FSD v3 Section 6.2)	Context for the approver — approving this request may push stock below the reorder trigger

15A.4.2 Worked Example

Figure	Value	Notes
Current Stock	100	Physical qty_on_hand
Reserved	20	Already committed elsewhere
Available	**80**	100 minus 20 — what this request can draw from

15A.4.3 Validation Rule

Hard rule	requested_qty (or the reduced approved_qty being set by the current approval step)

	must be <= qty_available at the time of that approval action. If the approver attempts to set approved_qty greater than current qty_available, the system blocks the action with an inline error showing the live availability figures from Section 4.1 — the approver must reduce the quantity or wait for availability to change.

This check is re-evaluated fresh at each approval step, not only once at MIR submission — availability can change between the Supervisor's approval and the Store Manager's approval if other requests are approved in the interim, so each step re-validates against current figures.

15A.5 Stock Reservation

This section formalises the reservation mechanism referenced throughout this addendum and is the consumption-side counterpart to how Addendum 5A handles inbound quantity tracking.

15A.5.1 Reservation Trigger and Effect

Rule	On final MIR approval (last required step in the resolved workflow completes), the system reserves stock: for each approved line, InventoryItem.qty_reserved += approved_qty for the (product_id, warehouse_id) pair. qty_on_hand is NOT changed at this point — the stock is still physically in the warehouse. A StockReservation record (Section 5.3) is created per line to track the reservation independently of the eventual issue.

15A.5.2 Worked Example

State	Physical (qty_on_hand)	Reserved / Available
Before MIR approval	100	Reserved 20 / Available 80
MIR for 25 units approved	100 (unchanged)	**Reserved 45 / Available 55**

15A.5.3 StockReservation — Field Definitions

Field	Type	Required	Notes
reservation_id	UUID	Yes	Primary key
request_id	FK -> MIR	Yes	
request_line_id	FK -> MIRLine	Yes	
product_id	FK -> Products	Yes	
warehouse_id	FK -> Warehouses	Yes	
reserved_qty	Decimal	Yes	Set at approval. Decrements as issues occur (Section 6).
status	Lookup	Yes (auto)	ACTIVE / PARTIALLY_RELEASED / RELEASED / CANCELLED
reserved_at	DateTime	Yes	
released_at	DateTime	No	Set when reserved_qty reaches 0 (fully issued or cancelled)

15A.5.4 Reservation Release Rules

- Reservation is released (qty_reserved decremented) at the moment of actual Material Issue (Section 6) — by the issued quantity, not the full reserved quantity, supporting partial issues (Section 7).
- If an approved MIR is cancelled before any issue occurs, the full reserved_qty is released immediately back, restoring qty_available.
- A reservation does not expire automatically by default — it remains ACTIVE until issued or explicitly cancelled. A configurable Hangfire job may optionally flag (not auto-cancel) reservations older than a configurable number of days for Store Manager review, preventing silent indefinite stock-locking.
- Partial issue against a reservation reduces StockReservation.reserved_qty by the issued amount and sets status = PARTIALLY_RELEASED; when reserved_qty reaches 0, status = RELEASED.

15A.6 Material Issue Voucher (MIV)

The Material Issue Voucher is the warehouse's physical fulfilment document against an approved MIR — the consumption-side counterpart to the GRN. It is the only document that actually decrements qty_on_hand.

15A.6.1 MIV Header — Field Definitions

Field	Type	Required	Notes
issue_no	Text(20)	Yes (auto)	MIV-YYYY-NNNNN
issue_date	Date	Yes	Default: today
warehouse_id	FK -> Warehouses	Yes	Inherited from the source MIR
request_id	FK -> MIR	Yes	Which approved request this issue is against
issued_by	FK -> Users	Yes (auto)	Current user — Warehouse Operator
received_by	FK -> Users / Text(150)	Yes	The person physically taking custody — internal user if known, else free-text name (e.g. site labourer not in the system)
status	Lookup	Yes (auto)	DRAFT / ISSUED / CANCELLED
created_at	DateTime	Yes	

15A.6.2 MIV Line — Field Definitions

Field	Type	Required	Notes
issue_line_id	UUID	Yes	Primary key
issue_id	FK -> MIV	Yes	
request_line_id	FK -> MIRLine	Yes	Which MIR line this fulfils
product_id	FK -> Products	Yes	
reserved_qty	Decimal	Yes (auto)	Snapshot of StockReservation.reserved_qty at time of issue, for reference
issued_qty	Decimal	Yes	Quantity physically handed over on this voucher — may be less than reserved_qty

Field	Type	Required	Notes
			(Section 7)
unit_cost	Decimal	Yes (auto)	Current weighted-average or FIFO cost per InventoryItem.unit_cost (FSD v3 Section 6.3) at time of issue
line_value	Decimal	Yes (computed)	issued_qty x unit_cost — feeds project/department costing (Sections 9, 10)
batch_number	Text(50)	Conditional	Required if Products.is_batch_tracked = true
serial_numbers	Text[] / JSON	Conditional	Required if Products.is_serial_tracked = true — one or more specific serials per Section 8
bin_id	FK -> Bins	No	Specific bin/shelf/rack the item was picked from, per Addendum 19 Section 7 structure

15A.6.3 Issue Posting — Atomic Effects

On MIV status = ISSUED (the action that finalises a voucher, analogous to GRN approval), the following happen in one transaction:

- InventoryItem.qty_on_hand -= issued_qty for each line's (product_id, warehouse_id)
- InventoryItem.qty_reserved -= issued_qty (releasing the matching portion of the reservation)
- StockReservation.reserved_qty -= issued_qty; status updated per Section 5.4
- MIRLine.issued_qty += issued_qty; line_status recalculated (OPEN / PARTIALLY_ISSUED / FULLY_ISSUED)
- MIR header status recalculated: PARTIALLY_ISSUED if any line not fully issued, FULLY_ISSUED if all lines fully issued
- One StockLedger entry written per line: transaction_type = MATERIAL_ISSUE, quantity_out = issued_qty (Section 15)
- Project or Department cost ledger entry posted per line_value (Sections 9, 10)

15A.7 Partial Issue

Common in construction and project environments where the warehouse cannot always fulfil a full request in one go.

15A.7.1 Worked Example

Field	Value	Notes
Requested	100 Cement Bags	MIRLine.requested_qty
Available at issue time	60	Warehouse cannot fulfil in full today
Issued (this voucher)	**60**	MIV line issued_qty = 60
Pending	40	MIRLine.pending_qty = approved_qty minus issued_qty

15A.7.2 Behaviour

- The MIR line remains OPEN (specifically line_status = PARTIALLY_ISSUED) — it is not closed or cancelled by a partial issue.
- A second (and subsequent) Material Issue Voucher can be raised against the same MIR line whenever more stock becomes available, up to the remaining pending_qty.
- The MIR header status remains PARTIALLY_ISSUED until every line reaches FULLY_ISSUED, mirroring the multi-GRN pattern already established for inbound POs in Addendum 5A.
- The original reservation (Section 5) is reduced incrementally with each partial issue, not released in full on the first voucher — the remaining 40 units stay reserved (qty_reserved) until issued or the MIR is explicitly cancelled.
- There is no maximum number of partial issue vouchers against a single MIR line.

15A.8 Batch / Serial Selection

For controlled inventory items — laptops, generators, medicine, electrical equipment — the Material Issue Voucher requires the warehouse staff to select the specific batch or serial number(s) being issued, not just a quantity.

15A.8.1 Behaviour by Tracking Type

Product flag	MIV line behaviour
is_batch_tracked = true (FSD v3 Section 6.2)	On entering issued_qty, the system presents available InventoryItem rows for this product+warehouse grouped by batch_number and expiry_date (oldest expiry first, suggesting FEFO — First-Expired-First-Out). Warehouse staff selects which batch(es) to draw from; if quantity spans multiple batches, multiple sub-rows are created on the same MIV line, each with its own batch_number.
is_serial_tracked = true	The system presents individually available serial numbers (InventoryItem rows with a populated serial_number and qty_on_hand = 1 each, per FSD v3 Section 6.3) for this product+warehouse. Warehouse staff selects exactly issued_qty distinct serial numbers from the available list — cannot select more serials than issued_qty or fewer.
Neither flag set	Standard quantity-only entry, no batch/serial selection UI shown.

15A.8.2 Audit Trail

Every batch or serial selected on an MIV line is recorded permanently against that line (not just the aggregate quantity) and is queryable by batch/serial number across its full lifecycle: which GRN it arrived on, which warehouse/bin it sat in, which MIR/MIV issued it, and to which project or department — a full chain of custody, essential for high-value or regulated items (e.g. medicine batches, asset-tagged equipment).

15A.9 Project Cost Allocation

When request_type = PROJECT, every Material Issue automatically posts a cost entry against that project, enabling real-time project costing without a separate manual journal entry.

15A.9.1 Worked Example

Field	Value	Notes
Project	Project A	From MIR.project_id
Issued	20 Cement Bags	MIV line issued_qty
Unit Cost	1,200	InventoryItem.unit_cost snapshot
Cost Posted	**24,000**	20 x 1,200, posted to Project Cost Ledger

15A.9.2 ProjectCostLedger — Field Definitions

Field	Type	Required	Notes
ledger_id	UUID	Yes	Primary key
project_id	FK -> Projects	Yes	
transaction_type	Lookup	Yes	MATERIAL_ISSUE / MATERIAL_RETURN (negative) / WASTAGE
reference_type	Text(20)	Yes	MIV / RETURN / WASTAGE
reference_id	UUID	Yes	FK to the source document
product_id	FK -> Products	Yes	
quantity	Decimal	Yes	Positive for issue/wastage cost, negative for return credit
unit_cost	Decimal	Yes	Snapshot at time of transaction
amount	Decimal	Yes	quantity x unit_cost
posted_at	DateTime	Yes	

15A.10 Department Cost Allocation

Mirrors Section 9 exactly, for request_type = DEPARTMENT and request_type = EMPLOYEE (an employee request is allocated to that employee's department), posting to a DepartmentCostLedger table with the same structure as ProjectCostLedger, substituting department for project_id and cost_center for any GL mapping.

15A.10.1 Worked Example

Field	Value	Notes
Department	IT Department	From MIR.department
Issued	5 Laptops	Serial-tracked, Section 8
Cost Center	**IT**	System posts departmental expense automatically

15A.10.2 DepartmentCostLedger — Field Definitions

Identical structure to ProjectCostLedger (Section 9.2) with department (Lookup) and cost_center (Text(50)) replacing project_id. Both ledgers share the same posting trigger logic — a single internal

CostAllocationService.PostIssueCost(miv_line) call branches on MIR.request_type to write to the correct ledger.

15A.11 Material Return

Issued material that is not fully consumed can be formally returned to the warehouse, reversing the stock and cost effects of the original issue proportionally.

15A.11.1 Worked Example

Field	Value	Notes
Issued	50	Original MIV issued_qty
Used	40	Recorded via Consumption, Section 12/13
Returned	**10**	Material Return Voucher

15A.11.2 MaterialReturn — Field Definitions

Field	Type	Required	Notes
return_no	Text(20)	Yes (auto)	MRV-YYYY-NNNNN
return_date	Date	Yes	
issue_id	FK -> MIV	Yes	Original issue voucher being returned against
warehouse_id	FK -> Warehouses	Yes	Receiving warehouse for the return
returned_by	FK -> Users / Text(150)	Yes	Mirrors MIV.received_by
received_by	FK -> Users	Yes (auto)	Warehouse Operator accepting the return
reason	Lookup	Yes	EXCESS_QUANTITY / WORK_CANCELLED / WRONG_ITEM / OTHER
status	Lookup	Yes (auto)	DRAFT / RECEIVED / CANCELLED

15A.11.3 MaterialReturnDetail — Field Definitions

Field	Type	Required	Notes
return_line_id	UUID	Yes	Primary key
return_id	FK -> MaterialReturn	Yes	
issue_line_id	FK -> MIVLine	Yes	Which original issue line this return is against
product_id	FK -> Products	Yes	
returned_qty	Decimal	Yes	Must not exceed (issued_qty minus previously_returned_qty) for the source line
condition	Lookup	Yes	GOOD / DAMAGED — DAMAGED returns post to wastage rather than back to

Field	Type	Required	Notes
			usable stock (Section 14)
batch_number / serial_numbers	Text/JSON	Conditional	Same tracking rules as Section 8, must match what was originally issued

15A.11.4 Posting Effects on Receipt

- If condition = GOOD: InventoryItem.qty_on_hand += returned_qty (stock physically usable again). Stock Ledger entry: type MATERIAL_RETURN, quantity_in.
- If condition = DAMAGED: stock is NOT returned to usable qty_on_hand. Instead a Wastage record (Section 14) is created automatically, and the Stock Ledger entry is type WASTAGE rather than MATERIAL_RETURN.
- Project/Department Cost Ledger (Sections 9/10) receives a credit entry (negative amount) for returned_qty x the original issue's unit_cost, reducing the net cost charged to that project/department.
- MIRLine.issued_qty is NOT reduced by a return — the historical issue record remains accurate; the return is its own offsetting transaction, consistent with the never-update-directly principle (Section 15).

15A.12 Direct Consumption Entry

Some organisations, and some material categories even within an organisation that does track project stores, do not need the issue-then-consume-later distinction — the item is issued and consumed in the same act. Cleaning material, office supplies, and fuel are typical examples.

15A.12.1 Behaviour

Rule	A Product can be flagged is_direct_consumption = true (new field on Products, FSD v3 Section 6.2). When an MIR line references such a product, the resulting Material Issue Voucher automatically creates a matching MaterialConsumption record (Section 13) in the same transaction as the issue posting — there is no separate manual consumption-entry step for these items. The Issue then Consume Immediately sequence happens as one atomic action from the user's perspective, though internally it remains two related records (MIV + MaterialConsumption) for reporting consistency.

15A.12.2 Effect on Reporting

Because consumption is recorded immediately, direct-consumption items show consumed_qty = issued_qty and balance_qty = 0 at all times in the Material Consumption Register (Section 13) — there is no pending-consumption state to track for these categories, simplifying reporting for high-volume, low-value consumables.

15A.13 Material Consumption Register

Tracks the full lifecycle of issued material from issue through to final disposition (consumed, returned, or wasted), giving a complete picture per item per project/department.

15A.13.1 Site Warehouse / Project Store Flow

For request_type = PROJECT requests where Projects.site_warehouse_id is populated, issued material is conceptually transferred to that site warehouse before consumption, using the same transaction primitives already defined:

Transfer Out	Main warehouse InventoryItem.qty_on_hand decreases by issued_qty (the MIV posting itself, Section 6.3, IS the transfer-out when a site warehouse exists)
Transfer In	Site warehouse InventoryItem.qty_on_hand increases by the same quantity — a second Stock Ledger entry, type TRANSFER_IN, written in the same transaction as the MIV posting
Consumption	Recorded against the site warehouse's stock as material is actually used on-site — decrements the site warehouse qty_on_hand further and creates the MaterialConsumption record

Where no site warehouse is configured for the project (the common case for departments and most smaller projects), the material is treated as consumed directly from the main warehouse's perspective — issue and eventual consumption tracking both reference the main warehouse, and the Transfer Out/In pair above is simply skipped.

15A.13.2 MaterialConsumption — Field Definitions

Field	Type	Required	Notes
consumption_id	UUID	Yes	Primary key
issue_line_id	FK -> MIVLine	Yes	Which issue this consumption is reducing
project_id / department	FK / Lookup	Conditional	Inherited from the source MIR
product_id	FK -> Products	Yes	
consumed_qty	Decimal	Yes	
consumed_date	Date	Yes	
recorded_by	FK -> Users	Yes (auto)	
notes	Text(300)	No	

15A.13.3 MaterialConsumptionDetail — Register View

The register itself is a computed/reporting view joining Issue, Consumption, Return, and Wastage per item per request, not a separately maintained table — each column below is derived as specified:

Item	Issued	Consumed	Returned	Wastage	Balance	Source
Cement	100	90	5	5	**0**	

Balance Qty = Issued minus Consumed minus Returned minus Wastage. A non-zero balance indicates material still on-site/in-department awaiting consumption, return, or wastage recording — useful for identifying stale issues that need follow-up.

15A.14 Wastage Tracking

A construction-environment necessity — broken tiles, damaged cement, and similar loss must be formally recorded with approval, both for cost accuracy and for identifying recurring quality or handling issues.

15A.14.1 Wastage — Field Definitions

Field	Type	Required	Notes
wastage_id	UUID	Yes	Primary key
wastage_no	Text(20)	Yes (auto)	WST-YYYY-NNNNN
project_id / department	FK / Lookup	Conditional	Where the wastage occurred
issue_line_id	FK -> MIVLine	Yes	Which issued material this wastage is against
product_id	FK -> Products	Yes	
wastage_qty	Decimal	Yes	
reason	Lookup	Yes	BROKEN / DAMAGED / EXPIRED / HANDLING_LOSS / OTHER
reason_detail	Text(300)	No	Free-text elaboration
approved_by	FK -> Users	Yes	Required before wastage is posted — see Section 14.2
approved_at	DateTime	No (auto)	
status	Lookup	Yes (auto)	PENDING_APPROVAL / APPROVED / REJECTED

15A.14.2 Approval and Posting

- Wastage requires approval (Supervisor or Store Manager, configurable) before it is posted — it cannot be self-recorded and immediately finalised by the same person who issued the material, to prevent unreviewed loss write-offs.
- On approval, a Stock Ledger entry is written: type WASTAGE, quantity_out = wastage_qty (the material is permanently gone, not returned to usable stock).
- The cost of approved wastage posts to the Project/Department Cost Ledger as an additional charge — wastage is a real cost to the project, not a free write-off.
- Wastage arising automatically from a DAMAGED Material Return (Section 11.4) is pre-populated with reason = DAMAGED and still requires the same approval step before posting.

15A.15 Stock Ledger Impact

Foundational rule of this addendum	Stock is never updated directly. Every module — GRN, Material Issue, Material Return, Stock Adjustment, Transfer — writes through one shared service that updates InventoryItem and the StockLedger together, in the same transaction, every time. This is the same InventoryLedgerEntries design already specified in earlier inventory work and is formally extended here to cover every consumption-side transaction type introduced by this addendum.

15A.15.1 Transaction Types and Their Ledger Effect

Transaction type	Quantity In	Quantity Out	Triggering document
GRN_RECEIPT	Y (e.g. +100)	—	GRN approval (Addendum 5A)
MATERIAL_ISSUE	—	Y (e.g. -20)	Material Issue Voucher posting (Section 6.3)
MATERIAL_RETURN	Y (e.g. +5)	—	Material Return receipt, condition = GOOD (Section 11.4)
STOCK_ADJUSTMENT	Y or —	Y or —	Manual correction (FSD v3 Section 6.4) — e.g. -2 for a count variance
TRANSFER_OUT	—	Y (e.g. -10)	Site warehouse transfer, source side (Section 13.1)
TRANSFER_IN	Y (e.g. +10)	—	Site warehouse transfer, destination side (Section 13.1)
WASTAGE	—	Y	Approved wastage posting (Section 14.2) or damaged return (Section 11.4)
RETURN_PENDING / RETURNED	—	Y (on dispatch)	Supplier Return Order, unchanged from Addendum 8A

15A.15.2 Worked Ledger Sequence

Continuing the cement example used throughout this addendum, the full ledger sequence for one product at one warehouse might read:

Transaction	Qty In	Qty Out	Running Balance
GRN	**+100**		**100**
MATERIAL_ISSUE		**-20**	**80**
MATERIAL_RETURN	**+5**		**85**
STOCK_ADJUSTMENT		**-2**	**83**
TRANSFER_OUT		**-10**	**73**
TRANSFER_IN	**+10**		**83**

Note the last two rows: TRANSFER_OUT and TRANSFER_IN are written against two different InventoryItem rows (source warehouse and destination warehouse respectively) but both fire from the same originating action — the running balance shown here is for the source warehouse only; the destination warehouse has its own independent running balance starting from 0 or its prior figure.

15A.16 Reports

Report name	Key metrics	Primary access
Material Issue Register	All MIVs with status, warehouse, requester, value, date range	Store Manager, Inventory Manager
Material Consumption Report	Issued vs consumed vs balance by item, per the	Project Manager, Department Head

Report name	Key metrics	Primary access
	register view in Section 13.3	
Project Consumption Report	Total material cost by project, broken down by item, sourced from ProjectCostLedger (Section 9.2)	Project Manager, Finance
Department Consumption Report	Total material cost by department/cost centre, sourced from DepartmentCostLedger (Section 10.2)	Department Head, Finance
Stock Movement Report	All transaction types in/out by product/warehouse over a date range	Inventory Manager
Stock Ledger Report	Full chronological ledger per product+warehouse with running balance (Section 15)	Inventory Manager, Auditor
Material Return Report	All returns by reason, condition (good/damaged), and project/department	Store Manager
Wastage Report	Wastage by project, item, reason, approver, with cost impact	Project Manager, Finance, Auditor
Reserved Stock Report	Current StockReservation records by product/warehouse, age of reservation, linked MIR	Inventory Manager, Store Manager

15A.17 New Tables Summary

All tables introduced by this addendum reside in a new material_schema, owned by a MaterialDbContext, following the same schema-per-module isolation pattern as every other module in this system.

Table	Purpose	Section
Projects	Minimal project master supporting MIR and project costing	2.2
MaterialIssueRequest	MIR header	2.1
MaterialIssueRequestDetail	MIR lines	2.3
StockReservation	Tracks approved-but-not-yet-issued stock holds	5.3
MaterialIssue	MIV header	6.1
MaterialIssueDetail	MIV lines, including batch/serial	6.2
MaterialReturn	Return voucher header	11.2
MaterialReturnDetail	Return voucher lines	11.3
MaterialConsumption	Consumption events against issued material	13.2

Table	Purpose	Section
ProjectCostLedger	Project-level cost postings from issue/return/wastage	9.2
DepartmentCostLedger	Department-level cost postings, mirrors ProjectCostLedger	10.2
Wastage	Approved wastage records	14.1

Note: the requirement list also referenced a standalone StockLedger table. Per this system's existing design (carried over from earlier inventory work), this is implemented as InventoryLedgerEntries in inventory_schema — already specified and shared across all modules including this one, rather than duplicated as a separate material_schema table. All Material Issue, Return, and Wastage postings write to that same shared ledger, per Section 15.

— End of Addendum —

16 Appendix — System-Wide Lookup Master

The following lookups are shared across multiple modules and maintained in a central lookup administration screen (Admin only CRUD).

Lookup: Country (ISO 3166 — sample, CRUD: Yes)

Code / ID	Display value	Active	Notes
PK	Pakistan	Yes	—
AE	United Arab Emirates	Yes	—
US	United States	Yes	—
GB	United Kingdom	Yes	—
CN	China	Yes	—
DE	Germany	Yes	—
SA	Saudi Arabia	Yes	—

Lookup: Currency (ISO 4217 — sample, CRUD: Yes)

Code / ID	Display value	Active	Notes
PKR	Pakistani Rupee	Yes	Base currency (default)
USD	US Dollar	Yes	—
AED	UAE Dirham	Yes	—
EUR	Euro	Yes	—
GBP	British Pound	Yes	—
SAR	Saudi Riyal	Yes	—

Lookup: Tax Code (CRUD: Yes)

Code / ID	Display value	Active	Notes
GST17	GST 17%	Yes	Pakistan standard GST
ZERO	Zero-rated (0%)	Yes	Exported goods
EXEMPT	Exempt	Yes	Exempt category goods
WHT	Withholding Tax	Yes	WH tax on services

Additional lookups introduced by the addenda are defined alongside the capability they belong to rather than repeated here: PO Status (full list) in Section 5A.10; SRO Type, Return Reason, and Debit Note Reason in Section 8A.5; and the Material Issue & Consumption lookups (Transaction Type, Wastage Reason, MIR/MIV status values, etc.) within Section 15A.

17 Non-Functional Requirements

17.1 Performance

- All list screens must load within 2 seconds for up to 10,000 records.
- Search and filter must return results within 1 second.
- Bulk operations (import, export) should handle up to 5,000 records per batch.
- Dashboard KPI widgets must render within 3 seconds including cache warm-up.

17.2 Security

- All data in transit must be encrypted via TLS 1.2+.
- Passwords must be hashed using bcrypt (min 12 rounds).
- Session timeout after 30 minutes of inactivity (configurable).
- All API endpoints must require authentication tokens (JWT or OAuth 2.0).
- PII fields (bank accounts, contact details) encrypted at rest using AES-256.
- Penetration testing required before go-live and annually thereafter.

17.3 Usability

- System must be responsive and accessible on desktop and tablet (min 768px width).
- All forms must support keyboard navigation.
- Mandatory fields highlighted with red asterisk (*).
- Inline validation on form fields before submission.
- Confirmation dialogs required for all delete and approve actions.
- Multi-language support architecture required; English as default.

17.4 Data Integrity Rules

- PO cannot be raised against an Inactive or Blacklisted supplier.
- GRN cannot be created without a corresponding approved PO.
- Invoice amount must not exceed PO grand total by more than the configured variance threshold (default 5%).
- Payment cannot be processed without a matched and approved invoice.
- Stock adjustments above a configurable threshold require a second-level approval.
- Audit trail records are immutable — no update or delete permitted by any user.
- All approval workflow state transitions must be atomic — no partial state on failure.

— End of Document —

Appendix B — Consolidation Source Map & Change Log

This appendix records where every source document's content landed in this consolidated edition, so the merge can be audited item by item against the original files.

B.1 Base Document Lineage

Supply_Management_System_FSD.docx (v1) and Supply_Management_System_FSD_v2.docx (v2) were verified section-by-section against Supply_Management_System_FSD_v3.docx: every heading and field definition present in v1 and v2 is present in v3 in equal or greater detail (v3 adds Sections 12–17 and expands Sections 3–4 with no deletions). v1 and v2 are therefore fully superseded and their content is not separately reproduced — this consolidated document is built on v3 as the base layer.
SMS_Development_Task_Register_v3.docx was excluded from this merge: it is a development/sprint-planning register (task IDs, effort estimates, unit-test counts), not a functional specification, and does not describe user-facing behaviour.

B.2 Addendum 5A — PO Delivery Lifecycle

Fully incorporated as Section 5A (5A.1 through 5A.10), inserted after Section 5 (Procurement), unchanged in content and sequence from the source addendum.

B.3 Addendum 8A — Supplier Returns Management

Fully incorporated as Section 8A (8A.1 through 8A.7), inserted after Section 8 (Warehouse Management), unchanged in content and sequence from the source addendum.

B.4 Addendum 17 — Category Fixes & Role CRUD

Item summary, verbatim from the source addendum:

This addendum addresses five items raised against the live Inventory Management and User Administration screens. Four are defect remediations against functionality already specified in FSD v3 Section 6 (Product

Category) — the navigation, view, edit, and delete behaviours did not match the original specification. One is a new feature — Role CRUD — which did not exist anywhere in FSD v3 and must be added as a new screen under the User & Access menu.

#	Type	Item	Resolved in section
1	**Defect**	Sub-Category has no separate, dedicated navigation link	Section 2
2	**Defect**	'View Sub Categories' page does not load / function from Product Categories	Section 3
3	**New Feature**	Role CRUD screen does not exist under User & Access menu	Section 4
4	**Defect**	Edit and Delete actions on Product Categories are non-functional	Section 5
5	**Defect**	Sub-Category has no Edit or Delete action at all	Section 6

Root cause note	Items 1, 2, 4, and 5 all trace back to the same gap: Sub-Category was originally modelled in FSD v3 as a child attribute accessed only through the parent Category record, with no independent menu entry, no independent list view, and no edit/delete action of its own. This addendum promotes Sub-Category to a first-class, independently manageable entity — given its own menu link, its own list screen, and full CRUD — which is also what resolves items 1, 2, and 5 simultaneously.

Mapping into this document: Item 1 -> Section 6.7; Item 2 -> Section 6.8; Item 3 (Role CRUD) -> Section 2.2; Item 4 -> Section 6.9; Item 5 -> Section 6.10.

B.5 Addendum 18 — Supplier RFQ Self-Service Portal

Fully incorporated as Section 4.11 (4.11.1 through 4.11.8), extending Section 4 (Demand Planning & Purchase Requisition), unchanged in content and sequence from the source addendum.

B.6 Addendum 19 — RFQ Award Gate, PO Creation UX, Warehouse Structure, GRN Inspection

Item summary, verbatim from the source addendum:

This addendum groups 12 raised items into four thematic groups by the module they affect. Items within a group share data model or workflow dependencies and are specified together.

#	Item	Group / Section
1	'Compare and Award' button must be disabled when no quotation has been submitted	Group A — Quotation (Sec. 2)
2	Import Lines control repositioned to top of Create PO screen with a clearer label	Group B — PO Creation UX (Sec. 3)
3	Awarded quotation supplier auto-selected and PO supplier field disabled when converting From	Group B — PO Creation UX (Sec. 4)

#	Item	Group / Section
	Quotation	
6	Per-item warehouse override on PO lines, with a single global warehouse as default	Group B — PO Creation UX (Sec. 5)
4	New Google Maps location field on Warehouse Master	Group C — Warehouse Structure (Sec. 6)
5	Warehouse Rack/Shelf/Bin management — full structural CRUD	Group C — Warehouse Structure (Sec. 7)
7	'Search Purchase Order' on Add New GRN — searchable dropdown of awarded, not-fully-received POs, with free-text fallback	Group D — GRN Workflow (Sec. 8)
8	GRN must capture receiving date AND time	Group D — GRN Workflow (Sec. 9)
9	Inspection requirement made optional, decided per-GRN at receiving time	Group D — GRN Workflow (Sec. 10)
10	Radio button on GRN screen to flag whether inspection is required for this receipt	Group D — GRN Workflow (Sec. 10)
11	Dedicated inspection-results section for all items against the PO	Group D — GRN Workflow (Sec. 11)
12	Final approval posts quantities to stock; blocked if inspection was required but not completed	Group D — GRN Workflow (Sec. 12)

Mapping into this document: Item 1 (Group A) -> Section 4.10; Item 2 (Group B) -> Section 5.7; Item 3 (Group B) -> Section 5.8; Item 6 (Group B) -> Section 5.9; Item 4 (Group C) -> Section 8.7; Item 5 (Group C) -> Section 8.8; Item 7 (Group D) -> Section 13.6; Item 8 (Group D) -> Section 13.7; Items 9 and 10 (Group D) -> Section 13.8; Item 11 (Group D) -> Section 13.9; Item 12 (Group D) -> Section 13.10.

B.7 Addendum 20 — Material Issue & Consumption Module

Fully incorporated as Section 15A (15A.1 through 15A.17), inserted after Section 15 (Caching), unchanged in content and sequence from the source addendum.